Sourcing scale for 2025

CBS Case Competition Invitational 2019

Case Me If You Can

MAERSK

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Executive summary

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Situation
A.P. Møller Maersk is a Danish transport and logistics conglomerate.
Since 1996, they have been the worlds largest container shipping company.

Opportunity
Maersk recently piloted their end-to-end logistics product in Philadelphia.
The pilot was successful and represents a significant PoC of the provision of end-to-end cold chain logistics solutions.

Question
How can Maersk leverage it's position as a leader in the shipping market to create valuable propositions for customers in the cold chain market?
And what should Maersk global cold chain strategy be for the next five years?

Source: Team analysis

With a foundation in Maersk's vision of becoming the global logistics integrator, we propose the following three initiatives

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> "We wish to become the global integrator of container logistics, connecting and simplifying our customers' supply chains."

Wrap the core

1

LAUNCH DIGITAL PLATFORM TO SOURCE NON-CORE SERVICES

Digitalize the Philadelphia pilot to enable scalability of your end-to-end offering whilst empowering customers through greater clarity and stability

Where to score

2

STRENGHTEN POC IN TOP ROUTES TO ROLL OUT GLOBALLY BY 2025

Target top 10 cold chain origins and destinations measured on average TEUs, to extend PoC before rolling out in remaining locations

Make customers want more

3

ENSURE ADOPTION FROM PARTNERS AND CUSTOMERS

Pull initial partners onto the platform whilst converting current cold chain customers from Maersk

Global consumer trends provide the foundation for a large and growing cold chain logistics market

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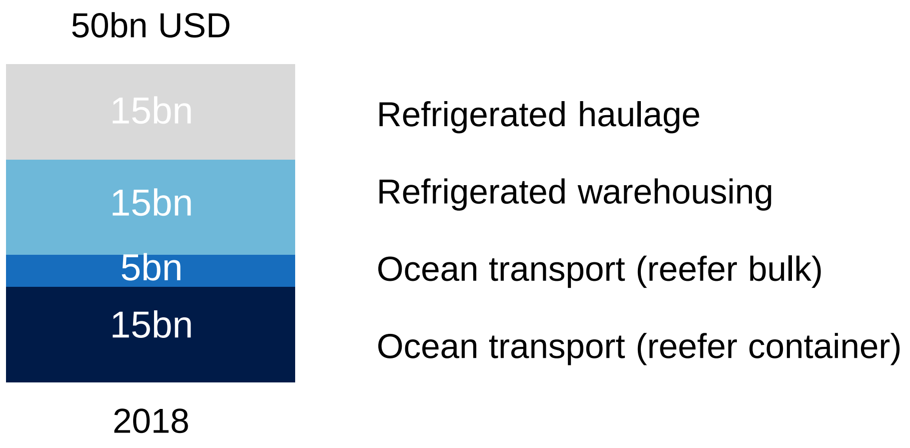
Consumers from the global middle class increase demand…

Growing purchasing power in the global middle class

This increases global demand for fresh, high quality food with fewer preservatives and longer shelf life

… Which drives expected market growth

Addressable cold chain market size development forecast



~10% CAGR

Expected by 2025

Source: Case material

The combination of reefer capacity leadership and the recent successful PoC puts Maersk in a unique position of opportunity

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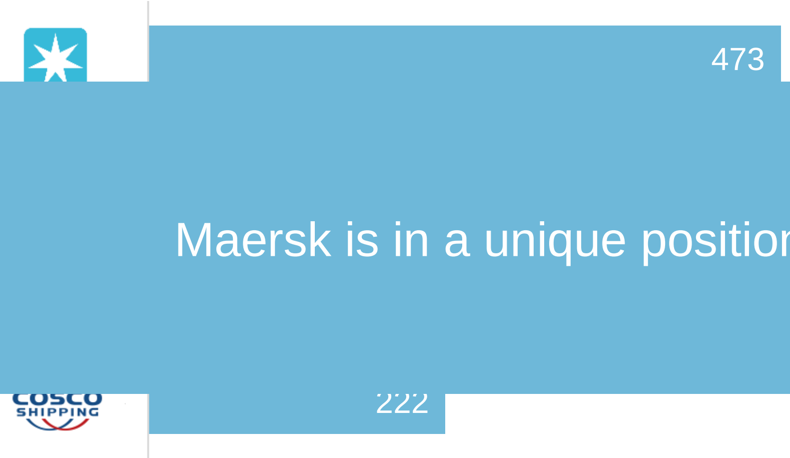
Maersk holds the largest reefer capacity amongst global international carriers…

Reefer Capacity (number of plugs in ‘000)



… and the recent Philadelphia pilot proved successful

Customer pain points successfully addressed

EXTENSIVE PAPERWORK

Standard logistics services are not optimal

Maersk is in a unique position to tap into the market for end-to-end cold chain logistics

TAKEAWAY

Maersk is positioned to grab cold chain market share

TAKEAWAY

Cold chain end-to-end was deemed worth paying a premium for

Source: case material

To address the end-to-end cold chain market, Maersk however currently relies on third-parties in the supply chain

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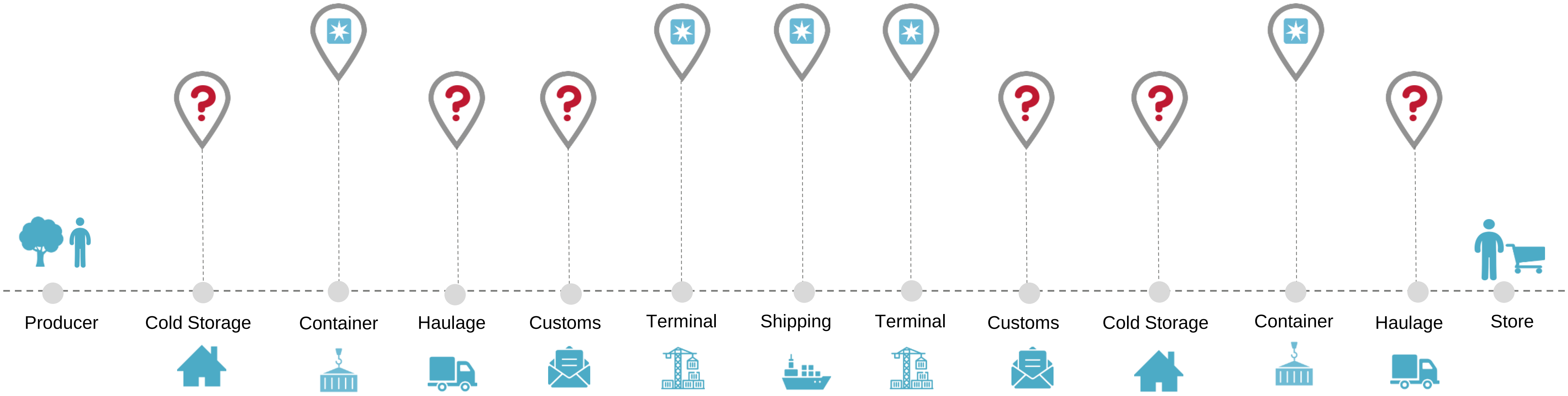

To provide an end-to-end solution, Maersk currently relies on partners

As to align with your vision of becoming the global integrator of logistics, scale is argued to be of the utmost importance

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IN-HOUSE

PARTNERSHIPS

An in-house solution would result in the following benefits…

Partnerships would pose several benefits…

To attain optimum scalability, Maersk should serve the end-to-end cold chain market in collaboration with partners

- Strong entry barriers
- Not based on core capabilities
- Your most traded commodities have high seasonality
- Scalable impact requires heavy investment
- Long time-to-benefit

- Third party risk due to lack of control
- Increased brand risk
- Reduced oversight and control throughout supply chain

Source: case dataset & cate material

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Pull initial partners onto the platform whilst converting current cold chain customers from Mærsk and Damco whilst

Digitally sourcing non-core logistics services will allow for scaling the end-to-end pilot efficiently

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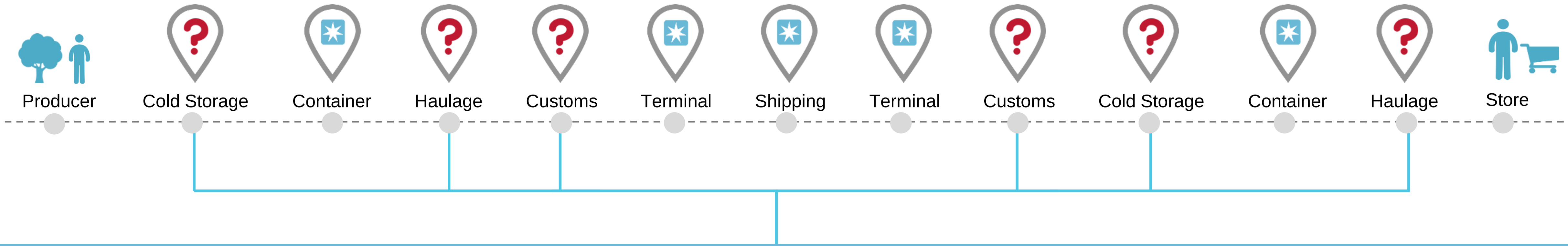

The platform allows Maersk to efficiently and scalably source partners for non-core logistics services from partners and scale their end-to-end cold chain solution

Digital sourcing platform for non-core supply-chain services

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Automatically recognized and bundled on relevant shipments

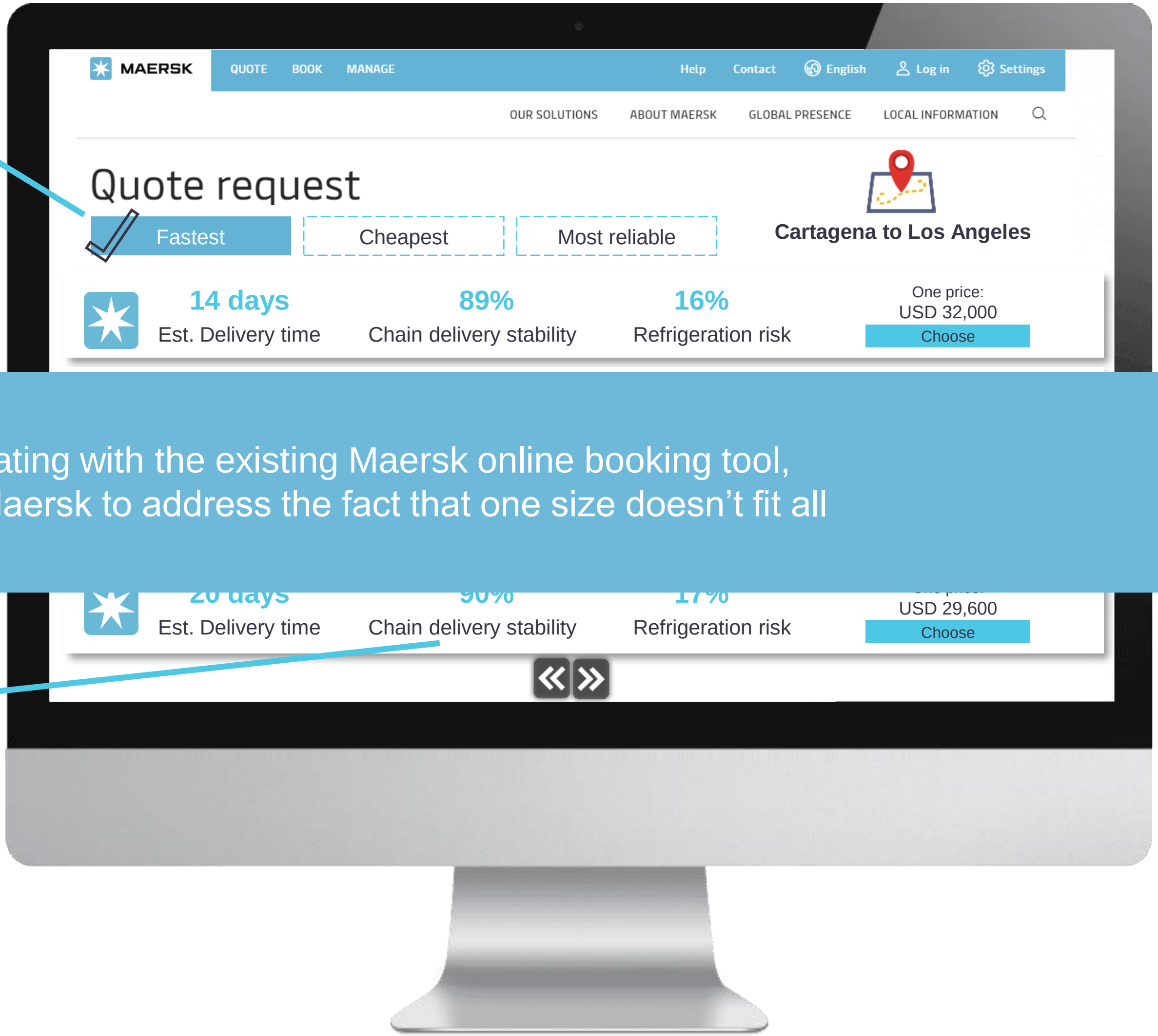
Based on location, track record and offering

Fastest, cheapest or most reliable

One size doesn’t fit all – customers get to choose between quotes

Customers gain transparency and the shipment options that fit them the best

Ascending supply-chain combinations based on preference



Integrating with the existing Maersk online booking tool, allows Maersk to address the fact that one size doesn't fit all

Comparison based on partners track-record

Wrap the core

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End-to-end solution requires focusing on combination of origin and destination countries through focusing on popular routes

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The key to success is products that are truly end-to-end…

- Søren Skou, CEO of A.P. Moller - Maersk

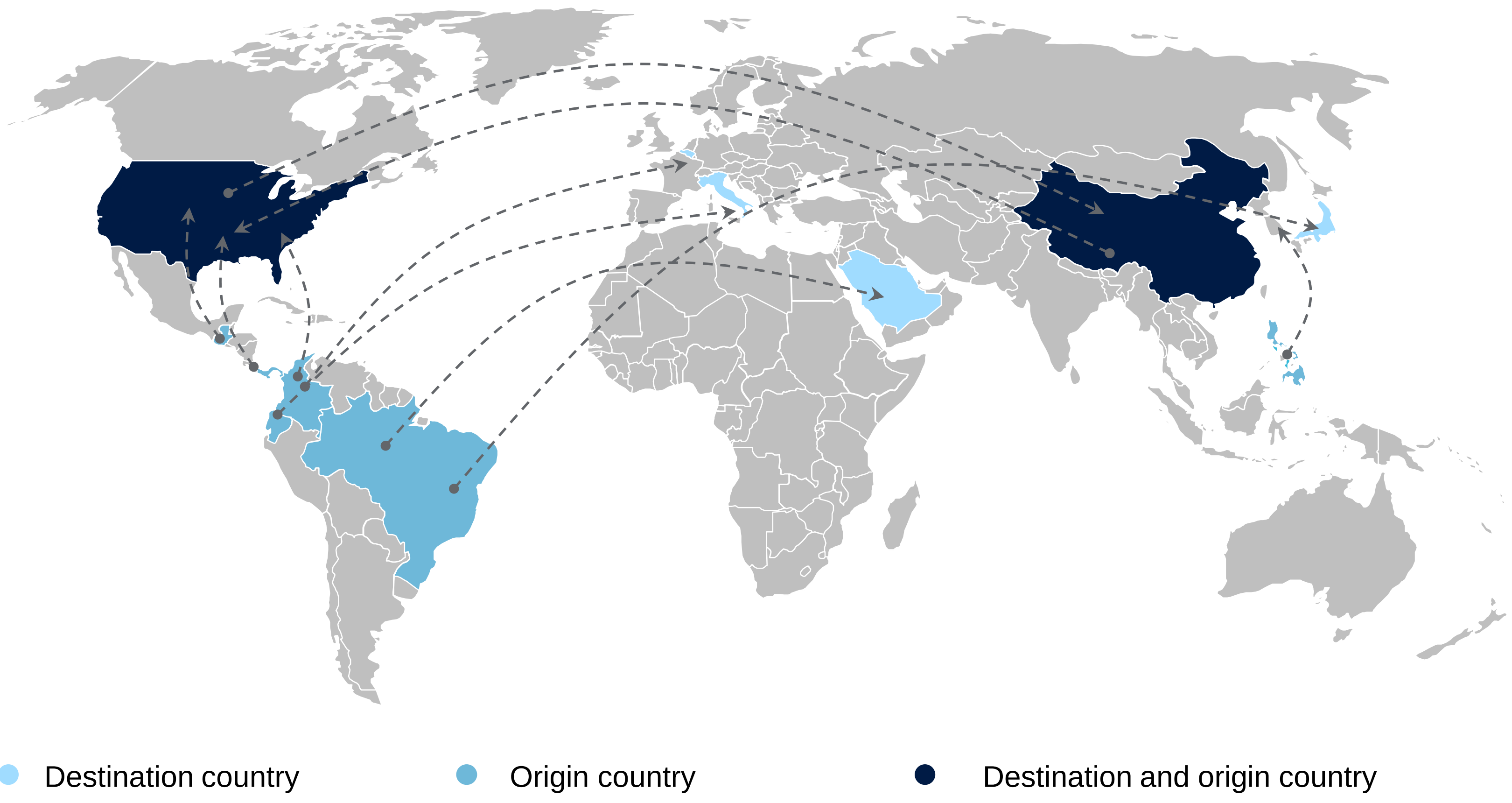
Truly end-to-end through partners

The goal of becoming truly end-to-end requires Mærsk to establish partnerships in both origin and destination countries

Customers in both origin and destination countries

As the customer can be both producers and end-customers, customer acquisition must take place in both origin and destination countries

Top 10 Maersk Cold Chain Routes



Source: Case Dataset

Maersk should initially target the top 10 cold chain routes to extend Proof of Concept before scaling globally

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2019-2020

1 **Top 10 routes** measured on TEUs

2 Extended **Proof of Concept** built on pilot from top 10 routes

2020-2025

3 **Scale partnerships globally** to facilitate any end-to-end shipment

Top 10 Maersk Cold Chain Routes

Origin countries		Destination countries
Philippines	→	*Korea*
Columbia	→	*US*
Costa Rica	→	*US*
Brazil	→	*Japan*
Guatemala	→	*US*
US	→	*Belgium*
Ecuador	→	*China*
China	→	*Italy*
Brazil	→	*Saudi Arabia*

2.6m Average yearly TEU for top 10 routes. **23.4% of total reefer TEUs**

Source: Case Dataset

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Maersk should drive partner onboarding through relationship managers, referral programmes and transition of current partners

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Maersk should use a diverse marketing mix…

Area	Component	Reason to use	Method
Customer acquisition and retention	UPSELL	○ Low hanging fruits ○ Trust in relationship manager	○ Upsell current Maersk customers by having their relationship manager reach out.
	DIGITAL	○ Maersk has previously seen success in using social media for B2B marketing	○ Promotion videos on Linkedin and Facebook
	LOYALTY LADDER	○ Retains customers ○ Increases stickiness	○ Every time a customer uses Maersk consecutively, they get a discounted shipment
Partner onboarding	RELATIONSHIP MANAGER	○ Many potential partners lack digitalization and prioritize relationships	○ Provide relationship manager for new partners
	REFERRAL PROGRAM	○ Many partnerships build on relationships incentivizing to refer partners to platform	○ 5% discount on your next shipment if you refer one of your partners to the platform
	TRANSITION OF CURRENT 3RD PARTY PROVIDERS	○ Provides a strong partnership base on the platform from the beginning	○ Onboard current 3rd party providers to the platform

…to attract customers and partners with different needs

Exponential partner attraction

Use of relationship managers to leverage on relationship

Retention that will provide long-lasting relationships

The end-to-end solution provides the customer with valuable insights and a seamless shipping experience

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Sources bananas from farms and ship to retailers globally through a highly complex supply chain

Contacted by current Mærsk point of contact about end-to-end solution and decides to use it

Chooses the most consistent delivery type in delivery to US

Local trucking company with Mærsk reefers picks up bananas and takes them to APM terminal

Chiquita is able to track products and temperatures through RCM

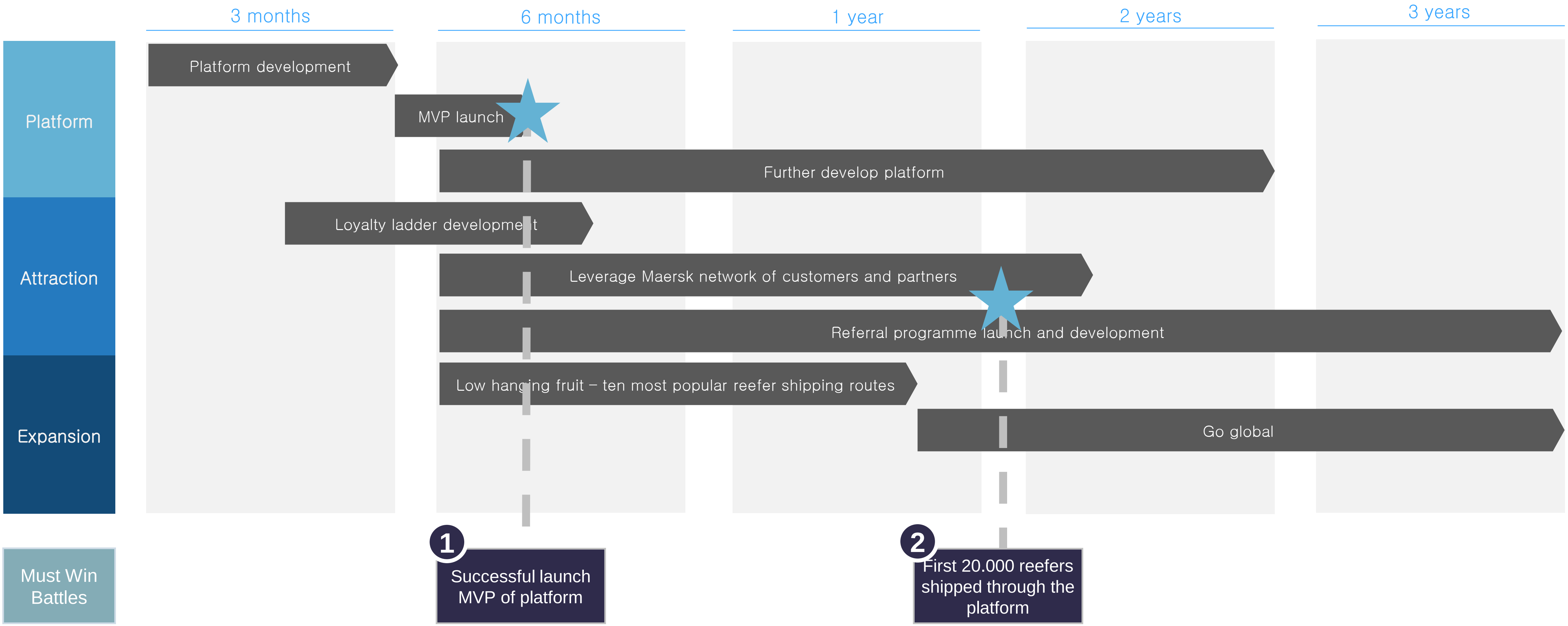
MBC customs brokers ensure a smooth customs inspection

Tekop refrigeration package bananas before they are trucked to the retailer

Chiquita receives one joint invoice and rates the process

Source: Case Material

Successful implementation relies on quickly developing a MVP and quickly securing a strong global footprint



Investing and expanding in a global end-to-end cold-chain solution will result in a positive NPV of $4,3 b. and increase revenue $7 b. by 2025

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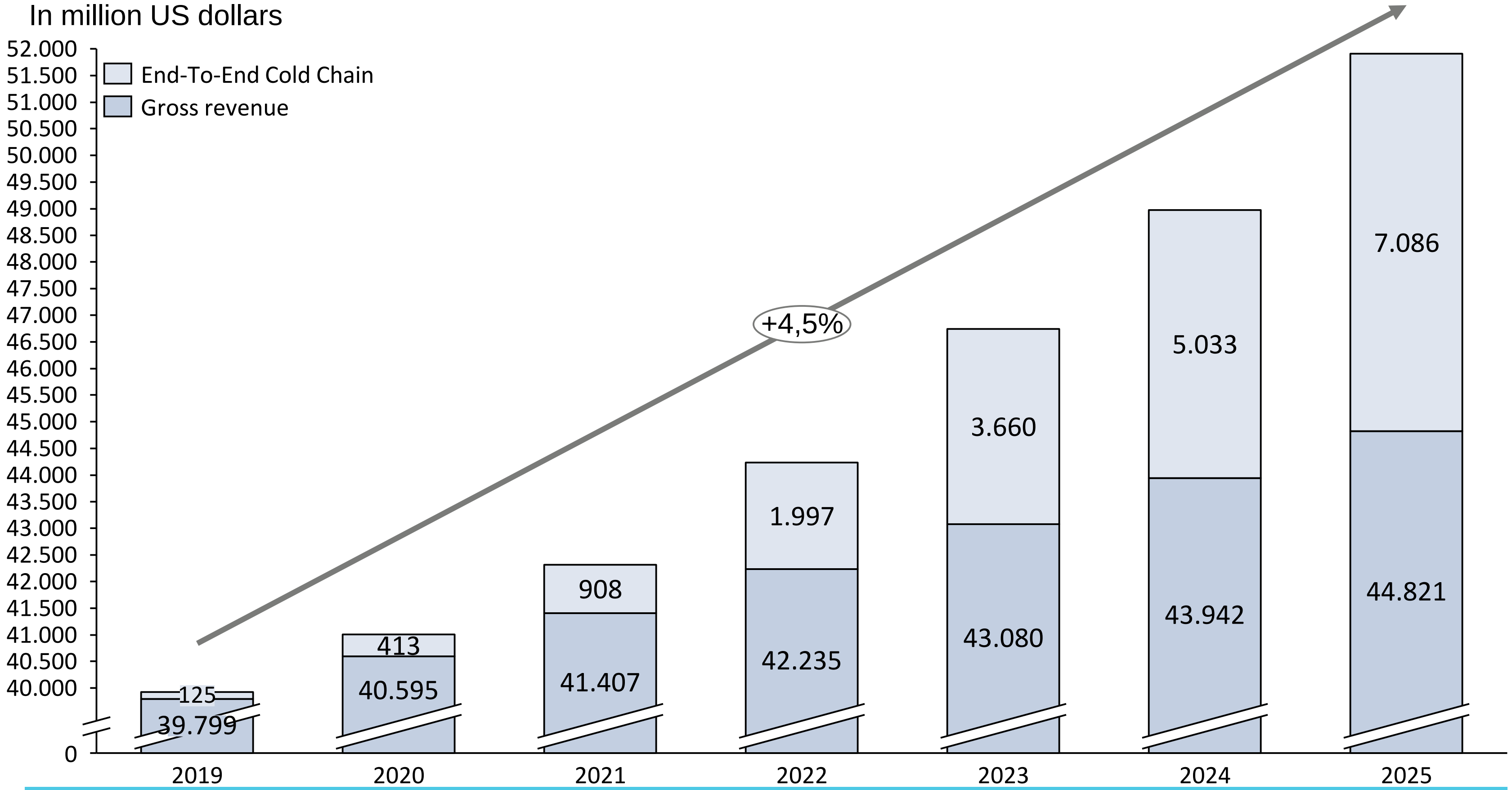

NPV of $4,3 billion

Assumptions

0,5% market share in 2019, growing to 16% by 2025

12% EBITDA margin

Terminal growth of 2%

Discount rate: 8,5%

General market growth: 2%

$100 m. CAPEX year 1: Marketing, development, FTE's

Investing in end-to-end cold chain poses some key risks for Maersk that will need mitigation

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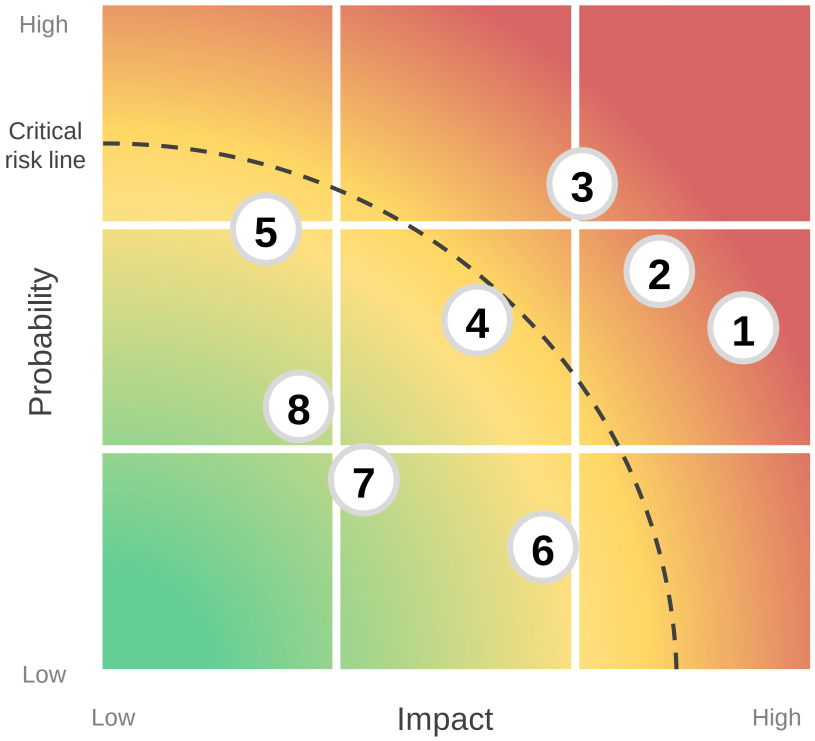

	Risk	Mitigation
1	Liability for partner mistakes in the cold chain	Proactively work with partners to anticipate delays and implement mitigation, along with monitoring and rating partner performance
2	Lower than expected demand for end-to-end offering	Cut investments and refocus on core capability
3	Shipment delays due to miscommunication	Ensure transparent communication with all stakeholders by establishing shared communication channels
4	Few partners signing up for platform	Incentivize via referral scheme
5	Risk of design flaws	Continuously work with customers and partners to improve platform
6	Financial risk	Relatively small upfront investment required, as partnerships is asset-light
7	Competition copying model	Lock up customers to ensure first-mover advantage

Synthesis

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INSIGHTS	SOLUTION	IMPACT

Successful cold chain pilot project

- Customers want to pay a premium for end-to-end service
- Pilot project proved ability to solve key customer pain points
- Large attractive market forecast to grow by CAGR of 10%
- Maersk is well positioned to take advantage of cold chain market as the leader in reefer capacity

Partnerships offer greatest scalability

- A need for quick scalability offered by sourced partnerships
- Acquisitions too expensive and slow for quick global presence

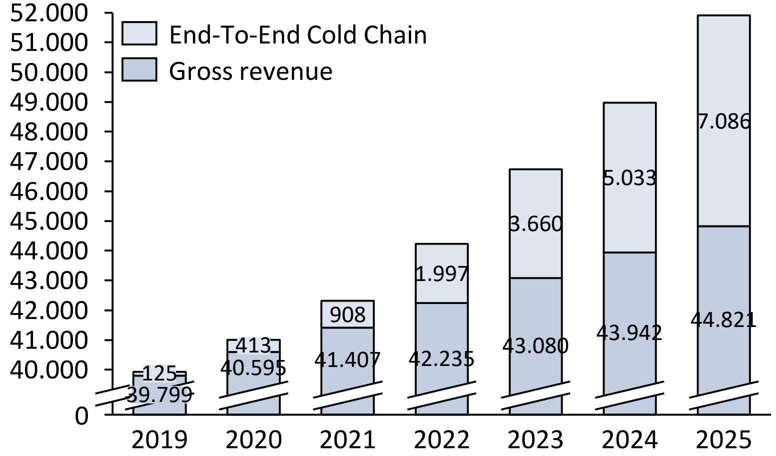
1 **WRAP THE CORE**
Launch digital platform to source non-core logistics services

2 **WHERE TO SCORE**
Strengthen PoC in top 10 routes to roll out globally by 2025

3 **MAKE THEM WANT MORE**
Ensure adoption from partners and customers

Estimated impact

- ✓ Allow for scalable partners acquisition to ensure end-to-end logistics
- ✓ Further improve delivery reliability and temperature consistency
- ✓ One size doesn't fit all – provide costumers with best fit shipments



Increased revenue by 7bn USD by 2025

Index

In the first year a large part of the revenue will come from upselling to existing Maersk customers

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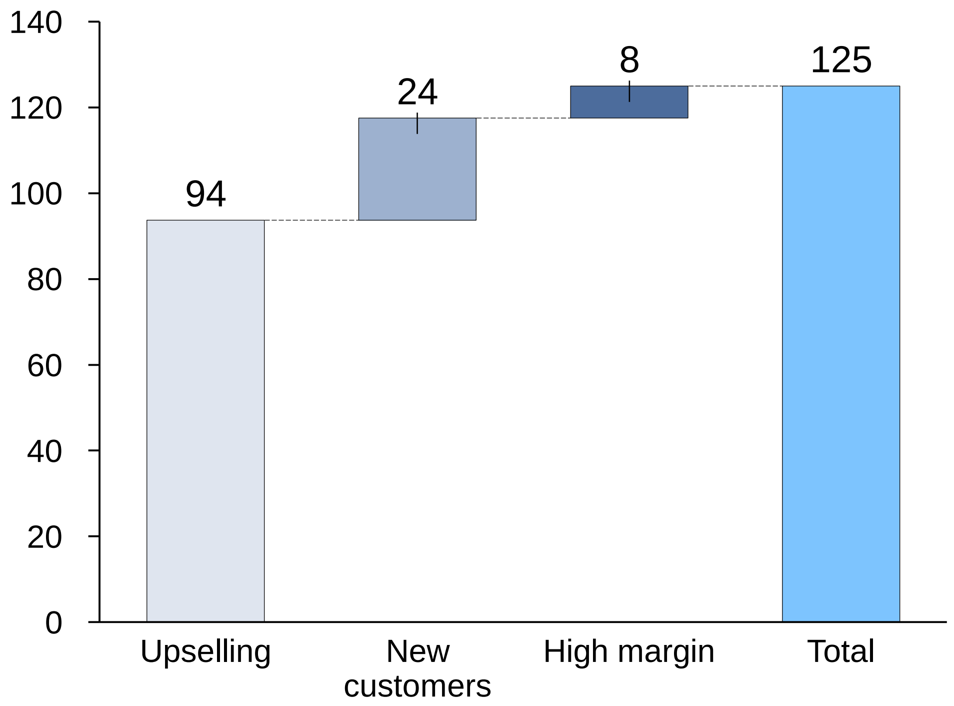
First year revenue breakdown, $M.



NPV sensitivity analysis

NPV sensitivity, $m.	Market share, 2025				
Discount rate	14%	15%	16%	17%	18%
6,50%	5.791	6.241	6.691	7.140	7.590
7,50%	4.767	5.136	5.505	5.874	6.243
8,50%	4.052	4.365	4.679	4.992	5.305
9,50%	3.524	3.796	4.068	4.341	4.613
10,50%	2.923	3.358	3.598	3.839	4.080

Market size and market share forecast

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Cold chain market size forecast	50.000.000.000	55.000.000.000	60.500.000.000	66.550.000.000	73.205.000.000	80.525.500.000	88.578.050.000
Market share	0,5%	1,5%	3,0%	6,00%	10,0%	12,5%	16,0%
Revenue	250.000.000	825.000.000	1.815.000.000	3.993.000.000	7.320.500.000	10.065.687.500	14.172.488.000
Mærsk share: 50% of revenue	125.000.000	412.500.000	907.500.000	1.996.500.000	3.660.250.000	5.032.843.750	7.086.244.000
Average number of reefers	16.667	55.000	121.000	266.200	488.033	671.046	944.833

Financial model

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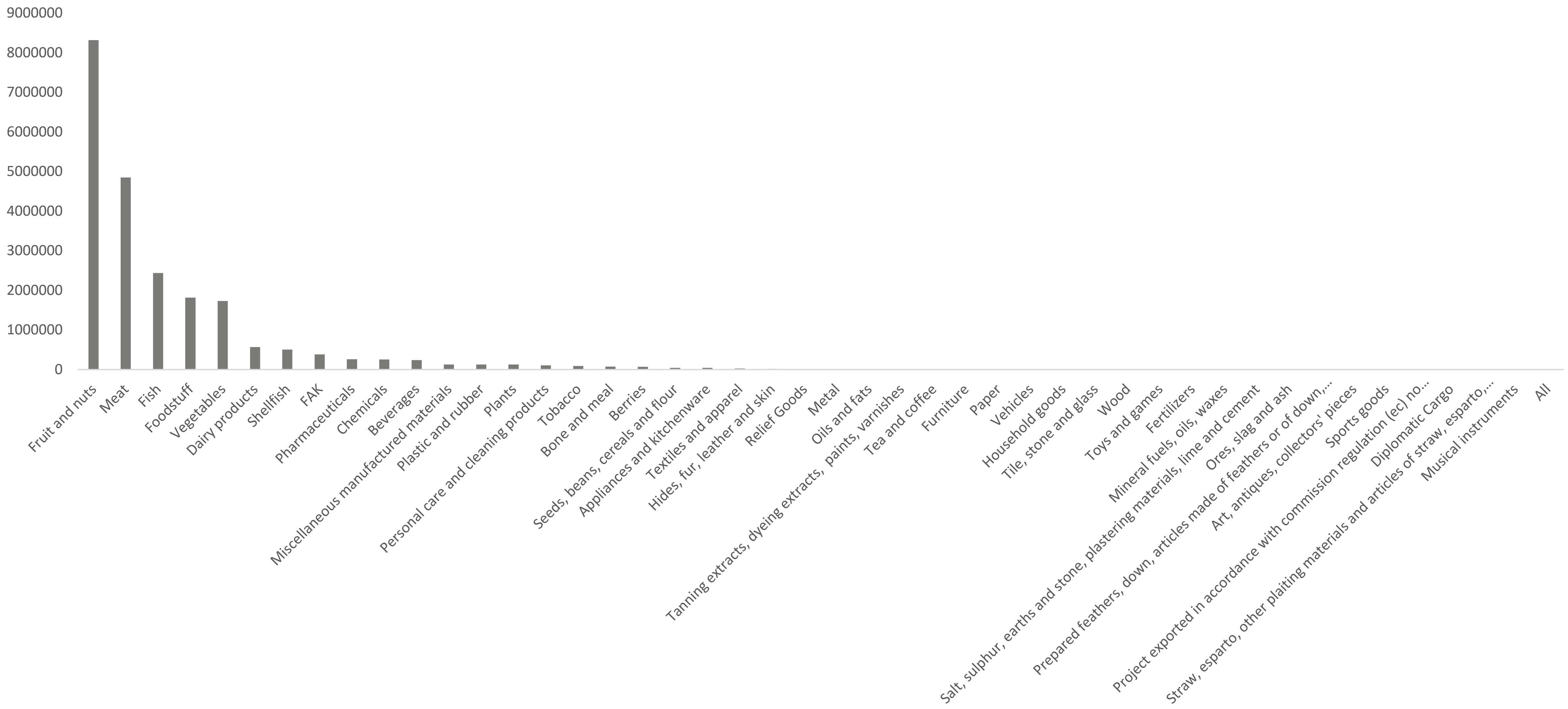
Financial model: Income statement

	Current numbers		Forecast						
	2018	2019E	2020E	2021E	2022E	2023E	2024E	2025E	
Revenue	39.019	39.799	40.595	41.407	42.235	43.080	43.942	44.821	
Cold-chain revenue		125	413	908	1.997	3.660	5.033	7.086	
Total revenue		39.924	41.008	42.315	44.232	46.740	48.975	51.907	
EBITDA	3806	4.791	4.921	5.078	5.308	5.609	5.877	6.229	
Amortisation	3325	3.358	3.392	3.426	3.460	3.495	3.530	3.565	
Other items, net	146	147	149	150	152	153	155	157	
EBIT	627	1.580	1.678	1.802	2.000	2.268	2.502	2.820	
Financial items, net	-389	- 393	- 397	- 401	- 405	- 409	- 413	- 417	
Profit before tax	238	1.187	1.281	1.402	1.595	1.859	2.089	2.403	
Taxes	386	261	282	308	351	409	460	529	
Net profit, continuing operations	-148	926	999	1.093	1.244	1.450	1.630	1.875	Terminal
Cash flow from project		- 21	- 10	9	51	117	175	262	288
		Model assumptions							
Market growth - cold chain		10%	10%	10%	10%	10%	10%	10%	
Organic revenue growth - overall		2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	2,00%	
Amortisation allocated		11	34	75	164	297	404	564	
EBITDA margin	9,75%	12%	12%	12%	12%	12%	12%	12%	

Total TEU's by commodity, reefers only



Split of TEU levels of top 10 cold-chain routes

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Origin country	Destination country	TEU Level
Philippines	Korea	2.14m
Columbia	US	632,000
Costa Rica	US	438,000
Brazil	Japan	419,000
Guatemala	US	399,000
Columbia	Belgium	321,000
US	China	259,000
Ecuador	Italy	254,000
China	US	237,000
Brazil	Saudi Arabia	227,000

Source: Case dataset

Appendix

Fruit and nuts, seasonality

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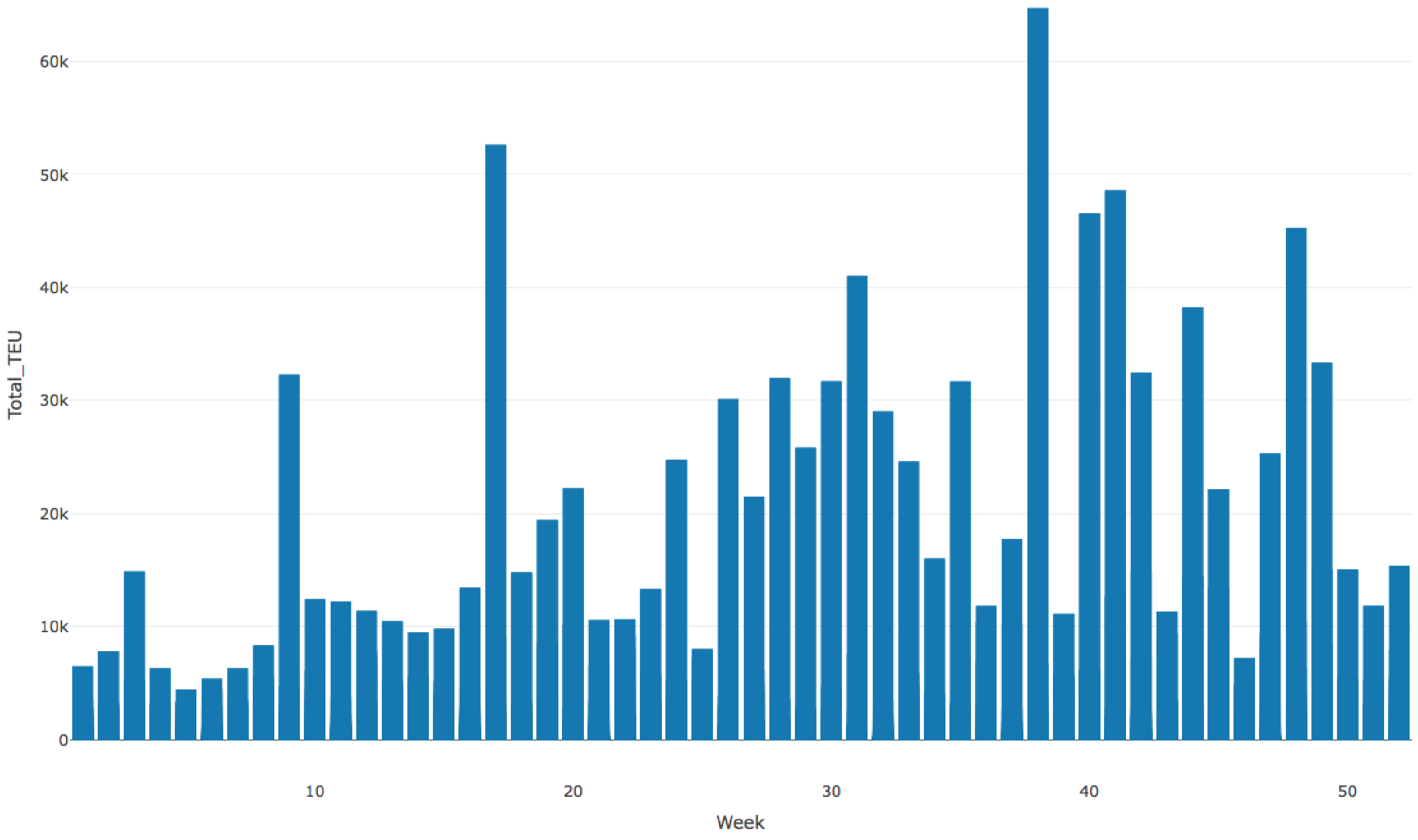
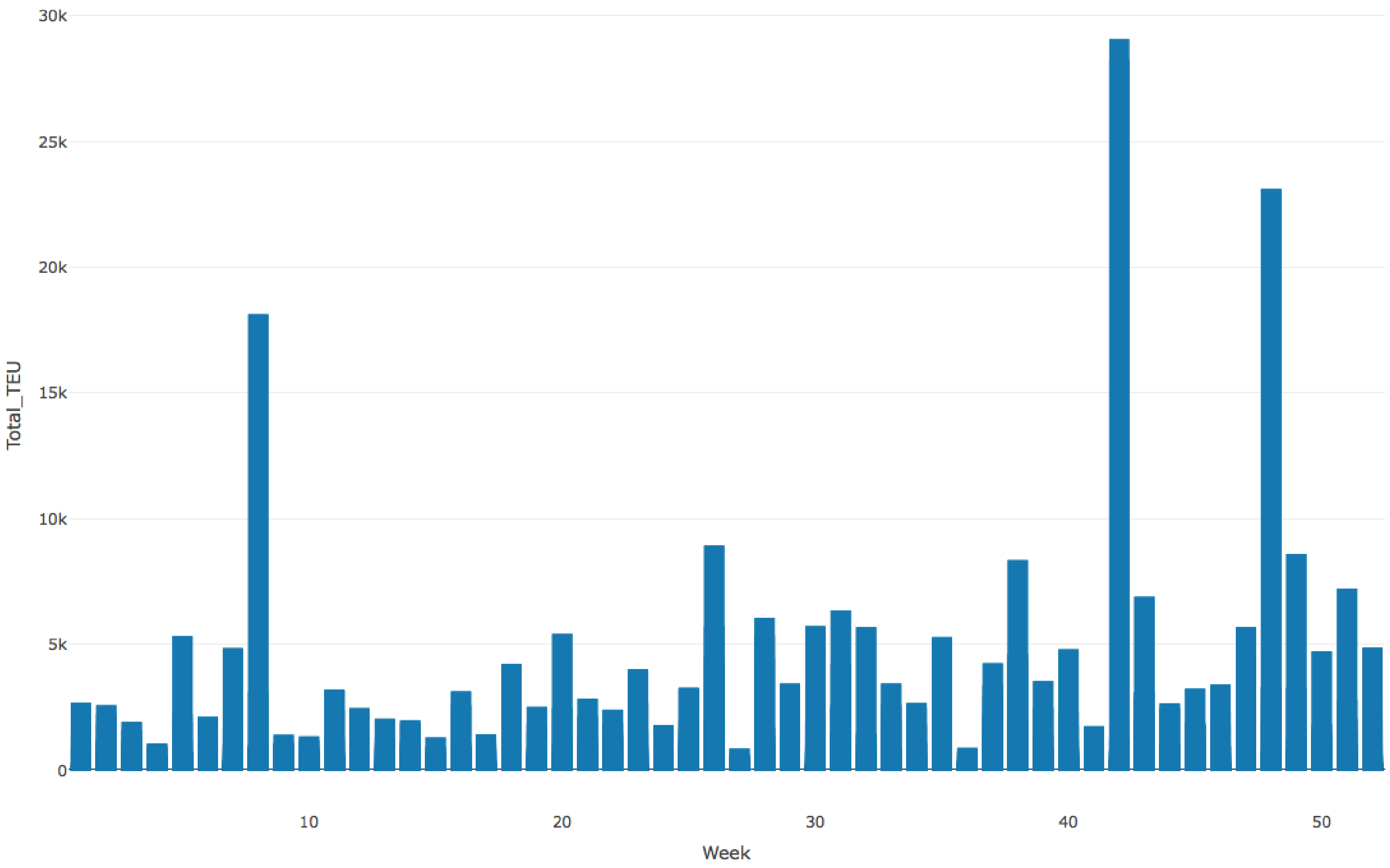


Source: Case data set

Meat seasonality



Source: Case data set

Fish seasonality

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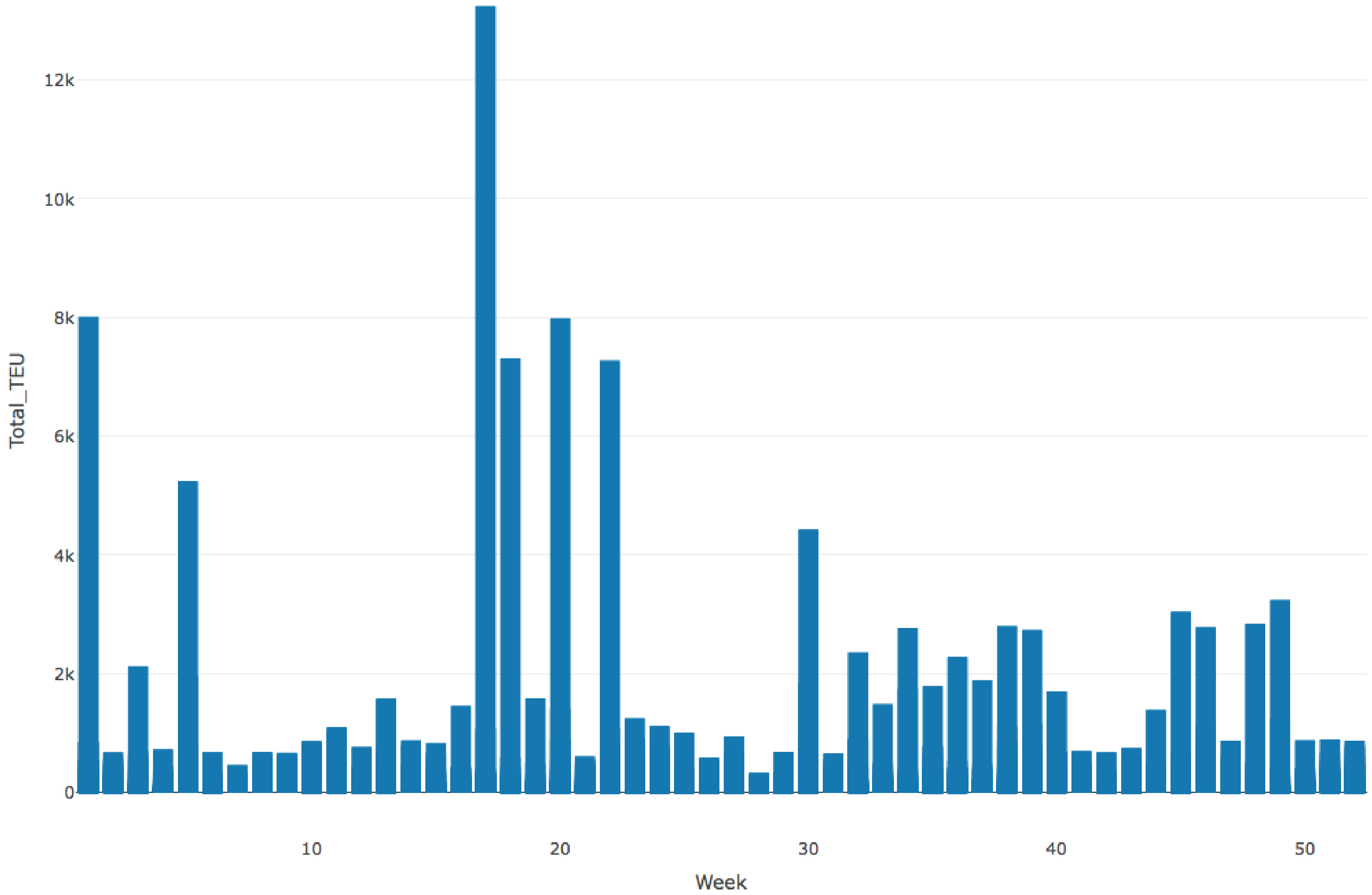
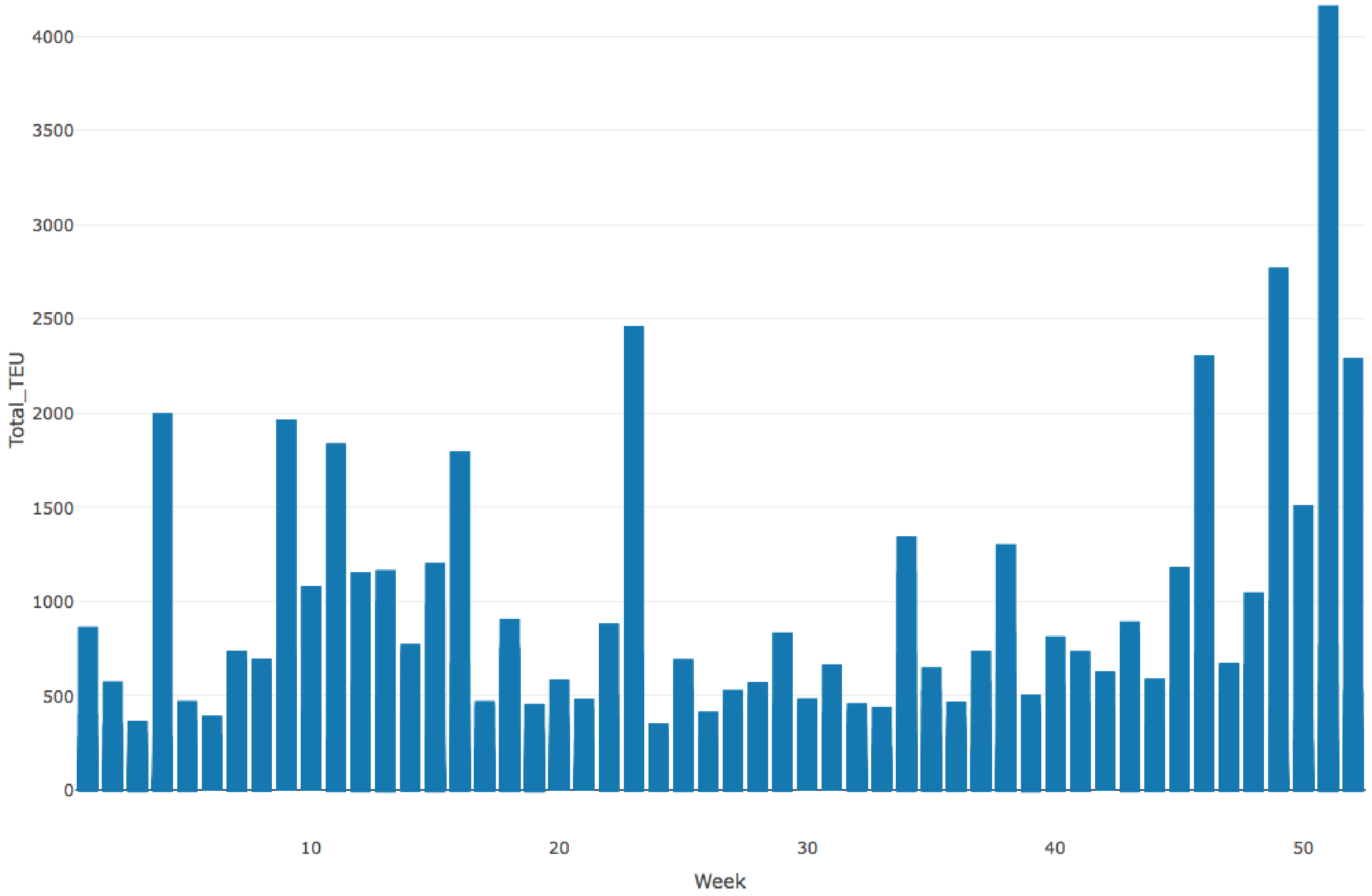


Source: Case data set

Foodstuff seasonality



Source: Case data set

Vegetables seasonality

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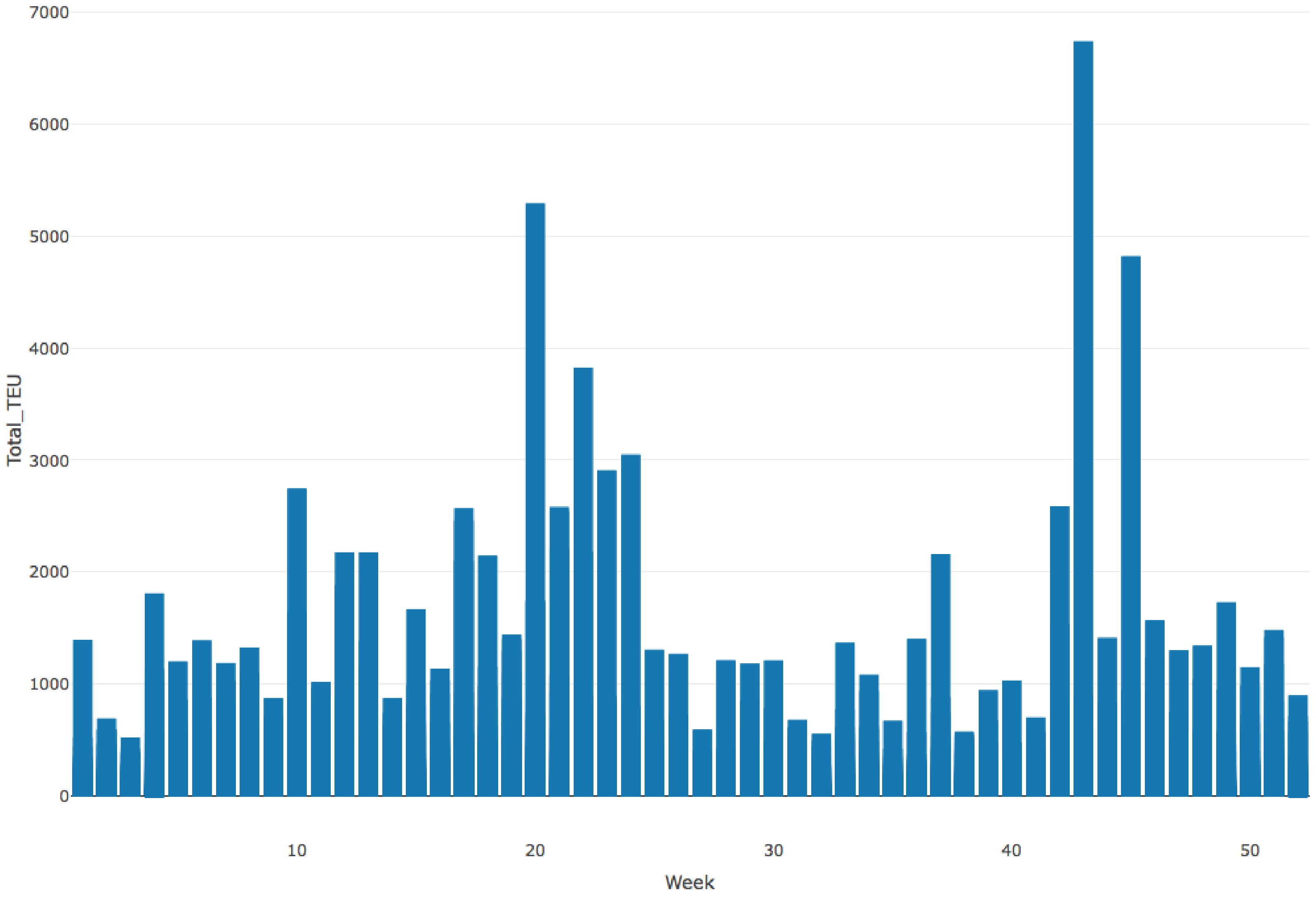

Source: Case data set

Why should we not transform Damco to digital freight forwarder?

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DAMCO

- They have never operated with cold chain
- They are a well established freight forwarder, but not digital at all
- This should not be a freight forwarding service, but a Maersk end-to-end solution

Why should Twill not expand into cold chain?

- Their size
- They do not have the capabilities currently
- They offer a standardized solution, and cold chain customers want specialized offerings
- They aim to be the leading freight forwarder for SMEs
- This should not be a freight forwarding service, but a Maersk end-to-end solution

Possibility to leverage on Tradelens' capabilities

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TRADELENS

- They digitalize the entire paper trail
- They aim at getting all parties throughout the supply chain on the platform
- This will enable Maersk to provide an even more seamless solution throughout the shipping process

The end-to-end cold chain solution will also have a positive impact on customers, thereby increasing their switching costs

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Scalable onboarding of non-core logistics providers to ensure end-to-end provision

Increased stability of delivery and termperature

One invoice

Increase on-time deliveries by 11% compared to market average

Reduce global food waste through a decrease in late deliveries

Reduce operational costs for customers by simplifying cold chain shipping

Moving Maersk and the World

Sapphire Consulting

David Hao, Katie Kirkconnell, Mark Laufert, Claire Uhm

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Our recommendations are informed by Maersk's strategic vision & goals for cold chain

Key Question

How can Maersk leverage its position as a leader in reefer shipping to create valuable propositions for customers in the cold chain market?

Our recommendations are informed by Maersk's strategic vision & goals for cold chain

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How can Maersk leverage its position as a leader in reefer shipping to create valuable propositions for customers in the cold chain market?

Strategic Vision & Goals

Develop a seamless, superior and simple end-to-end cold chain shipping solution

Our recommendations are informed by Maersk's strategic vision & goals for cold chain

Key Question

How can Maersk leverage its position as a leader in reefer shipping to create valuable propositions for customers in the cold chain market?

Strategic Vision & Goals

Develop a seamless, superior and simple end-to-end cold chain shipping solution

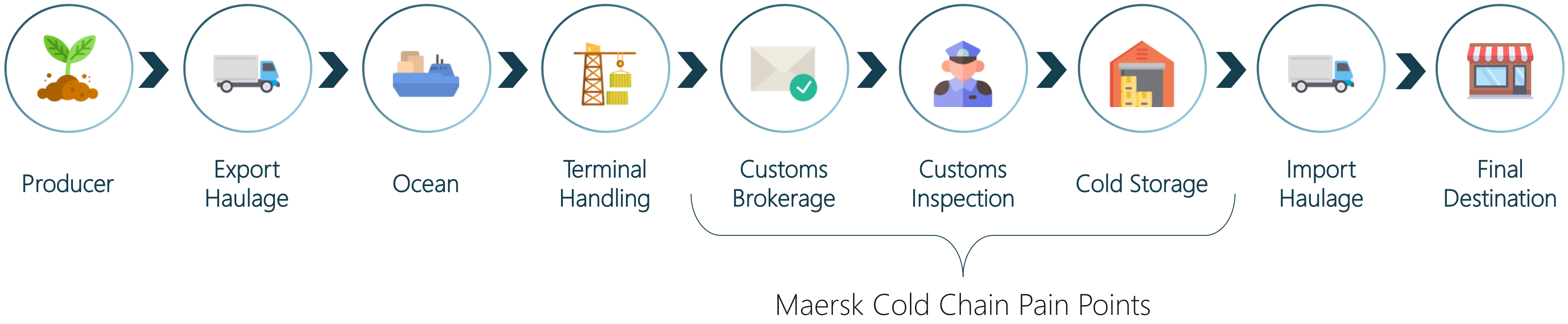
Increase "Stickiness"

Increase Margins

Increase Revenue

Analysis

Fresh Pass pilot revealed customers' prioritization of reliability



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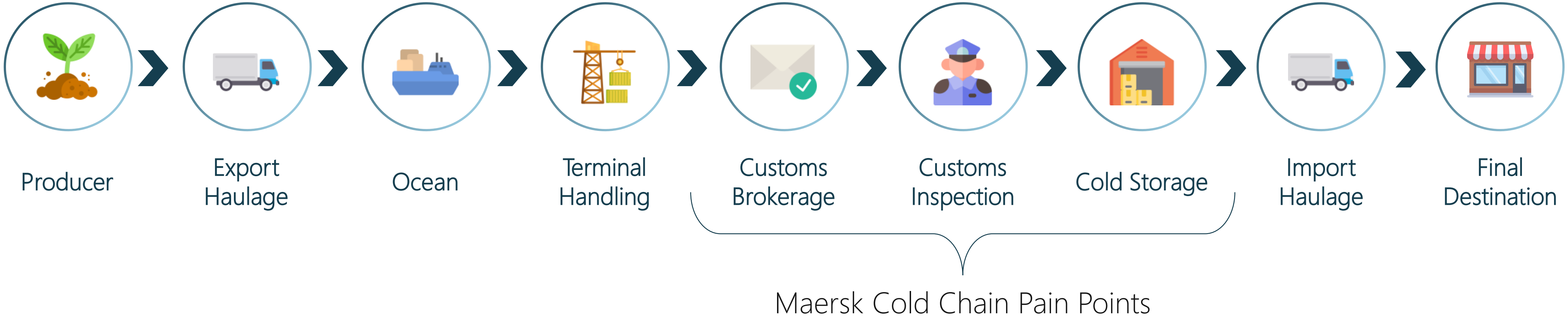
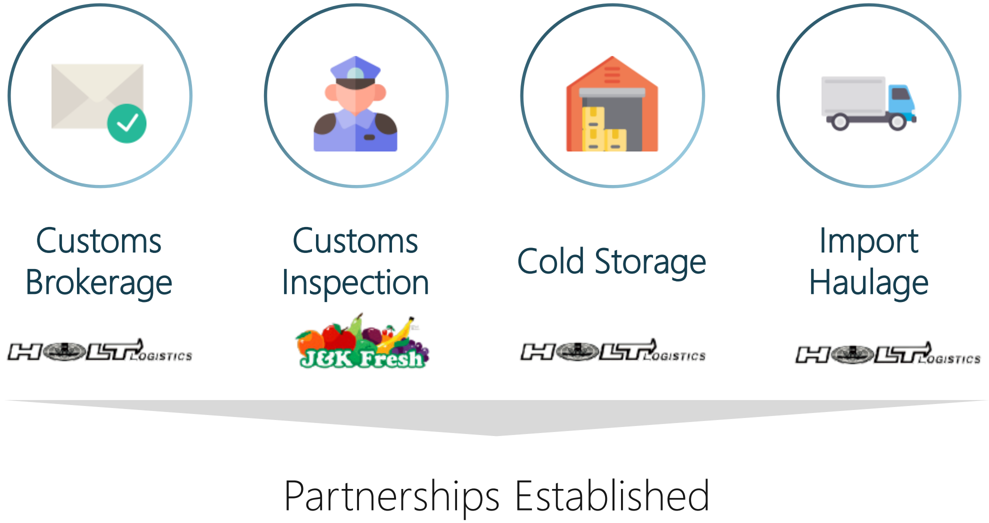


Fresh Pass Cold Chain Partnership Project



Analysis

Fresh Pass pilot revealed customers' prioritization of reliability

Producer → Export Haulage → Ocean → Terminal Handling → Customs Brokerage → Customs Inspection → Cold Storage → Import Haulage → Final Destination

$\underbrace{\text{Customs Brokerage} \quad \text{Customs Inspection} \quad \text{Cold Storage}}_{\text{Maersk Cold Chain Pain Points}}$

Fresh Pass Cold Chain Partnership Project

Customs Brokerage

Customs Inspection

Cold Storage

Import Haulage

Partnerships Established

"Unexpected feedback"

"What drives real value is not necessarily speed, but consistent delivery"

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What do the lessons learned from the Fresh Pass pilot mean for the global rollout?

We ruled out unproven technological additions to Maersk's cold chain business

Criteria

- Risk-Smart
- Scalable
- Implementable Quickly
- Profitable

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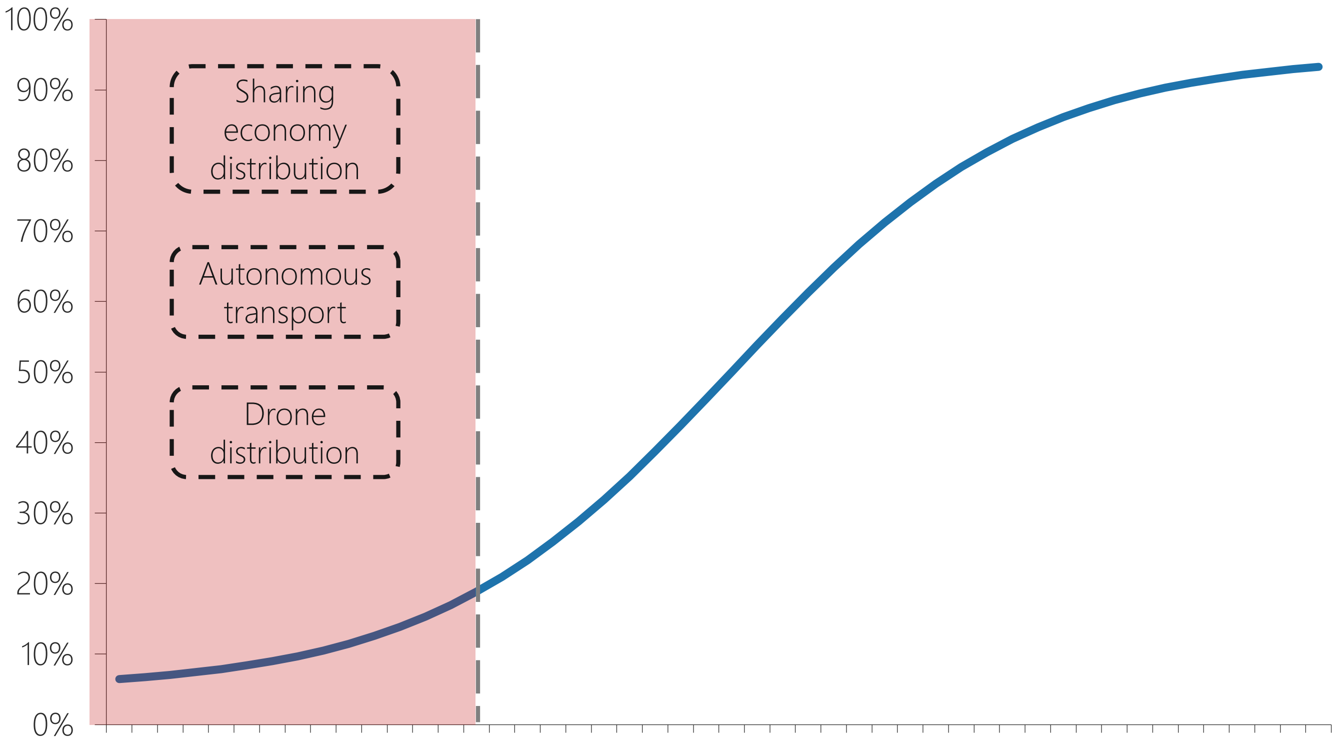
Technology Adoption S-Curve



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Analysis

Maersk's cold chain offerings should be on the cutting edge of **reliable** technology

Criteria

- Risk-Smart
- Scalable
- Implementable Quickly
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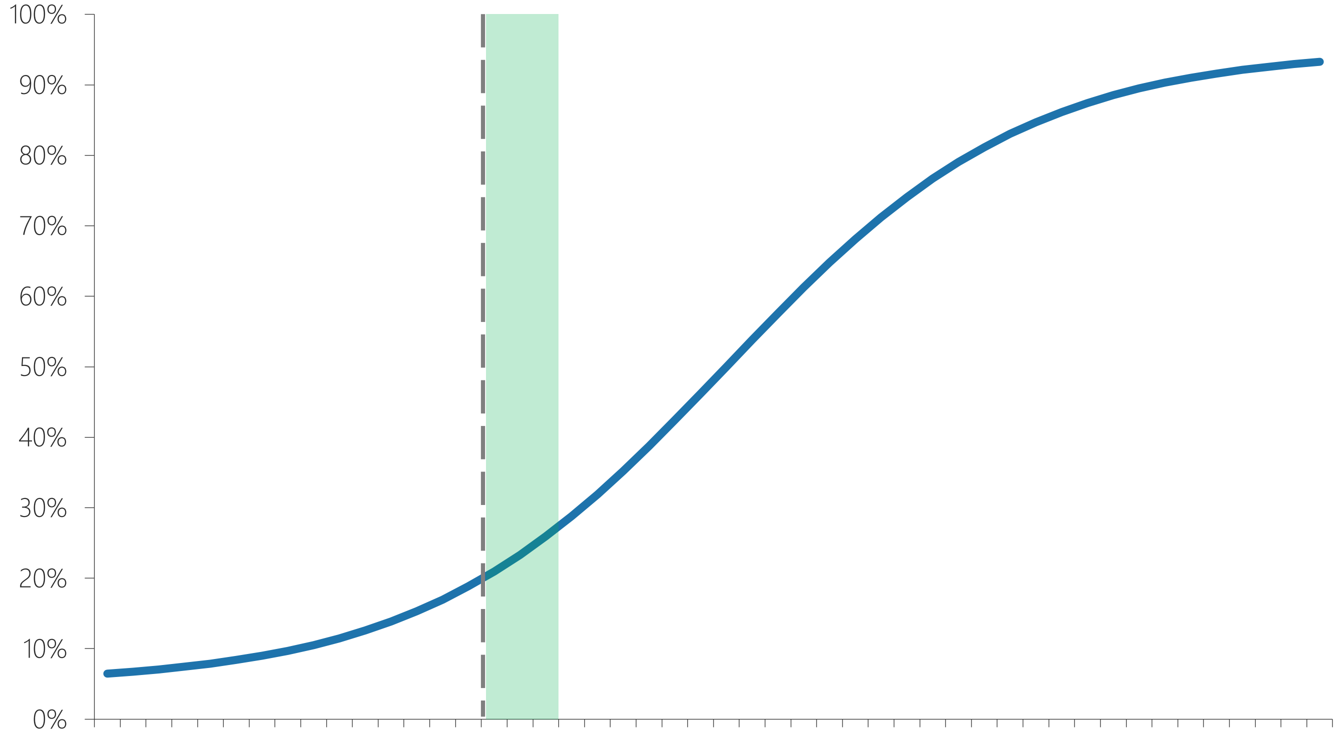
Technology Adoption S-Curve



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Building strength in cold chain

1

Buy & Partner

Acquire customs brokerages and cold storage facilities in major shipping hubs

Establish partnerships in less-trafficked ports

2

Analytics with a Human Touch

Introduce business intelligence API for supply chain management software

Pair Maersk supply chain consultant with Fresh Pass customers

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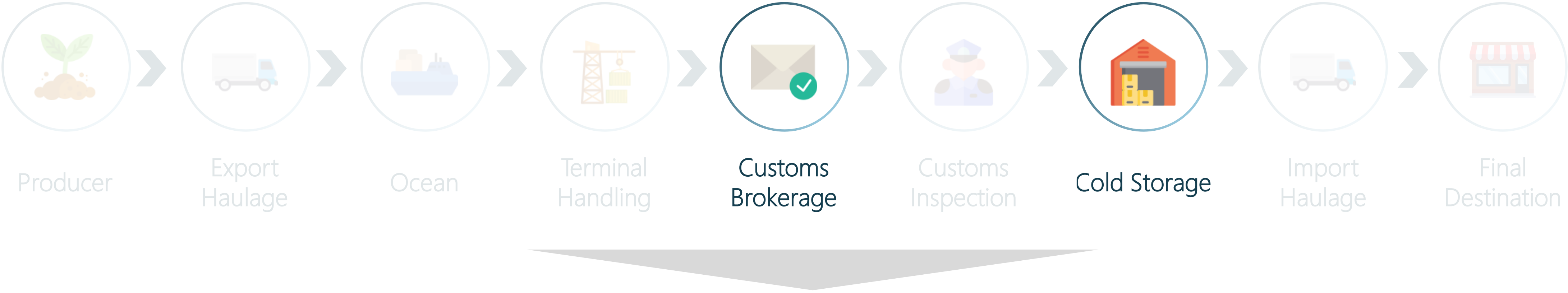
Supply Chain

Supply Chain



Option 1: Buy/Build

Option 2: Partner

Supply Chain

Producer
Export Haulage
Ocean
Terminal Handling
Customs Brokerage
Customs Inspection
Cold Storage
Import Haulage
Final Destination

Option 1: Buy/Build

Option 2: Partner

Ensure highest service quality and meet cost of capital with ownership in major destination ports

Issue of fragmentation by cold storage type is eliminated at most important locations

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Supply Chain

Producer | Export Haulage | Ocean | Terminal Handling | Customs Brokerage | Customs Inspection | Cold Storage | Import Haulage | Final Destination

Option 1: Buy/Build

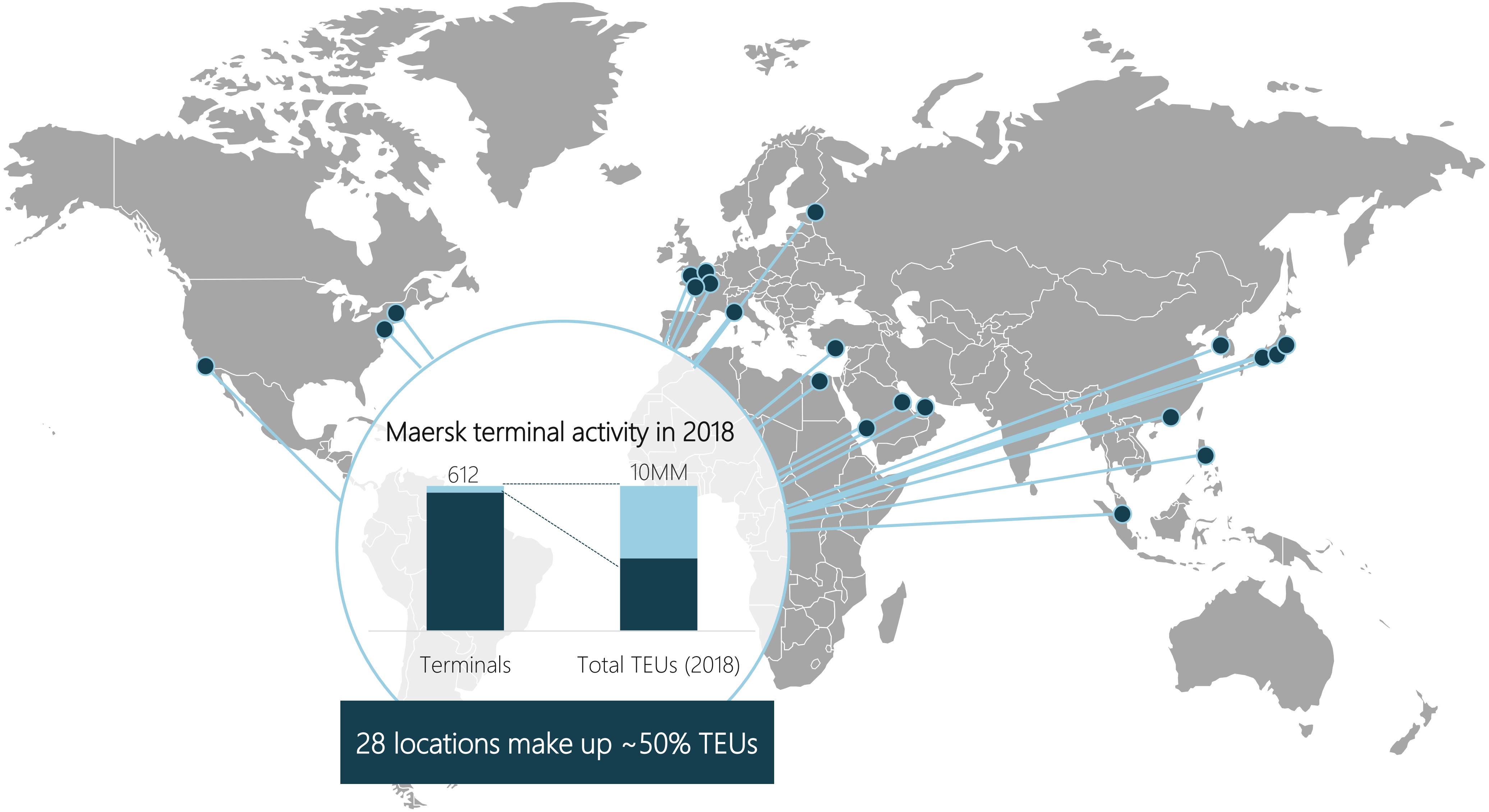
Option 2: Partner

Ensure highest service quality and meet cost of capital with ownership in major destination ports

Issue of fragmentation by cold storage type is eliminated at most important locations

Recommendation 1

Cold storage and customs brokerages should be purchased in key locations



28 locations make up ~50% TEUs

Seasonality issues can be eliminated through renovations and additional services

Renovate cold stores & drive off-season usage

Implement modularity to increase flexibility and combat seasonality

Offer at-cost services (e.g. packaging & food grading) build off-season demand

Key Benefits

Combats over-specialization of existing cold stores

Offsets seasonality

Supply Chain

Producer | Export Haulage | Ocean | Terminal Handling | **Customs Brokerage** | Customs Inspection | **Cold Storage** | Import Haulage | Final Destination

Option 1: Buy/Build

Option 2: Partner

Ensure highest service quality and meet cost of capital with ownership in major destination ports

Issue of fragmentation by cold storage type is eliminated at most important locations

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Recommendation 1

Establishing strong partnerships is crucial to wide-spread adoption of Fresh Pass

1 E-Sourcing	2 Referral Program	3 Probationary Period
		
Maersk e-Sourcing platform	RCM platform	Customer Feedback + Data Analytics
Established system for evaluating vendors and managing auctions	Feedback on prospective partners requested from >75% of customers	Customer satisfaction coupled with objective performance data

Combine customer preferences with first-party RFPs to select best possible partners

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Building strength in cold chain

1
Buy & Partner

Acquire customs brokerages and cold storage facilities in major shipping hubs

Establish partnerships in less-trafficked ports

2
Analytics with a Human Touch

Introduce business intelligence API for supply chain management software

Pair Maersk supply chain consultant with Fresh Pass customers

Recommendation 2

Maersk's position in the age of information

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Maersk owns vast amounts of data related to different stages of its cold chain operations

Data Type	Data Collected	Data Collection Method
	Routes, length of journey, origin, destination, terminal throughput, segments	Internal data
	Product type, volume	
	Temperature, humidity, gas levels (oxygen, CO_2)	Maersk Starcool and RCM

How can Maersk use data to provide additional value to customers and increase stickiness?

MAERSK

Recommendation 2

Introducing a two-pronged integrated business intelligence solution

Customer

Insights	Decisions
ReFresh	MAERSK Operations Consulting
A free inventory management CRM plug-in with a user-friendly dashboard that delivers real-time insights to Fresh Pass customers	A specialized supply chain operations consultancy that provides advisory services and a human touch to big data

Recommendation 2

Introducing ReFresh: A true end-to-end solution to increase value for customers

ReFresh is a free CRM plug-in with a user-friendly dashboard that integrates customer level data with Maersk supply chain data to provide real-time data insights

Feature	Description	Impact
Average temperature	Customers can track the temperature of containers over time	Increased agility with proactive monitoring
Pre-ripening meter	Ethylene biosensors can detect the ripening and spoilage of fruit	Reduced food waste costs and inventory holding costs to ripen fruit
Inventory forecasting	Prescriptive analytics can be applied to suggest reorder date	Reduced inventory holding and shortage costs

MAERSK

Recommendation 2

Leveraging customer data for cost optimization on both sides

Maersk also realizes significant cost savings from implementing ReFresh

Faster repairs of broken reefer cooling systems

Reduced spoilage and A/C costs through monitoring of ethylene gas

Shipping route optimization

MAERSK

Recommendation 2

Offer actionable insights: Provide advisory services to deepen customer relationships

Establish a **Maersk Operations Consulting** arm specializing in supply chain operations to help simplify complex insights for customers through expert human interaction to aid in decision making

Pair each customer... + ...to a dedicated consultant

Proactive Support	Reactive Support
Quarterly consultations with Maersk Operations Consulting	**Crisis support services** leveraging extensive Maersk data and Artificial Intelligence
Provide operational insights	Suggest recommended courses of action
Route optimization, sourcing suggestions, re-ordering points to minimize inventory holding costs, seasonality of shipping rates	Based on historical data, Maersk can leverage AI to suggest best courses of action based on what has been done in similar circumstances in the past

? "Why are my avocados always so ripe?"

MAERSK

“Do I have to do anything based on this temperature dip?”

“Why are my avocados always so ripe?”

“Do I have to do anything based on this temperature dip?”

“Why are my avocados always so ripe?”

“Why is my inventory projected to take so long?”

Question

"Why is my shipment projected to take so long to arrive?"

Answer

"The existing route is disproportionately long."

Recommendation

"Here is an optimized route our team has tailored specifically for you."

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Recommendation 2

Benefits for both sides

Cost savings from reduced spoilage and inventory

Learn about opportunities for operational efficiency

Receive customized recommendations for cold chain issues

Increased NPS

Increased customer stickiness

Upsell opportunities

Ability to make data-driven decisions

Increased transparency into supply chain operations

Implementing an integrated business intelligence solution will cultivate a **mutually beneficial relationship** between Maersk and its customers **through data**

Recommendation 2

Benefits for both sides

- Cost savings from reduced spoilage and inventory
- Learn about opportunities for operational efficiency
- Receive customized recommendations for cold chain issues

- Increased Net Promoter Score (NPS)
- Increased customer stickiness
- Upsell opportunities

Ability to make data-driven decisions

Increased transparency into supply chain operations

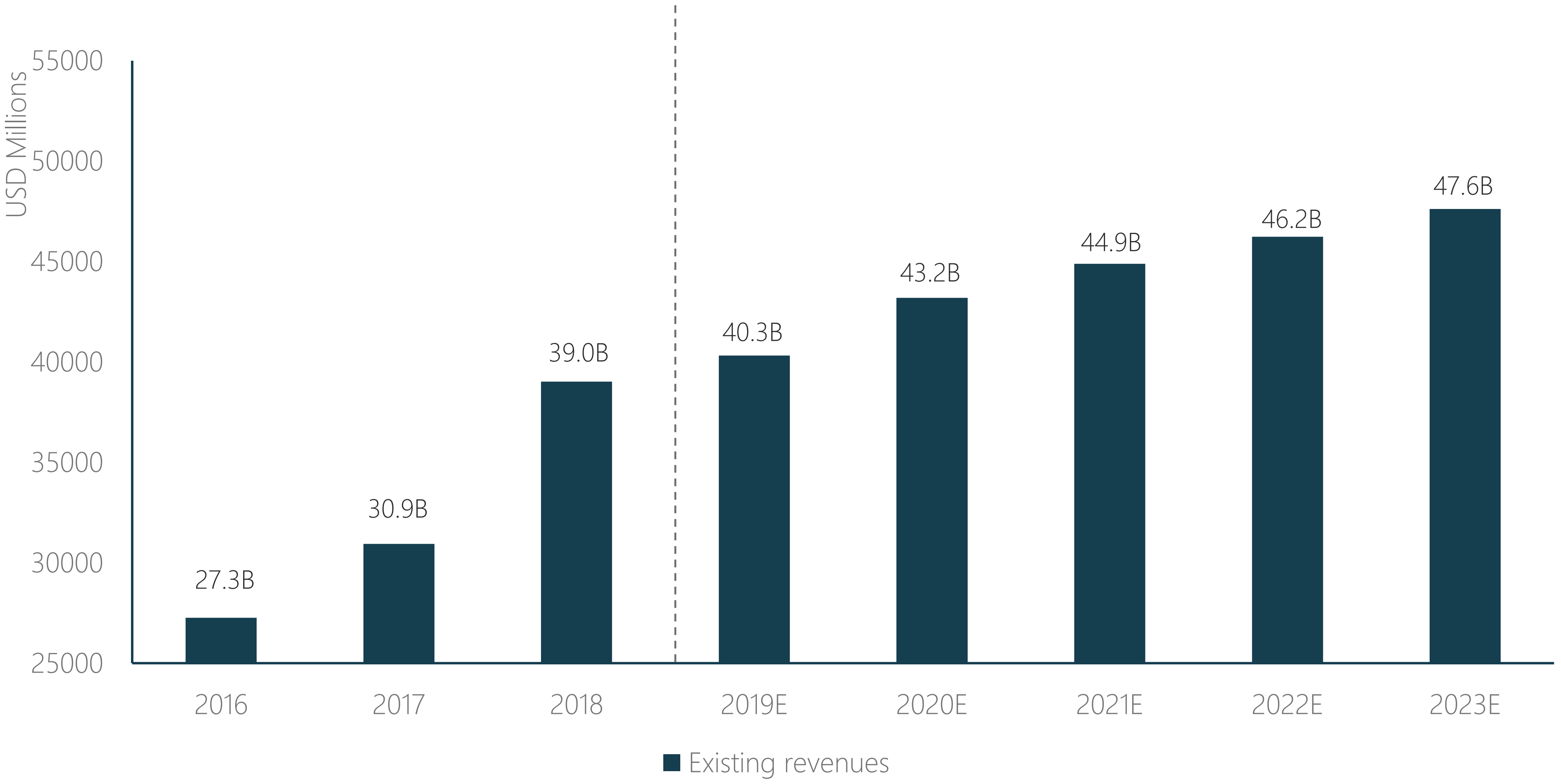
Implementing an integrated business intelligence solution will cultivate a **mutually beneficial relationship** between Maersk and its customers **through data**

What will be the impact of our recommendations?

What effect will there by on revenues?

Impact

Growth in the status quo environment is levelling off



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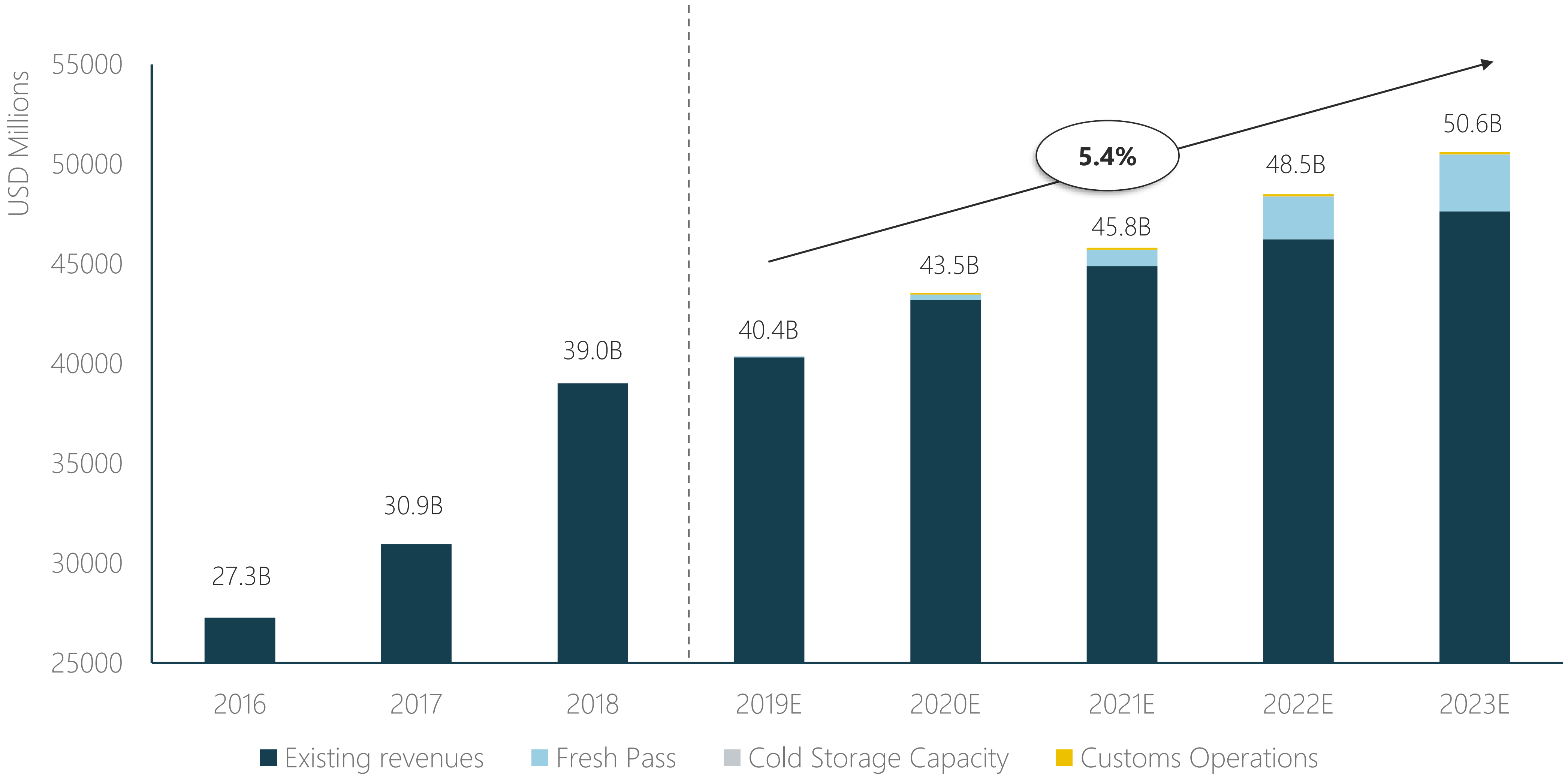
Impact

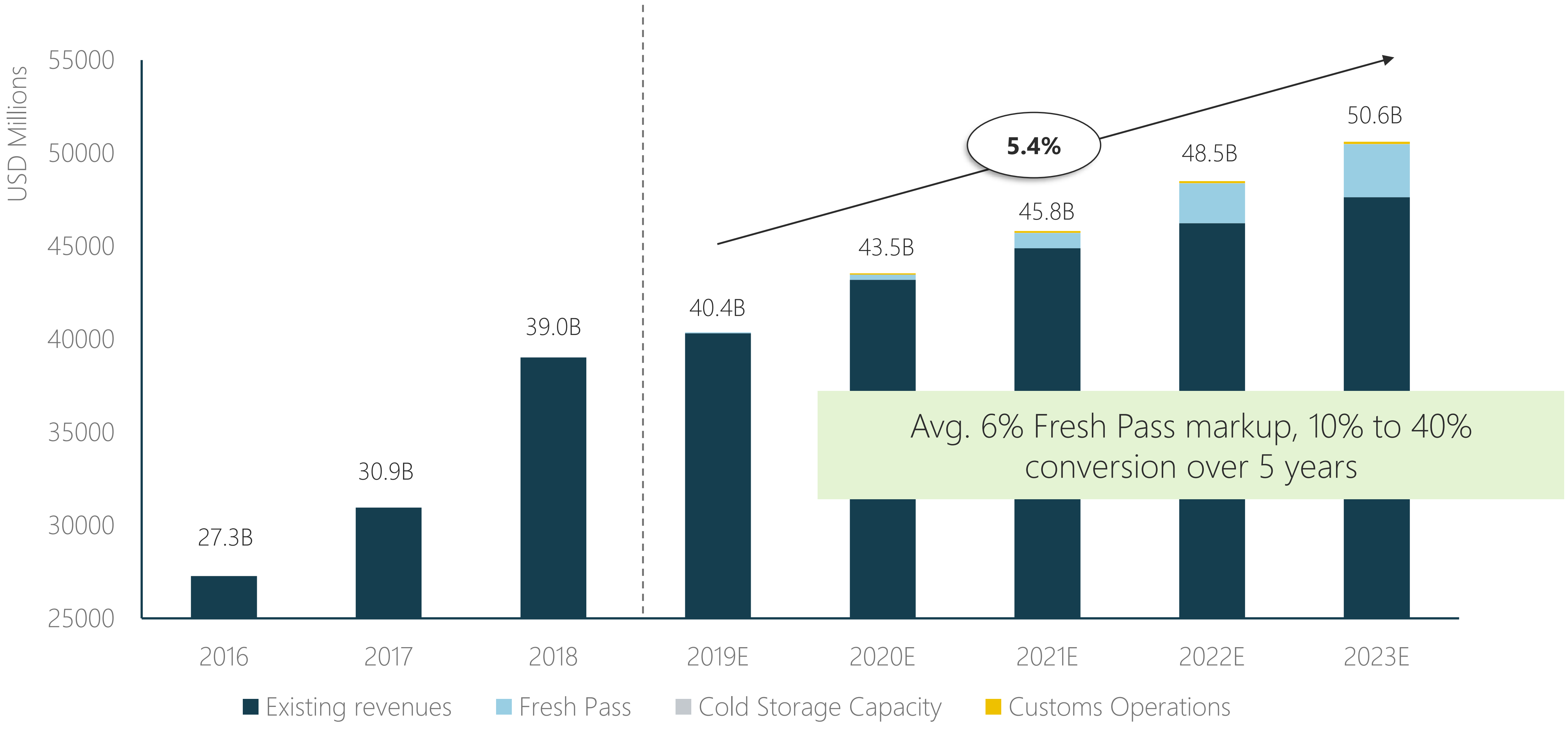
Proposed initiatives will produce a meaningful CAGR of 5.4%



MAERSK

Impact

Proposed initiatives will produce a meaningful CAGR of 5.4%



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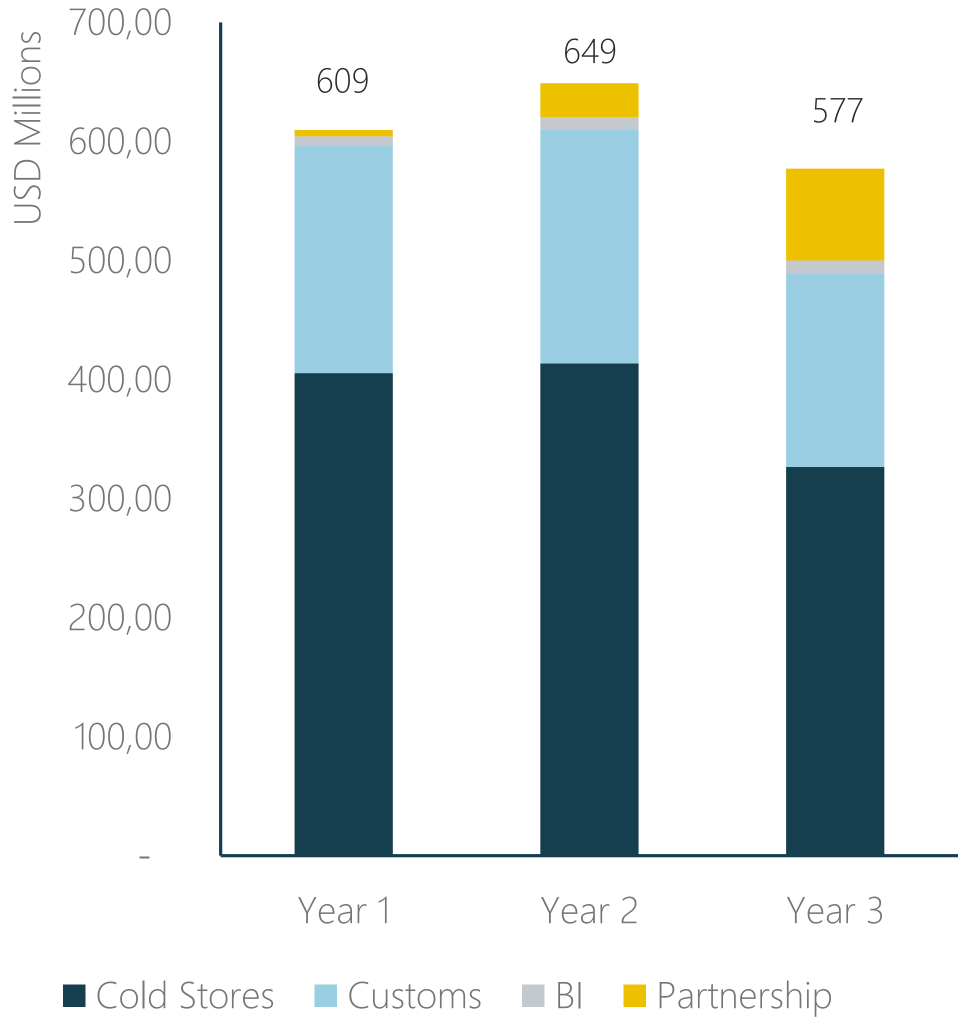
2

How much will this cost?

Impact

Investment costs within free cash flows, recouped by end of year 3



Upfront investment will incur costs of US$1.84B (DKK 12B) over 3 years

Free cash flow of $4.2B

CAPEX of $2.9B

Investment should be recouped by end of year 3.

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3

How risky is this investment?

Impact

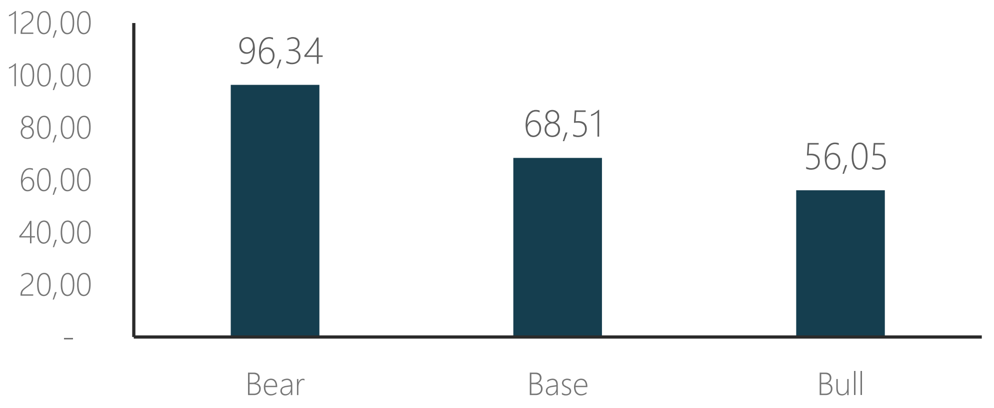
Our investment poses little risk in terms of break even

Incremental Acquisition Cost

Cost Buckets (See Appendix)

Cold Stores
Customs
Business Intelligence
Partnership
Payouts

Reefers Needed to Breakeven

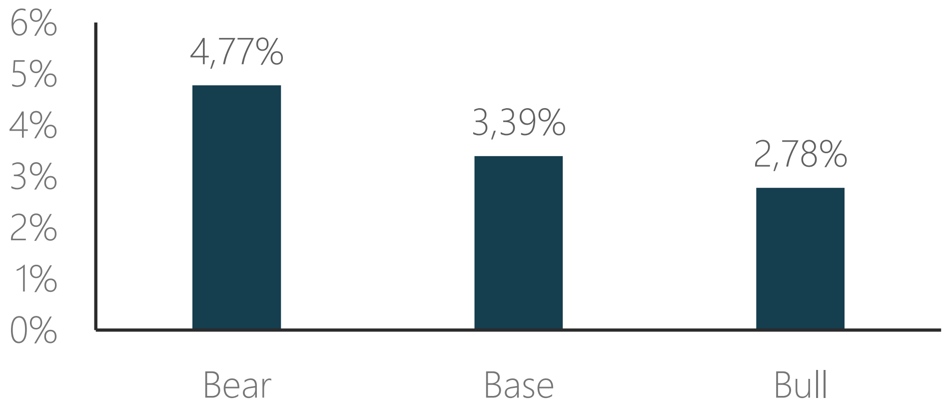


Maersk Reefer Ships of Pineapples

Inputs

Price per TEU of Pineapple Shipped
Number of TEUs Needed to Breakeven
Number of Unique Services

Increase Needed in Services



Just **3-5%** of our service routes need to be run one extra time for us to break even.

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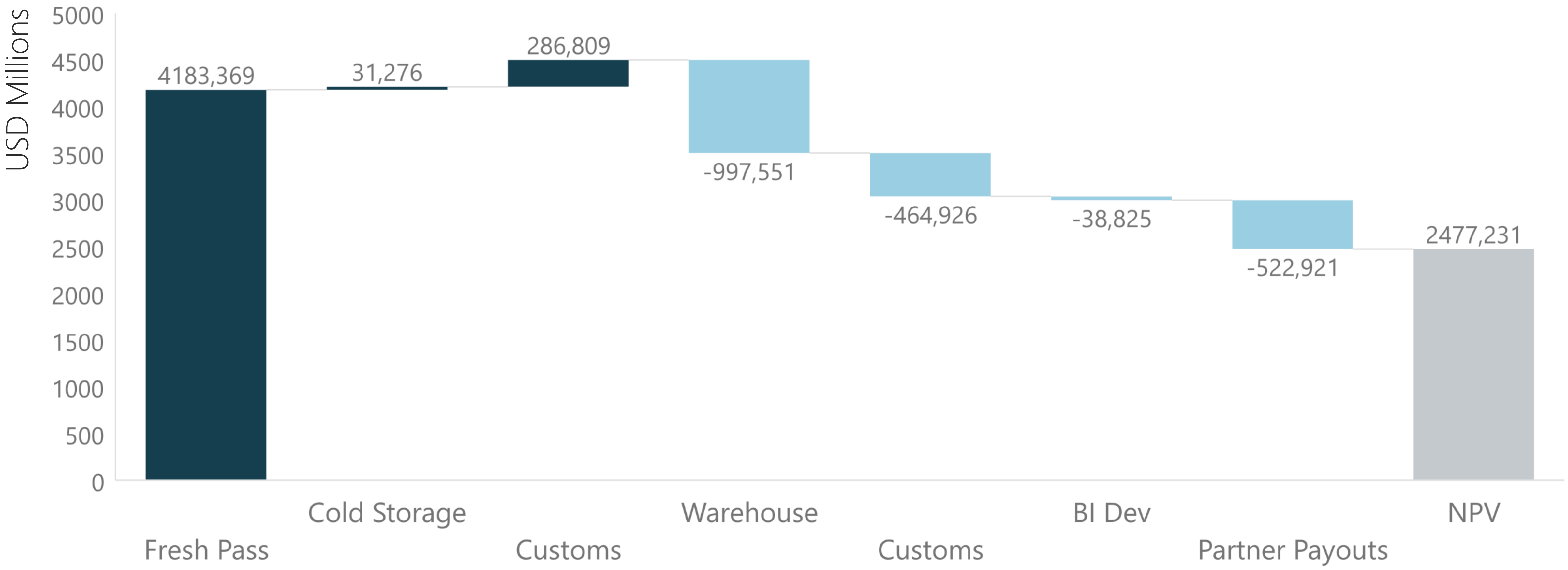
4

What is the overall NPV?

Impact

NPV of US$2.5B on a comparatively capital light project



5-year NPV of US$2.5B (DKK 16.4B)

WACC of 9.5%

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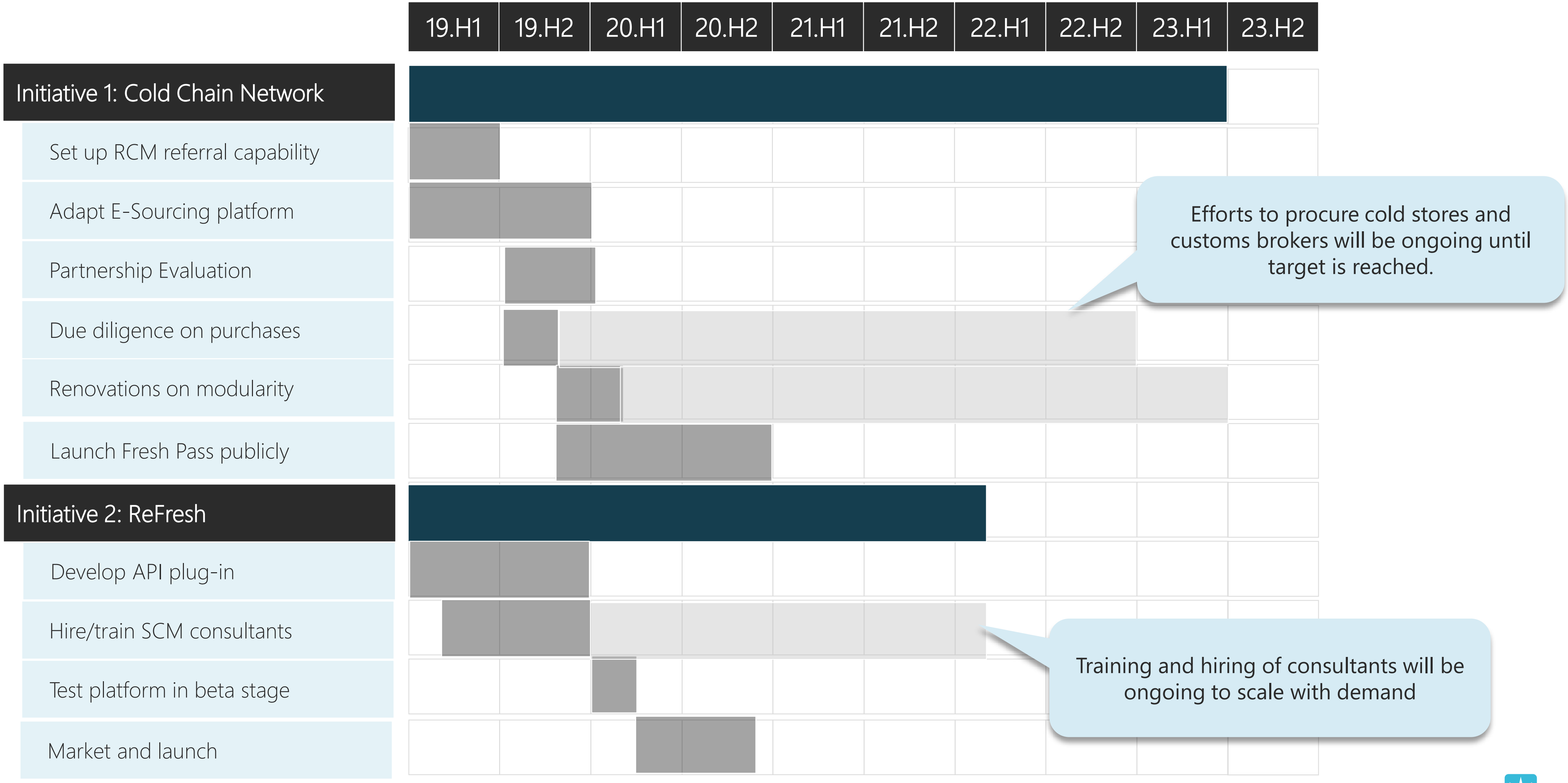
Feasibility

The ongoing roll-out will take place over a five year timeline



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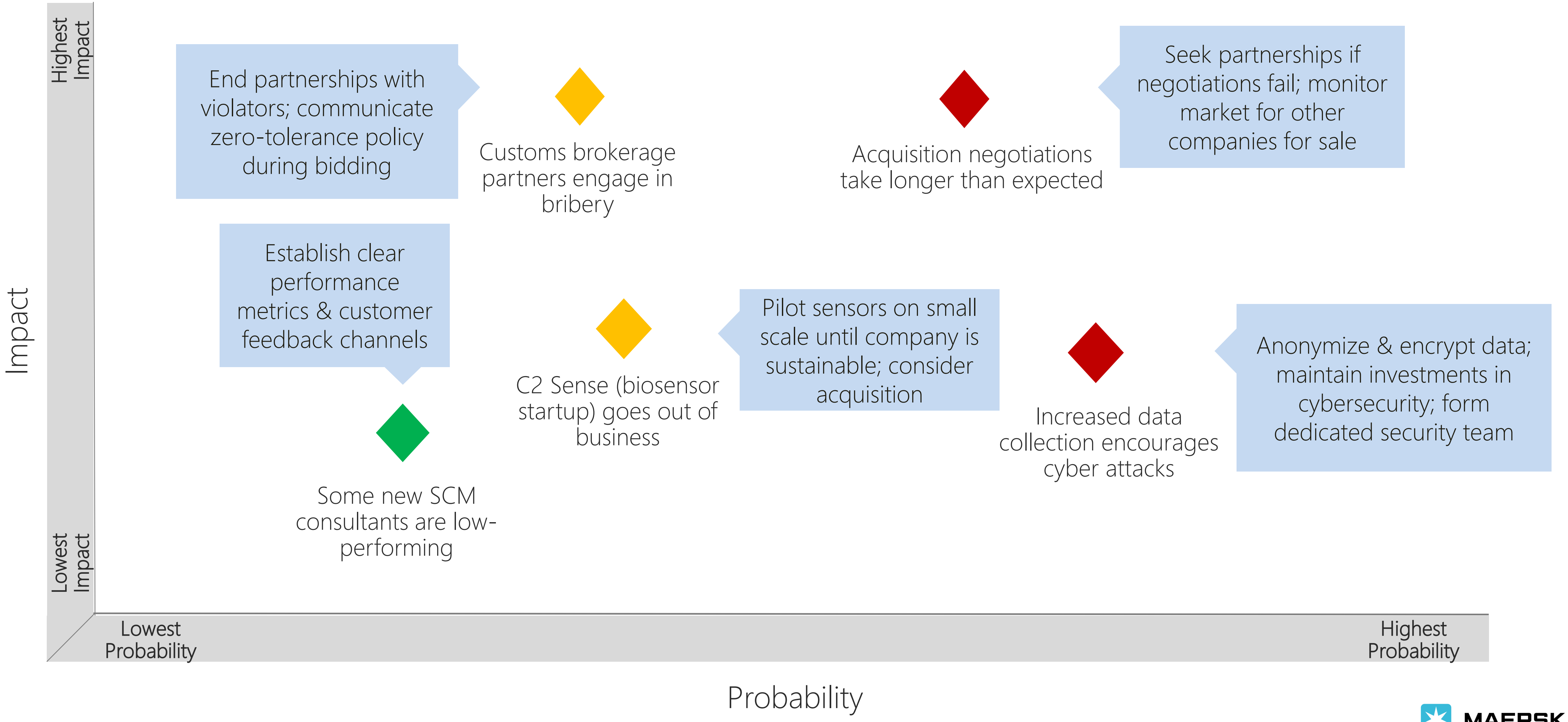
Feasibility

Project risks can be managed



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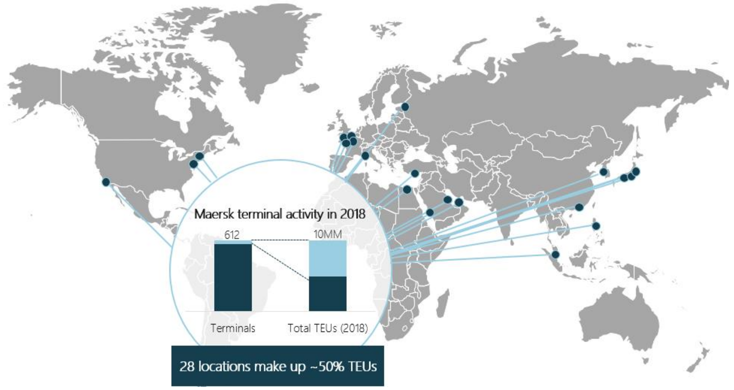
Maersk is well-positioned to revolutionize the cold chain segment

Buy & Partner

Establish seamless & reliable end-to-end cold chain infrastructure



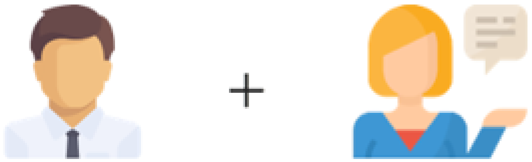
1. E-Sourcing
2. Referral Program
3. Probationary Period

Analytics with a Human Touch

Drive value for customers through personal consultation and analytics services

ReFresh

MAERSK Operations Consulting

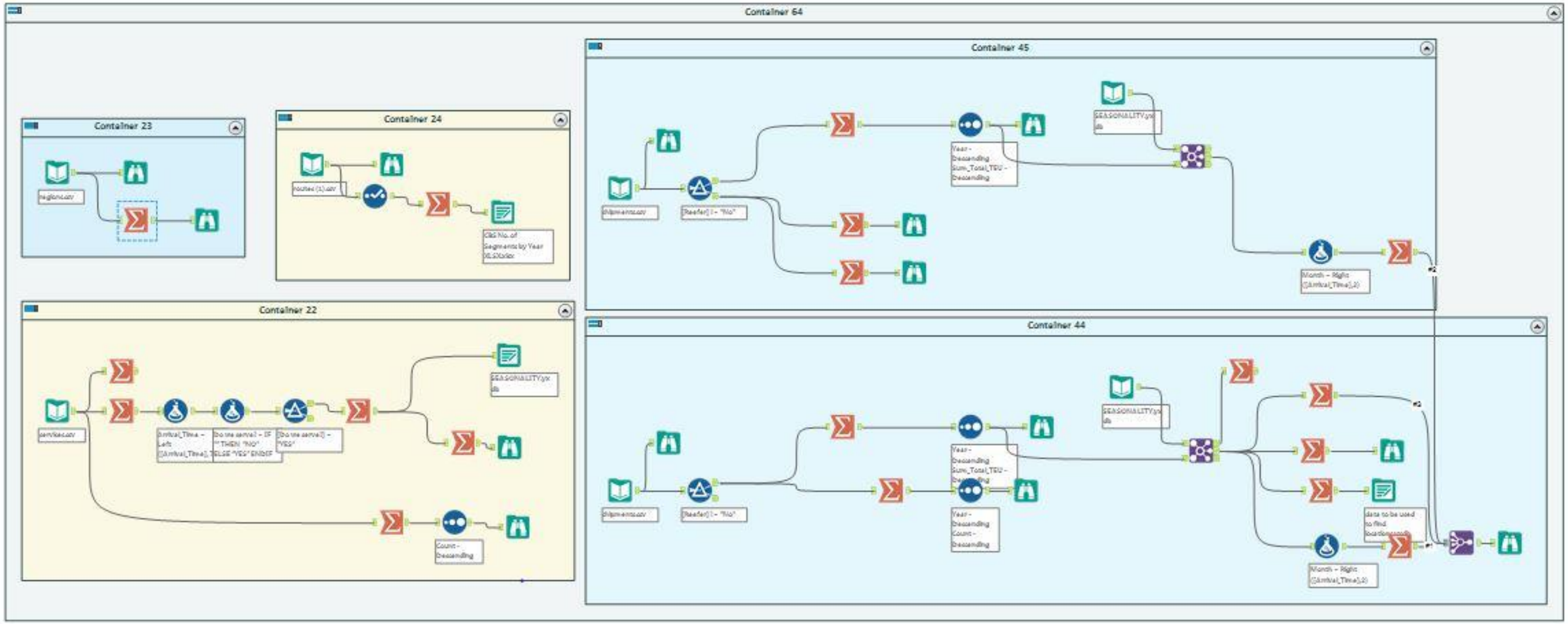


Impact: Net Present Value of **$2.48 Billion USD**

Appendix

Appendix

Alteryx manipulation of provided dataset



Fleet Overview for Ownership and Capacity

Fleet overview, year-end

	TEU 2018	TEU 2017	Number of vessels 2018	Number of vessels 2017
Own container vessels				
0 – 2,999 TEU	124,911	125,281	64	64
3,000 – 4,699 TEU	365,811	343,751	90	84
4,700 – 7,999 TEU	369,037	321,854	61	52
8,000 – 11,499 TEU	771,982	847,232	85	93
11,500 – 14,999 TEU	109,494	109,494	9	9
15,000 – 17,499 TEU	246,496	185,448	16	12
> 17,500 TEU	572,480	469,640	30	25
Total	**2,560,211**	**2,402,700**	**355**	**339**
Chartered container vessels				
0 – 2,999 TEU	388,314	463,887	189	240
3,000 – 4,699 TEU	290,950	339,628	74	85
4,700 – 7,999 TEU	272,428	395,913	45	67
8,000 – 11,499 TEU	289,467	313,176	31	34
11,500 – 14,999 TEU	207,168	211,522	16	16
Total	**1,448,327**	**1,724,126**	**355**	**442**
Total fleet	**4,008,538**	**4,126,826**	**710**	**781**
Newbuilding programme (own vessels)				
3,000 – 4,699 TEU	7,192	25,172	2	7
> 8,000 TEU	66,246	229,990	4	13
Total	**73,438**	**255,162**	**6**	**20**

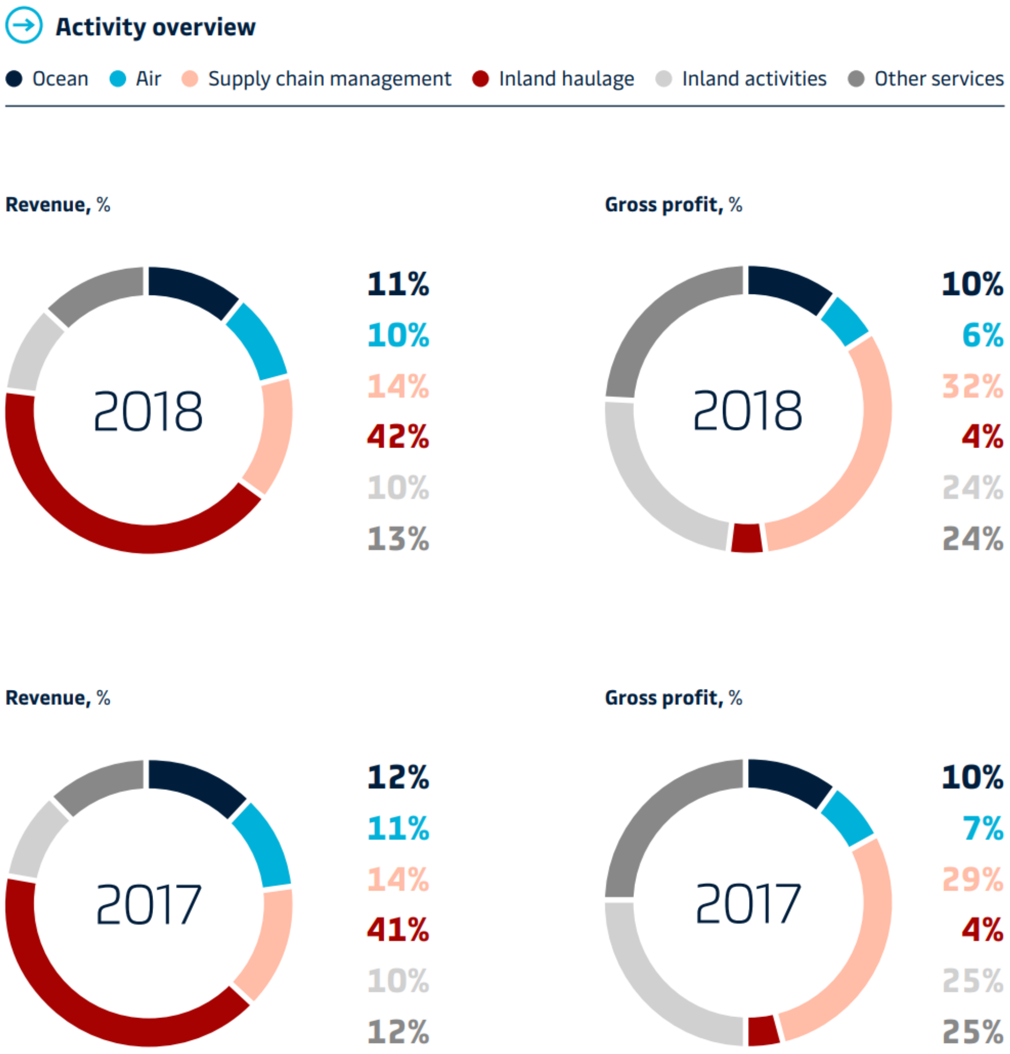
Ocean segment revenues and volume

Ocean highlights

USD million	2018	2017
Revenue	28,366	22,023
Profit/loss before depreciation, amortisation and impairment losses, etc. (EBITDA)	3,007	2,777
EBITDA margin	10.6%	12.6%
Gross capital expenditure, excl. acquisitions and divestments (CAPEX)	2,279	2,831
Operational and financial metrics		
Other revenue, including hubs (USD m)	3,441	2,547
Loaded volumes (FFE in '000)	13,306	10,939
Loaded freight rate (USD per FFE)	1,879	1,788
Unit cost, fixed bunker (USD per FFE incl. VSA income)	1,815	1,752
Hub productivity (PMPH)	81	73
Bunker price, average (USD per tonne)	424	321
Bunker cost (USD m)	5,042	3,341
Bunker consumption (tonne in '000)	11,894	10,395
Average nominal fleet capacity (TEU in '000)	4,115	3,456
Fleet, owned (end of period)	355	339
Fleet, chartered (end of period)	355	442

Appendix

Financial breakdown by Activities



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Appendix

CLTV Proxy

CLTV Proxy	Bear	Base	Bull
Price per TEU of pineapples	4640	5800	6380
TEUs Needed	395,577.62	316,462.10	287,692.81
TEU per Reefer	4,106.06	4619.3139	5,132.57
Reefers Needed	96.34	68.51	56.05
Unique Services	2,018.00	2,018.00	2,018.00
Reefers	5%	3%	3%
NPS Increase Needed	19 points	12 points	13 points

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Appendix

Cost Summary

Costs	Year 1	Year 2	Year 3
Cold Stores	405,242,857	413,450,857	326,474,286
Customs	191,006,864	196,737,070	162,507,446
BI	8,075,000	10,300,000	10,925,000
Partnership	5,313,338	28,271,396	77,176,038
Total	609,638,059	648,759,323	577,082,770
Total	1,835,480,151		

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Appendix

NPV (1/4)

	Year 1	Year 2	Year 3	Year 4	Year 5
Purchase of 28 warehouses					
Average US warehouse sq ft	114000	114000	114000		
Cost/WH ($300/sq ft)	34,200,000	35,910,000	37,705,500		
Total Cost	342000000	359100000	301644000		
Warehouses Owned	10	20	28		
Margin/sq ft	12	12	12		
Own Utilization	50%	70%	75%		
Annual profit	6,840,000	8,208,000	9,576,000		
Operating Costs	9,120,000	18,240,000	25,536,000	25,536,000	25,536,000
Annual Warehouse Cost	405,242,857	413,450,857	326,474,286	25,536,000	25,536,000
Present Value	370,084,801	344,822,549	248,660,397	17,762,185	16,221,174
Customs Inspection					
Price per TEU	29	30	31		
TEUs per ship	5,133	5,133	5,133		
Price per ship	149,345	153,825	158,440		
Ships processed/week	0.5	0.5	0.5		
Revenue/year	3,733,624.62	3,845,633.36	3,961,002.36		
Margin	50%	50%	50%		
Income/year	1,866,812	1,922,817	1,980,501		
Present value	18,914,005	19,481,425	20,065,868		
Number of broker firms	10	10	8		
Operating Cost	1,866,812	1,922,817	1,980,501	1,980,501	1,980,501
Annual Customs Cost	191,006,864	196,737,070	162,507,446	1,980,501	1,980,501
Present Value	174,435,492	164,080,874	123,774,422	1,377,586	1,258,069

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Appendix

NPV (2/4)

	Year 1	Year 2	Year 3	Year 4	Year 5
API Development Cost					
Software Engineer	125000	125000	125000	125000	125000
Number of Engineers	25	20	20	20	20
Annual Cost	3125000	2500000	2500000	2500000	2500000
Engineer Cost	3,125,000	2,500,000	2,500,000	2,500,000	2,500,000
Consultant Salary	80000	80000	80000	80000	80000
Number of Consultants	40	60	60	60	60
Consultant Cost	3200000	4800000	4800000	4800000	4800000
Systems Infrastructure					
Server Costs (AWS)	1250000	2500000	3125000	3125000	3125000
Hardware acquisition	500000	500000	500000	500000	500000
Annual BI Cost	8,075,000	10,300,000	10,925,000	10,925,000	10,925,000
Present Value	7,374,429	8,590,313	8,321,068	7,599,149	6,939,862
Partner Payouts					
Percentage Owed	25%	25%	25%	25%	25%
Revenues Attributed	23,272,421	135,592,441	405,307,677	1,065,644,373	1,428,474,790
Total Payout	5,818,105	33,898,110	101,326,919	266,411,093	357,118,698
Present Value	5,313,338.14	28,271,395.78	77,176,038.27	185,308,707.94	226,851,676.49
Total PV of Outflows	557,208,060	545,765,132	457,931,926	212,047,628	251,270,782
Sum of PV Outflows	2,024,223,527				

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Appendix

NPV (3/4)

	Year 1	Year 2	Year 3	Year 4	Year 5
Volume FFE	1568000	1646400	1728720	1815156	1905913.8
$/FFE	2679	2732.58	2787.2316	2842.976232	2899.835757
Enrolment rate	0.1	0.3	0.4	0.45	0.5
Enrolled FFE	156800	493920	691488	816820.2	952956.9
Fresh Pass Premium	0.05	0.055	0.06	0.065	0.07
Premium/FFE	133.95	150.2919	167.233896	184.7934551	202.988503
% Existing cust.	0.95	0.9	0.8	0.65	0.65
Total from existing	21003360	74232175.25	115640232.3	150943026.9	193439294.5
Total from new	23214240	158212009.9	510744359.2	1331692622	1592154193
Total Premium	44217600	232444185.1	626384591.5	1482635649	1785593488
Stickiness Bonus	1	1.05	1.1	1.15	1.2
Spoilage Factor	0.05	0.1	0.15	0.2	0.25
Total Incremental	46544842.11	271184882.6	810615353.7	2131288746	2856949581
Present Value FP	42506705.12	226171166.3	617408306.1	1482469664	1814813412

	Year 1	Year 2	Year 3	Year 4	Year 5
Cold storage					
External Utilization	0	0.3	0.3	0.3	0.3
Sq ft/warehouse	114000	114000	114000	114000	114000
Price/sq. ft	15	15.75	16.5375	17.364375	18.23259375
Utilization rate	0.6	0.65	0.7	0.75	0.75
Sq ft for rent	0	34200	34200	34200	34200
Revenue/warehouse	0	350122.5	395907.75	445396.2188	467666.0297
Warehouses	10	20	28	28	28
Total revenue	0	7002450	11085417	12471094.13	13094648.83
Present Value Cold Store	0	5840120.098	8443250.552	8674572.485	8318083.205

NPV (4/4)

	Year 1	Year 2	Year 3	Year 4	Year 5
Customs					
Price per TEU	28.25	28.815	29.3913	29.979126	30.57870852
TEUs per ship	5132.571	5132.571	5132.571	5132.571	5132.571
Price per ship	144995.1308	147895.0334	150852.934	153869.9927	156947.3926
Ships processed/week	0	0.5	0.5	0.5	0.5
Revenue/year	0	3697375.834	3771323.351	3846749.818	3923684.814
Brokers	10	20	28	28	28
Total Revenue	0	73947516.68	105597053.8	107708994.9	109863174.8
Present Value Customs	0	61673039.91	80428402.74	74919608.03	69788128.03

Revenue Breakdown	Year 1	Year 2	Year 3	Year 4	Year 5
Fresh Pass	46,544,842	271,184,883	810,615,354	2,131,288,746	2,856,949,581
Cold Storage Capacity	-	7,002,450	11,085,417	12,471,094	13,094,649
Customs Operations	-	73,947,517	105,597,054	107,708,995	109,863,175
Total Incremental	46,544,842	352,134,849	927,297,825	2,251,468,835	2,979,907,404
Present Value	42,506,705.12	293,684,326.28	706,279,959.42	1,566,063,844.04	1,892,919,623.16
Sum of PV Inflows	4,501,454,458.01				

NPV	2,477,230,930.63
WACC	9.50%

Appendix

Bridging the gap in Maersk's scope of data

Supply Chain

Producer › Export Haulage › **Ocean** › Terminal Handling › Customs Brokerage › Customs Inspection › Cold Storage › Import Haulage › **Final Destination**

C2 SENSE

C2 Sense is an MIT-based startup that produces low-cost biosensors with precise ethylene gas detection to monitor ripening and spoilage

ReFresh

Introduce an inventory management CRM plug-in with a user-friendly dashboard that enables access to customer level data

Appendix

Top 28 terminals by TEUs

Year	Site	Sum_Total_TEU
2018	KRPYOTM	1043101
2018	USPHLPT	616613
2018	SAJEDRGT	266500
2018	AEJALT2	233293
2018	NLROT01	204032
2018	RUPLPTM	176889
2018	JPTYO04	166927
2018	BEANTFT	160681
2018	HKHKGMO	147096
2018	ITC79RT	142812
2018	SGSINPS	128528
2018	EGALYAI	127855
2018	IQUMMGT	127849
2018	PHMNLTM	124607
2018	GBFXSTM	123670
2018	CNSGHY1	119071
2018	HKHKGHIT	104831
2018	JPYOKMY	103079
2018	USNWKTM	103006
2018	NLROTTM	100349
2018	RULEDTM	98279
2018	JPOSADI	96160
2018	VNHPHVG	91204
2018	TRMERMT	90964
2018	KWSWKTM	89393
2018	CNDAIDT	85964
2018	SADAMTM	84316
2018	USP1HPH	82706

Top 28 terminals by TEUs accounted for over 50% of TEUs across 612 terminals in 2018

How do we find the right talent for Maersk Operations Consulting?

Hiring Criteria

- A background in operations consulting or supply chain management
- At least 2 years of relevant work experience preferred
- Proficient in data analysis tools
- Strong in analyzing and improving processes to achieve business goals
- Purpose-driven – must be aligned with Maersk's values

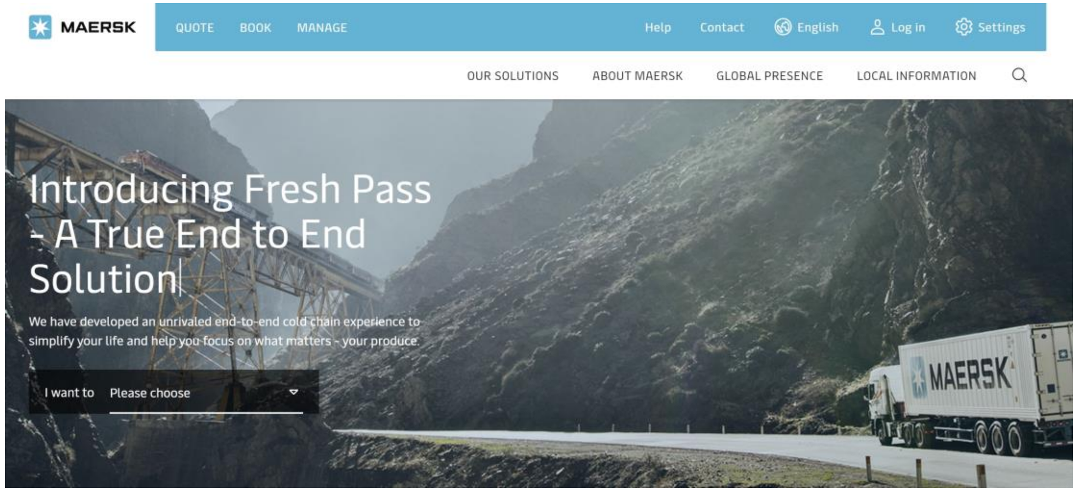
Finding Talent

- Postings on job boards with detailed description of Maersk and role
- Use LinkedIn recruiting tools to filter for specifications such as relevant educational background and work experience
- Campus recruiting events to generate interest and identify promising candidates
- *Start a relationship with* ***today's hire*** *as well as* ***tomorrow's talent***

Marketing

Prioritize targeting of existing customers first

Enrolling Existing Customers



Leverage digital platform (Maersk.com) for mass promotion and contact RCM customers

Push Fresh Pass at each customer touchpoint, offer trial priced at-cost

Existing customers easier to upsell and act as brand advocates.

Acquiring New Customers

Traditional channels like tradeshows and corporate conferences

Case studies and testimonials to boost perceived benefits

Drive further topline once value of Fresh Pass is proven with product-market fit.

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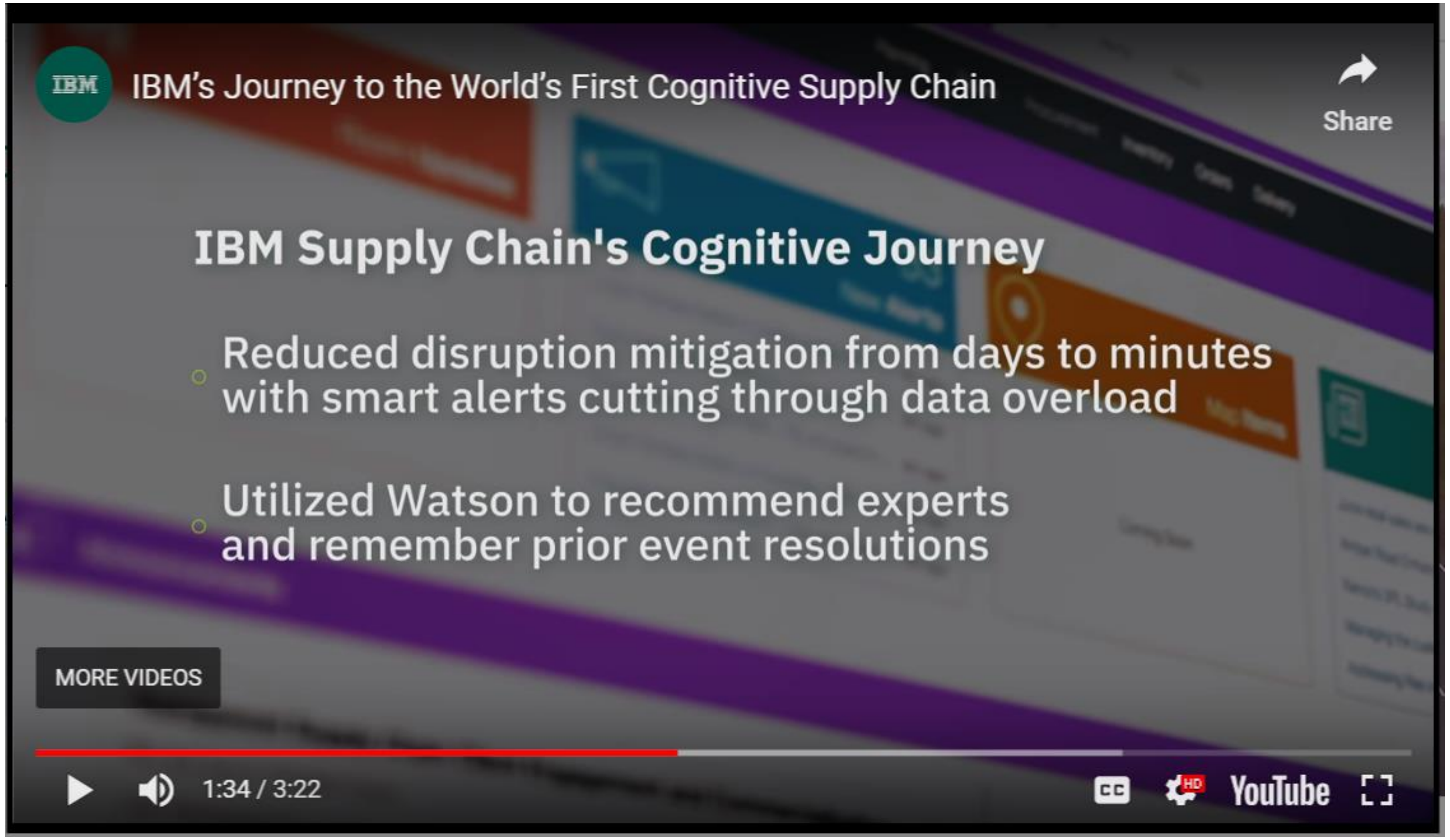
Case Study: IBM Watson cognitive supply chain



Provides reactive advisory services

Pairs customers with human subject matter expert (although they solely communicate via online channels)

https://www.ibm.com/supply-chain

Agile software development will deliver useful functionality quickly

	Agile Development	
Overview	*Cohesive teams including end users perform dev. tasks iteratively rather than sequentially*	
Styles	Scrum Teams	Extreme Programming
Tactics	Teams develop workable features ready for user testing in 2-4 week sprints	
Impact	Development is expedited; stakeholders are engaged; quality is improved	

Microsoft Azure partnership provides base upon which to build analytics capabilities

In 2017, Maersk commissioned Microsoft to power its app store using the Azure platform.

"To be successful, we need to partner with the people who have the knowledge, capabilities, and skills to help us on our digital journey. That's why we chose Microsoft Services."

- Ralf Weissbeck: Chief Information Officer of IT Infrastructure Services

Maersk Link Shipping Increases with increases NPS

"Maersk correlated a 4 point increase in Net Promoter Scores with a 1% increase in additional volume shipped by customers". Training improves Net Promoter Scores in local regions. "Maersk gave regional divisions the option of putting regional customer experience councils in place. The 55 regions that have set up local councils also received a three-day training course in customer experience improvement methods. The firm then did a study comparing regions with and without a council. The result: participating local offices score 10 points higher on their NPS than those offices that opted out". [Tweet this stat]

Appendix

Case Study: Shopify Plus and Merchant Success Managers

shopify plus

"Our Merchant Success Manager, Cassidy, treats me like a real person, not a client or a customer. He feels like a remote member of my team. Instead of being intimidated by Shopify Plus' platform, their support and the meetups with other merchants have inspired me to experiment and grow. I couldn't have done it without Shopify Plus."

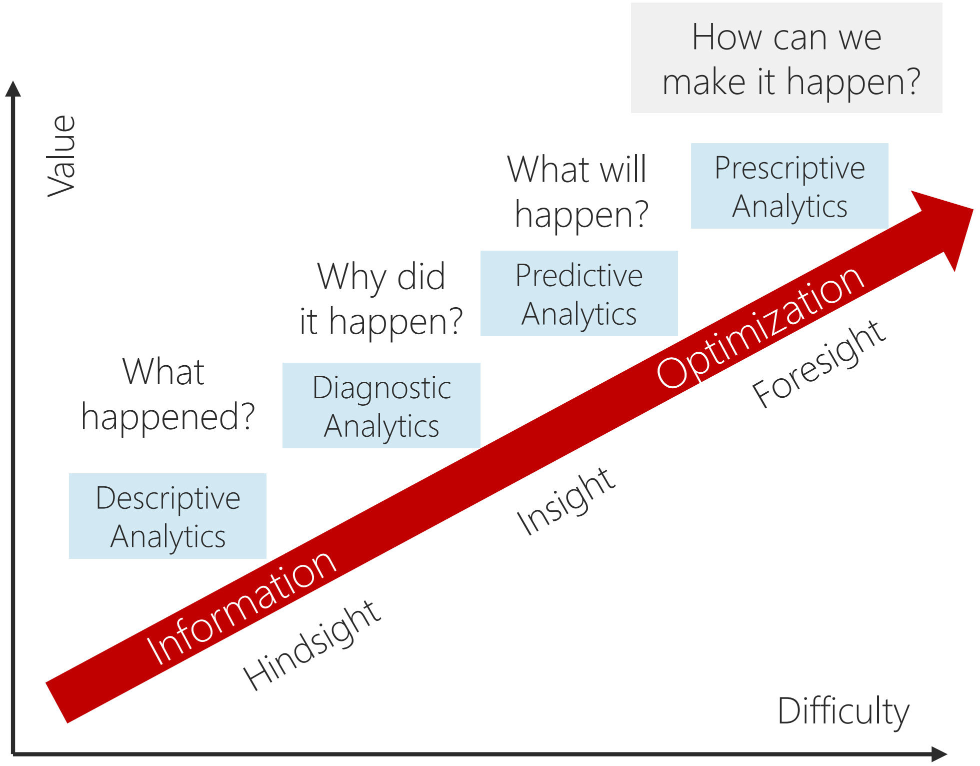
Cassie Smith

ECOMMERCE MANAGER AT JONES SODA

Appendix

Automated Suggestions – via Prescriptive Analytics

Equipped with customer shipping data, we are able to draw actionable insights and iteratively improve the customer shipping experience by addressing highlighted vulnerabilities.



Quality of Route

Assigned likelihood of route being recommended

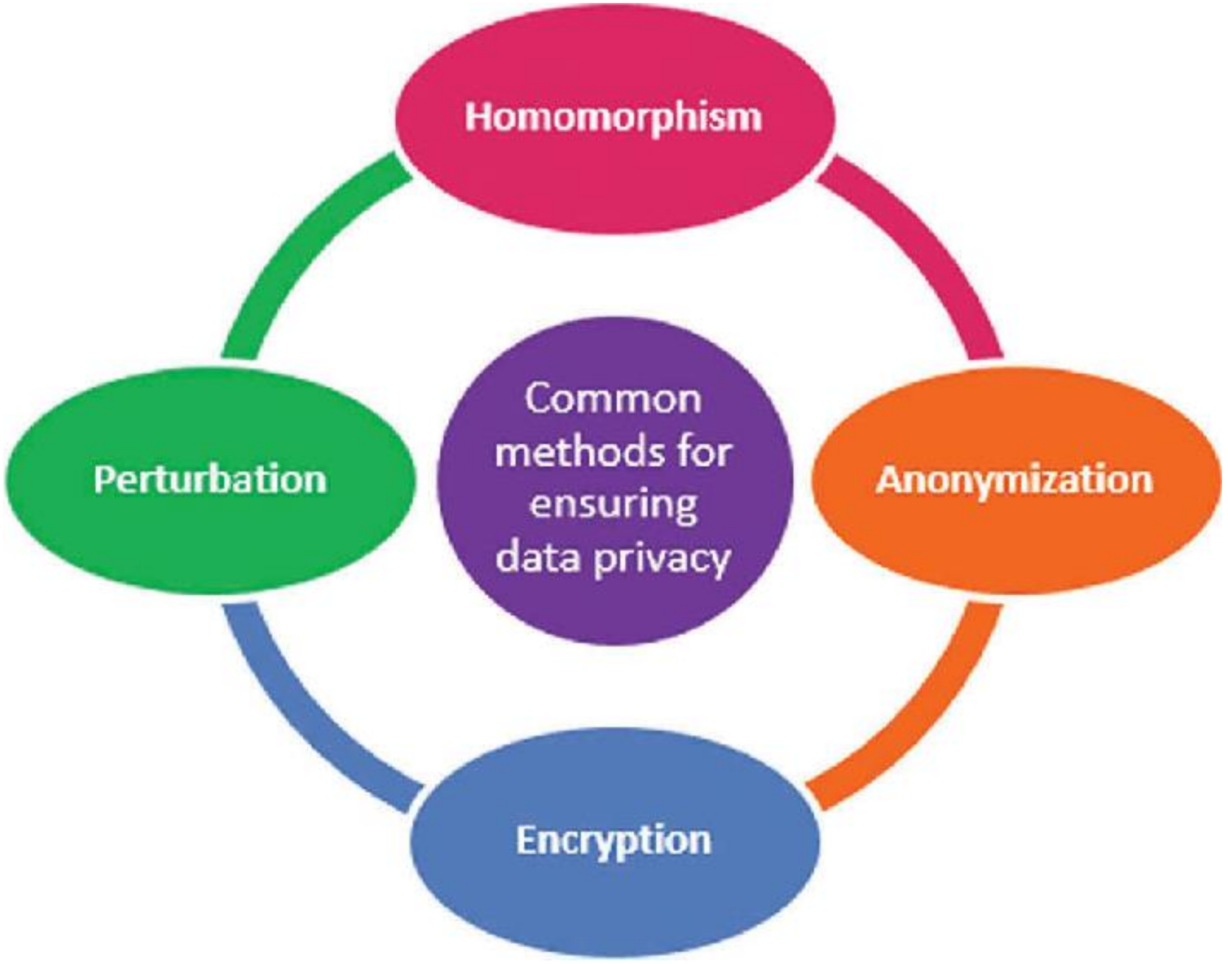
Specialist Delegation

Match customers with routes that optimize for price and risk

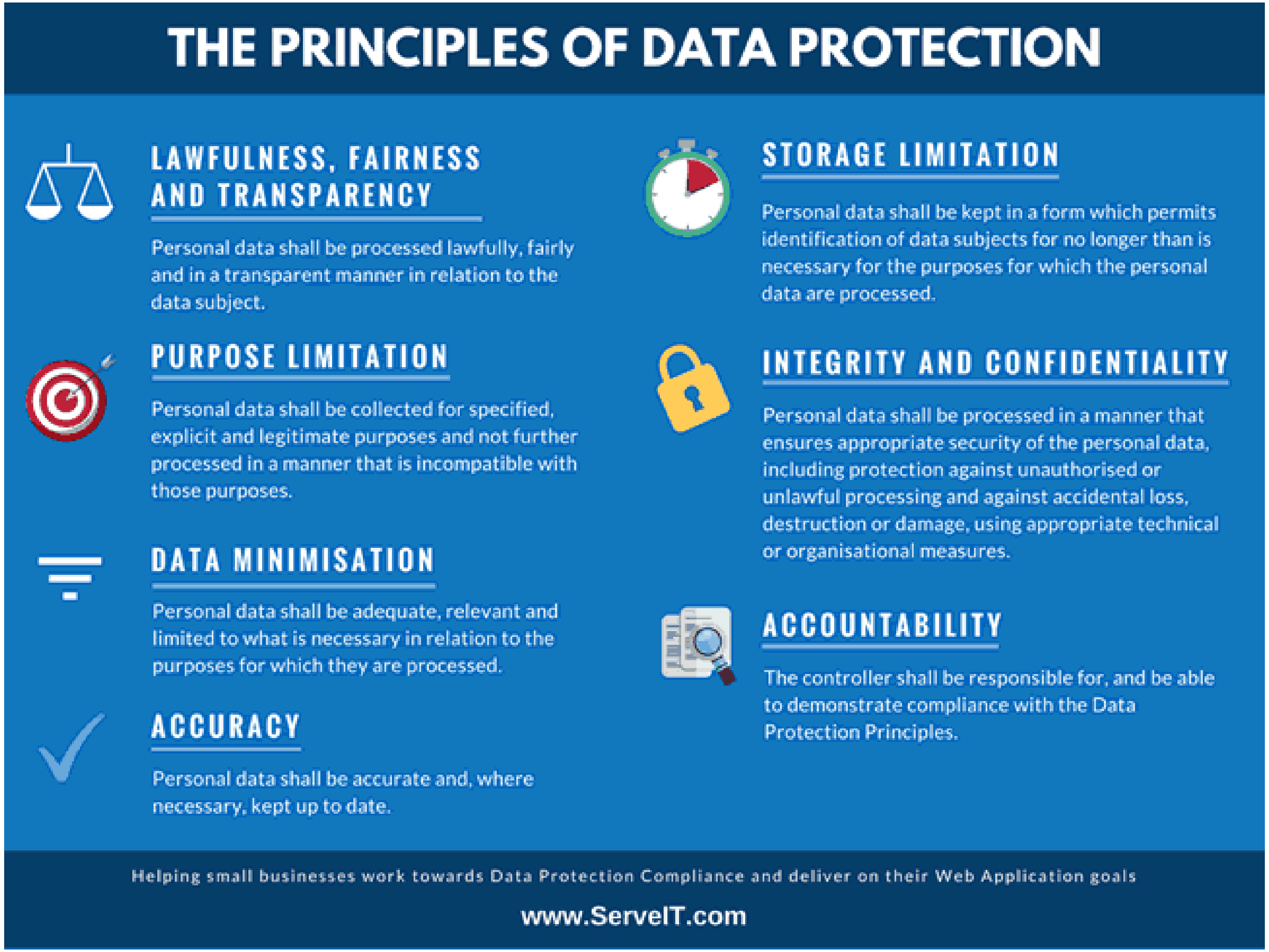
Knowledge Improvements

Understand and tailor insights to a customer's exact needs.

Ensuring data privacy for our customers



GDPR Key Principles



Appendix

Compliance with GDPR (1/2)

We can split GDPR into six privacy principles:

1. Lawfulness, fairness and transparency

Transparency: Tell the subject what data processing will be done.
Fair: What is processed must match up with how it has been described
Lawful: Processing must meet the tests described in GDPR [article 5, clause 1(a)]

2. Purpose limitations

Personal data can only be obtained for “specified, explicit and legitimate purposes”[article 5, clause 1(b)]. Data can only be used for a specific processing purpose that the subject has been made aware of and no other, without further consent.

3. Data minimisation

Data collected on a subject should be “adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed”.[article 5, clause 1(c)]
i.e. No more than the minimum amount of data should be kept for specific processing.

4. Accuracy

Data must be “accurate and where necessary kept up to date” [article 5, clause 1(d)]
Baselining ensures good protection and protection against identity theft. Data holders should build rectification processes into data management / archiving activities for subject data.

Compliance with GDPR (2/2)

5. Storage limitations

Regulator expects personal data is “kept in a form which permits identification of data subjects for no longer than necessary”. [article 5, clause 1(e)]
i.e. Data no longer required should be removed.

6. Integrity and confidentiality

Requires processors to handle data “in a manner [ensuring] appropriate security of the personal data including protection against unlawful processing or accidental loss, destruction or damage”. [article 5, clause 1(f)]

STORY

1904
Founded as a truly global company

1928
The first Maersk Line voyage launched linking people and products

1975
Entered the container shipping industry

2018
Refocused operations by selling oil & gas business

Now
Where to next?

Maersk

Building Links

EVA Consulting

Angela Guan

Joanna Ma

Kyla Tan

Sandaru Kam

Executive Summary

Analysis: Executive Summary

EVA Consulting

Question

How can Maersk leverage its global leadership in reefer shipping to pioneer the global cold chain market in the next 5 years?

Considerations

1. Threat of commoditisation in cold chain
2. Poor customer experience within existing value chain
3. Lack of end-to-end capability

Strategy

Connecting the Dots

- Digital analytics solutions across an end-to-end integrated supply chain
- New MyMaersk Metrics feature built onto the existing MyMaersk platform

Outside the Box

- Developing cold-storage capabilities in origin destinations
- Lease sharing model targeted towards SME customers within region

Impact

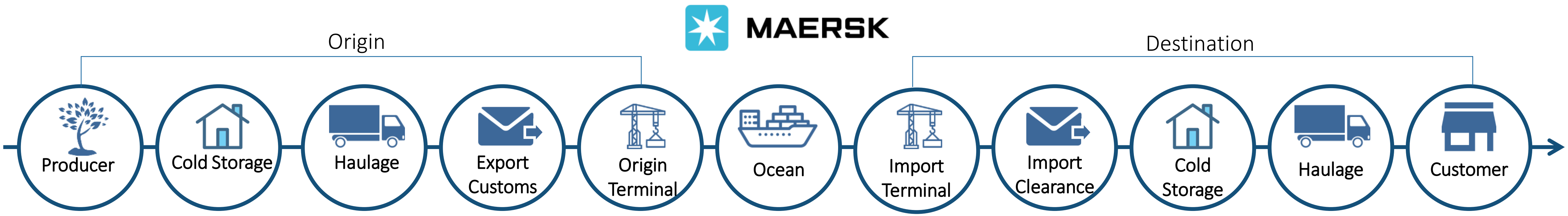
30,000 TEU
Cold Storage Capacity

31% EBITDA Margin
by 2023

1.8b DKK annual
Incremental Revenue

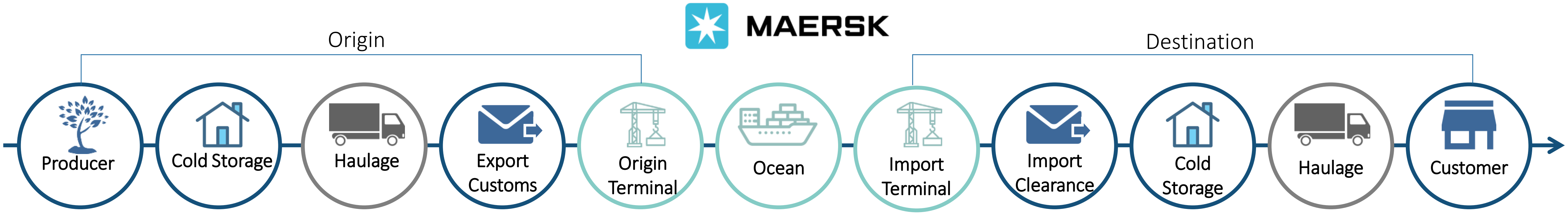
Maersk has a vision to create an end-to-end supply chain

Existing cold chain



Maersk's strengths are within the ocean segment where it owns and operates assets

Existing cold chain



~28% Market share in reefer shipping

69 Terminals

52 Countries

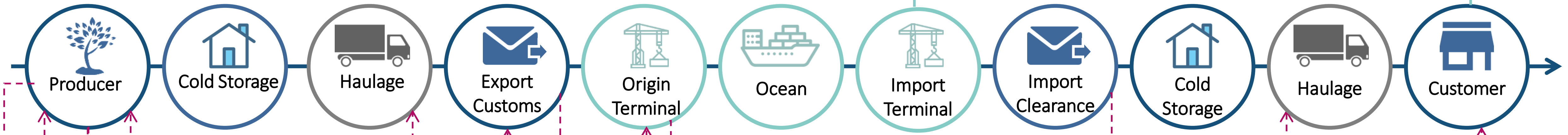
MAERSK

Customers face complexities in managing and financing payments to multiple stakeholders

Cold Chain Issues

EVA Consulting

Pain Points



1 Communicate to **multiple** stakeholders

2 Pay **multiple** stakeholders on **multiple** invoices

3 Significant time **delays** and transportation **costs**

Philadelphia Pilot **one-stop-shop** only helps reduce complexities in the communication and invoicing process but **does not combat commoditisation** where competitors are also…

- MSC: Investing in inland assets
- CMA CGM: Powered the major Ocean Alliance
- COSCO Shipping: Pursuing end-to-end visions in China

communicated to producer

Maersk must improve the stickiness of its customers with value-adding services by integrating itself throughout and beyond the cold chain

ayed

to trucking company

delivery date to end receiver

Analysis

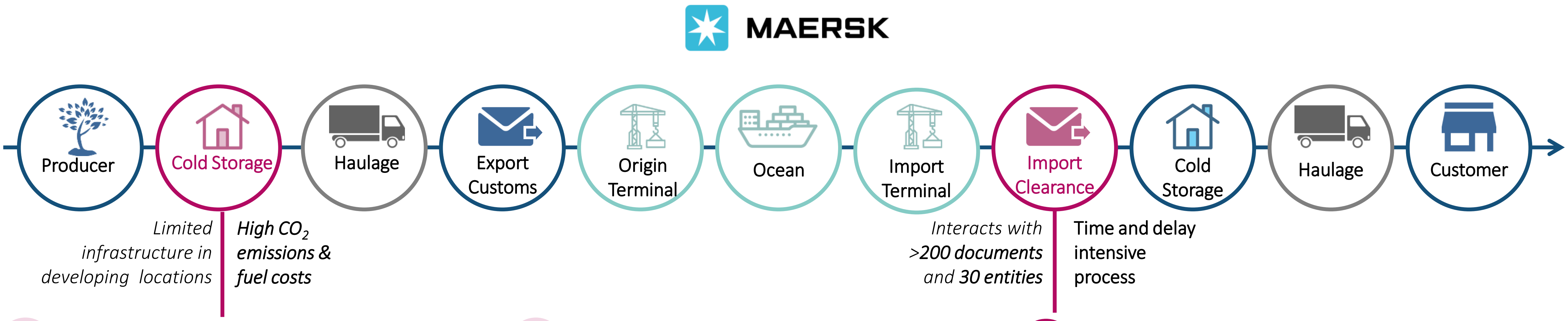
Source: McKinsey Digital Supply Chain Transformation Report, CMA GCM, joc.com

MAERSK

Maersk must build capability in all links within the supply chain to create a seamless customer experience

EVA Consulting

Cold Chain Issues



1 Communicate to multiple stakeholders

2 Pay multiple stakeholders on multiple invoices

3 Significant time delays and transportation costs

*"It is important to be **operationally in control of the critical assets in the supply chain.**"*
– Morten Bo Christiansen
Head of Strategy

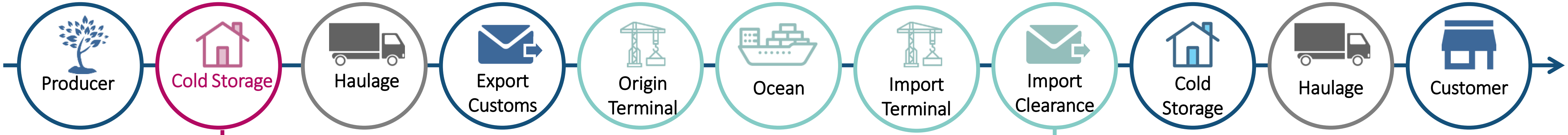
Source: McKinsey 2018 Report 'Travel and logistics: data drives the race for customers'

Maersk must invest into developing cold storage capabilities

Supply Chain Opportunity

EVA Consulting

MAERSK



Limited infrastructure in origin countries

High CO_2 emissions & fuel costs

?

Maersk has acquired capability in customs clearance

VANDEGRIFT

Why? Capability to **design and build individualised customs brokerage and compliance programs** to identify risk and provide solutions for customers

When? Acquisition announced Feb 2019

*"It is important to be **operationally in control of the critical assets in the supply chain."***
– Morten Bo Christiansen
Head of Strategy

Maersk's long term strategy should adopt five guiding principles

Long-term Strategy Guiding Principles

EVA Consulting

Maersk's five year cold chain strategy should follow these guiding principles…

L everage existing capabilities

I nnovate the cold chain

N avigate key challenges

K eep to our values

S tart Monday

Our strategies will enable Maersk to sustainably grow

Strategy Overview

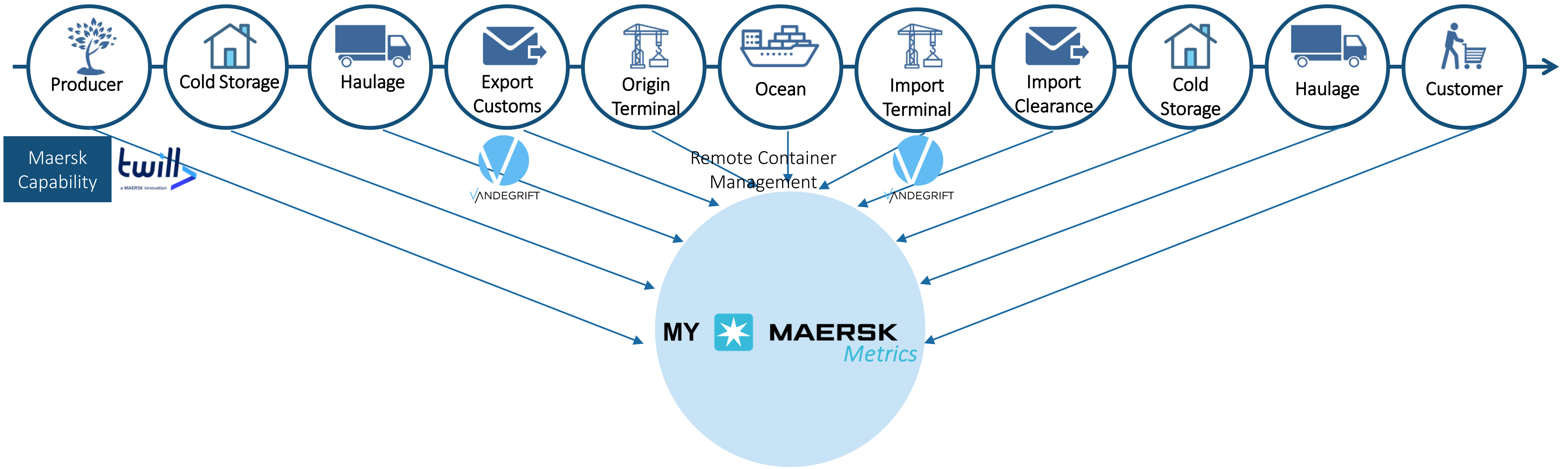
OBJECTIVE	STRATEGY	RECOMMENDATIONS	IMPACT
How can Maersk leverage its global leadership in reefer shipping to pioneer the global cold chain market in the next 5 years?	Create a value-adding end-to-end service to drive stickiness	***Connect the Dots*** Digital analytics platform across an end-to-end integrated supply chain transforming Maersk to a solutions provider 	12% CAGR
	Build end-to-end capabilities	***Outside of the Box*** *Building origin cold-storage capabilities and a lease sharing model targeted towards SME customers* 	30-60% energy saved

Connecting the Dots

Mearsk has existing capability to gain data from different parts of the supply chain

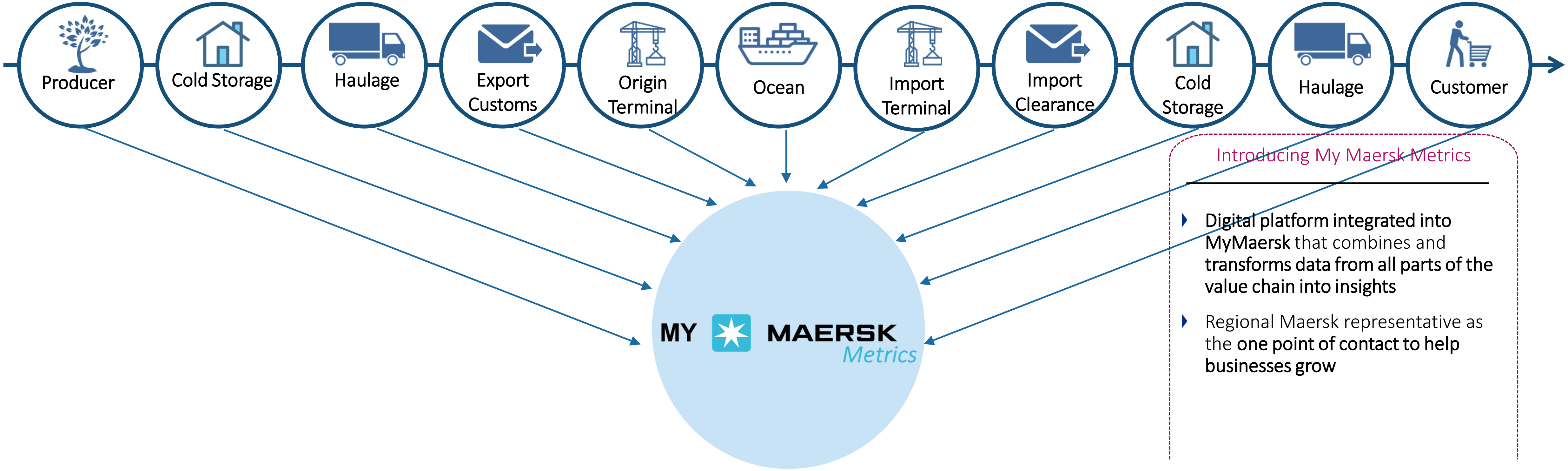
Connecting the Dots: Transforming the supply chain from linear to circular

EVA
Consulting



Introducing My Maersk Metrics: Helping businesses grow by transforming the supply chain

Connecting the Dots: Transforming the supply chain from linear to circular



Introducing My Maersk Metrics

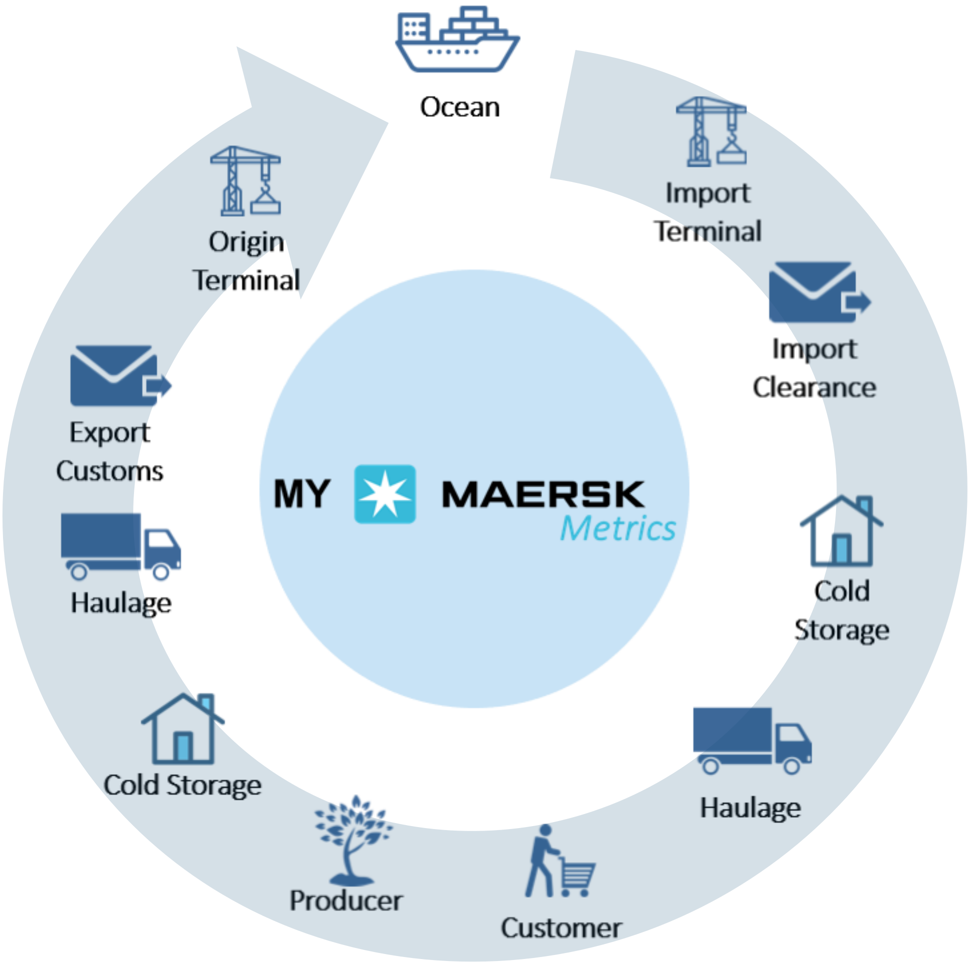
- **Digital platform integrated into MyMaersk** that combines and **transforms data from all parts of the value chain into insights**
- Regional Maersk representative as the **one point of contact to help businesses grow**

MAERSK

Introducing My Maersk Metrics: Helping businesses grow by transforming the supply chain

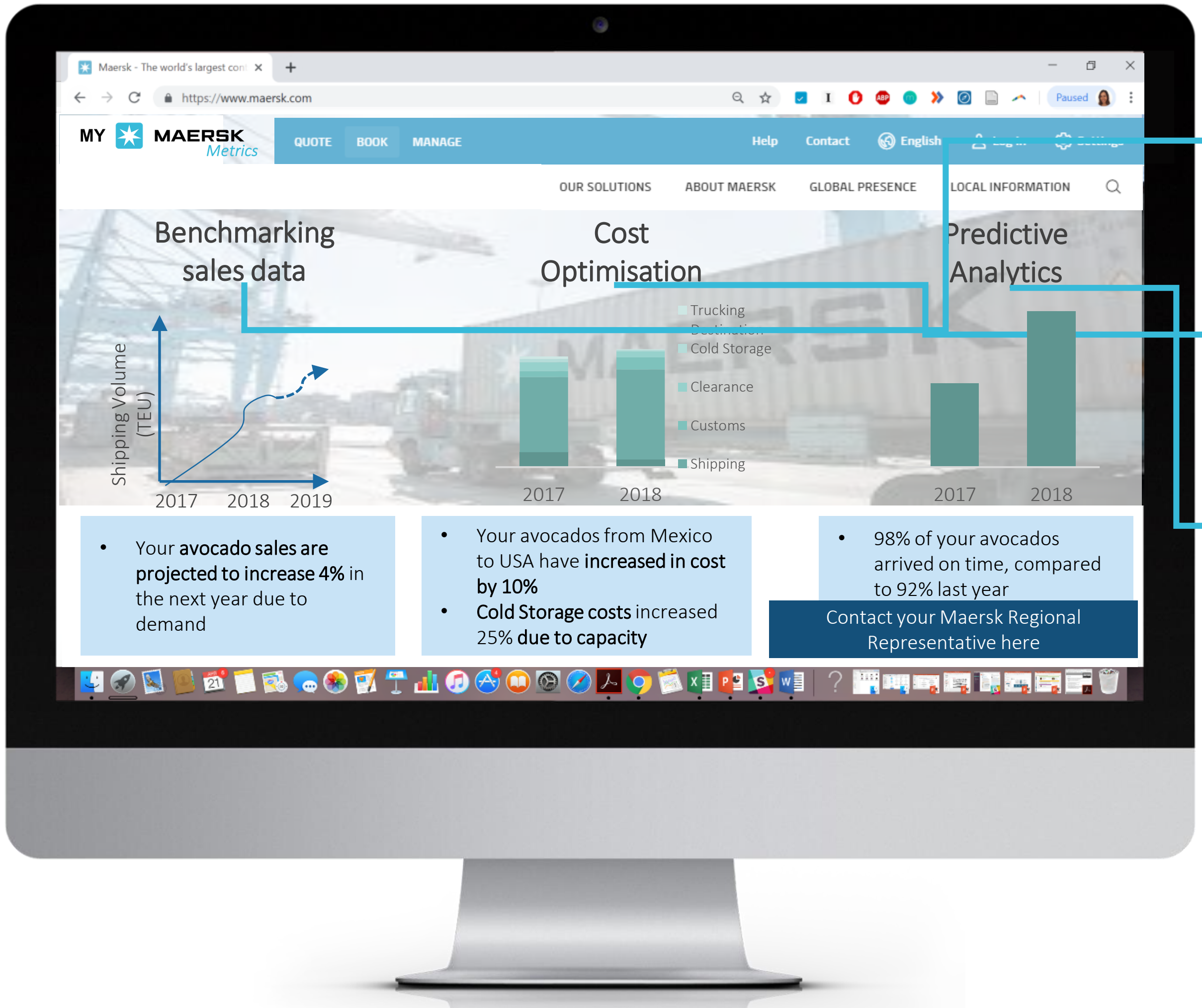
EVA Consulting

Connecting the Dots: Transforming the supply chain from linear to circular



MyMaersk Metrics creates a seamless end to end customer experience to build businesses

Connecting the Dots: What does Maersk Metrics look like to customers?



	MyMaersk Metrics Feature	Data Source	Description
1	Benchmarking Sales Data	Twill	• Displays **historical and current data against forecasted demand**
2	Cost Optimisation	RCM sensors Vendergrift	• Tracks transportation, customs, production costs to **optimize costs**
3	Predictive Analytics	RCM sensors	• **Predicts state of cargo** based on past data and macro events

Actionable insights to the customer

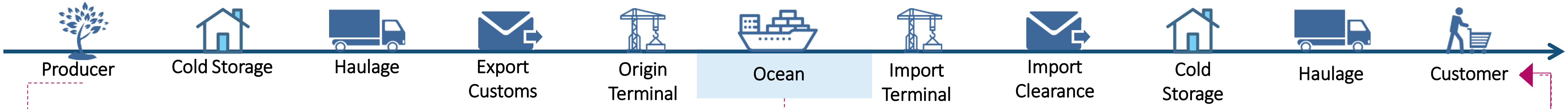
One Maersk point of contact

Improved transparency and customer friendliness

For Maersk, MyMaersk Metrics will leverage its data to optimise its network and assets

Connecting the Dots: What does Maersk Metrics look like to Maersk?

EVA Consulting



Data Source	Use of Data	Benefit to Maersk
Twill – Maersk's Digital Freight Forwarder	‣ Booking demand and tracking ‣ Consolidated paperwork	‣ **Predictive analytics on seasonality** in transit routes and booking demand across channels ‣ Ability to **feed data back into shipping network** to identify bottlenecks and busy periods for customers
RCM	‣ Data logs from reefers showing conditions during transport including temperature, humidity and CO_2 levels	‣ Become more proactive in planning to **predict the state of cargo assets on arrival** ‣ **Upsell** new remote-controlled services such as pre-cooling
Vandergrift – Customs Broker	‣ Access to trade customs filings, frequency and new compliance practices	‣ Manage shipping in response to delays, **predicting delays and planning for changes in regulations** ‣ Ability to build customized programs that **identify key regulations and trade compliance risks**

Maersk can leverage the data in its supply chain in a circular feedback way, ultimately improving the customer experience

Source: McKinsey Travel and Logistics 2018 Report

MyMaersk Metrics will be scaled across Maersk products in the long-term

Connecting the Dots: 5 Year Rollout Plan

Roll out plan

The Five Year Plan	Monday Morning March 2019	6 months September 2019	12 months March 2020	2 years March 2021	5 years March 2024
How to attain capability?	Leverage own digital capabilities	Test product with customers	Finalise platform product	Expand to rest of the business model	
How?	▸ Adapt customer service model of Philadelphia cold chain pilot ▸ Begin training Maersk customer single point of contact	▸ Rollout pilot program of digital platform to Philadelphia cold chain market as beta test ▸ Conduct research on value chain stakeholders data	▸ Finalise platform capability and incentives for stakeholders to use platform ▸ Examine other parts of business	▸ Conduct a roll out plan to other parts of business	
Resourcing	~2 FTEs (recruitment)	~50 FTEs	~160 FTEs (development, integration, operation)	200 FTEs	

Outside the Box

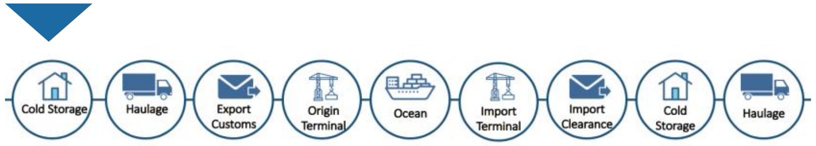
MAERSK

The need for cold storage arises in both origin and destination countries

EVA Consulting

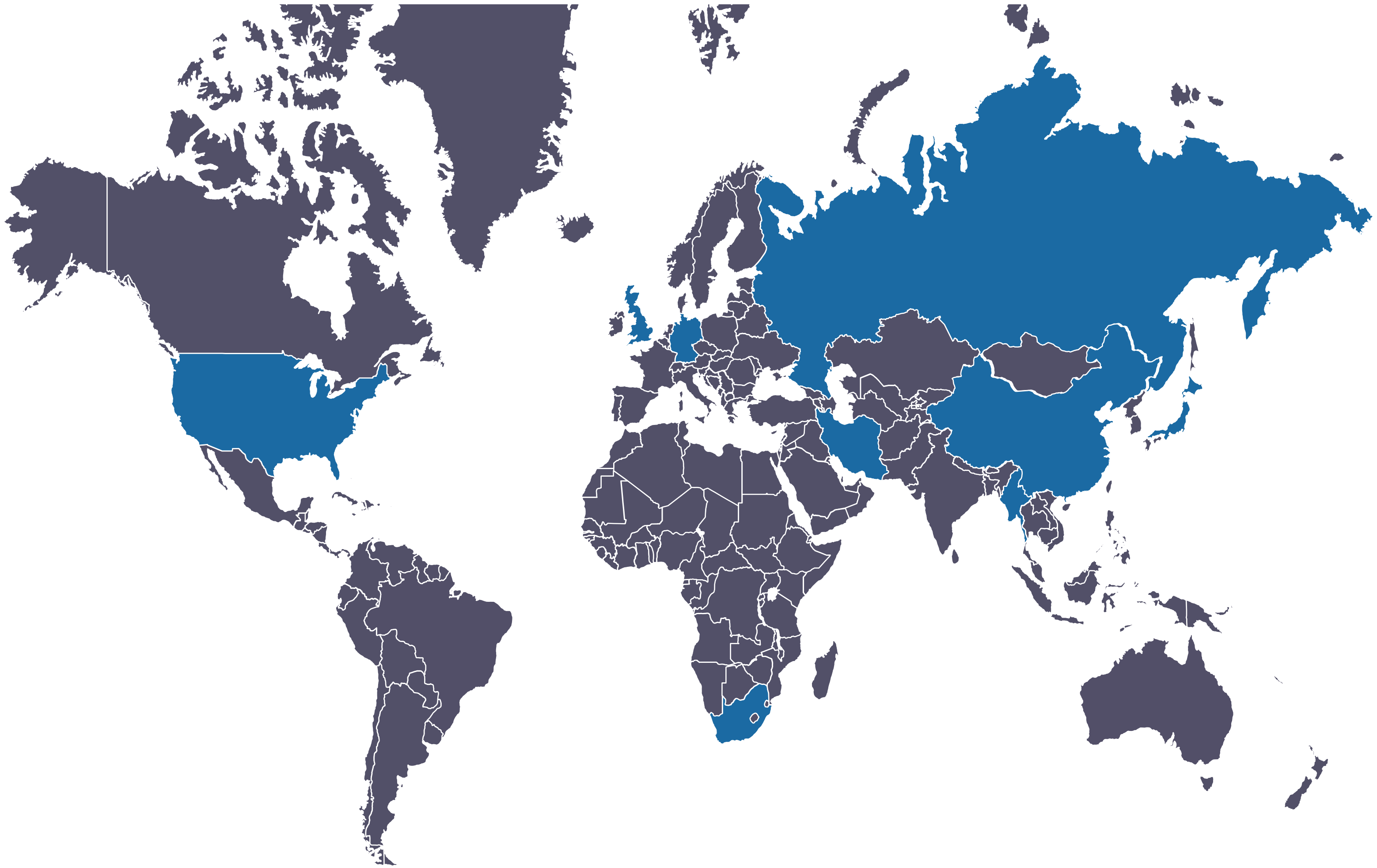
Outside the Box: Destination and Origin Countries

Value Chain



Analysis of Destination Countries

Country	TEU	Installed Capacity (m^3)
United States	3207650	130.965
Korea	2633342	71.3
China	1461561	105
Japan	1253704	37.612
Russia	794662	22
South Africa	772429	0.47
Netherlands	738017	13.7
Great Britain	621416	24.117
UAE	579367	25.6
Hong Kong	507531	26.4

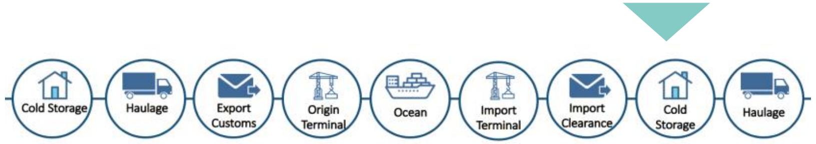


The destination countries do not present any growth opportunities due to XXXXXX

The need for cold storage arises in both origin and destination countries

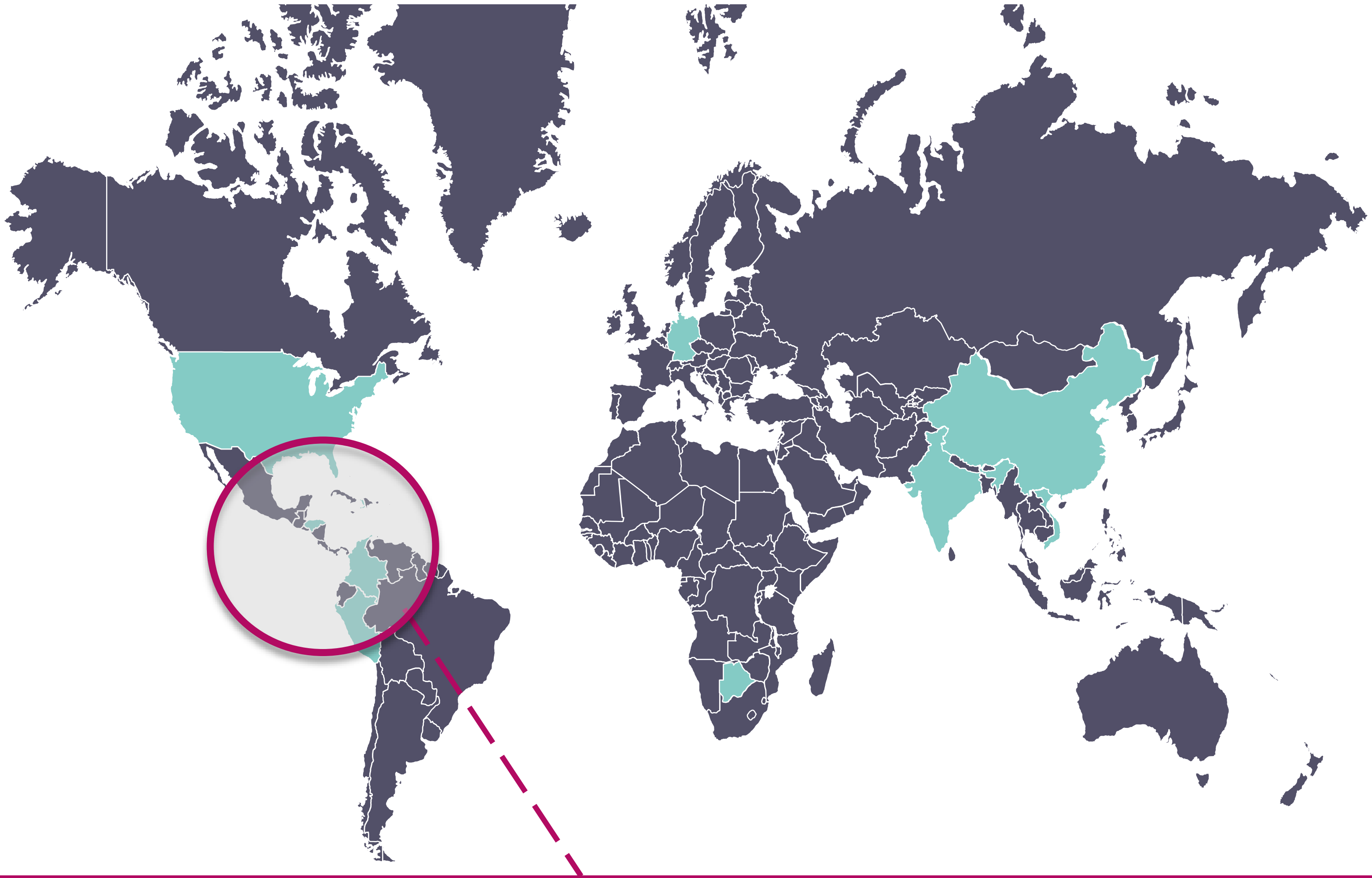
Outside the Box: Destination and Origin Countries

Value Chain



Analysis of Origin Countries

Country	TEU	Installed Capacity (m^3)
Philippines	2551839	0.100
United States	2223219	130.965
Brazil	1959223	19.057
China	1294633	105
India	1133232	150.229
Colombia	1070734	0.100
Costa Rica	823106	0.100
Netherlands	743474	13.7
Ecuador	741681	0.043
South Africa	665457	0.47



Therefore, Maersk should focus on developing its cold storage facilities in its top origin countries in South and Central America

Maersk Data

South and Central America present key growth opportunities

Outside the Box: Small to Medium Enterprises and Opportunities

Key Customer

Small to Medium Size Enterprises

Farmers, producers and Retailers

89%

Or total exporters in SAC

Needs of Small to Medium Enterprises

Flexibility

- Flexible contracts terms and prices
- Short term partnerships
- Ability to switch key processes

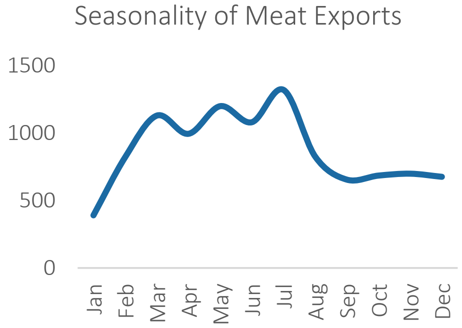
Low-cost

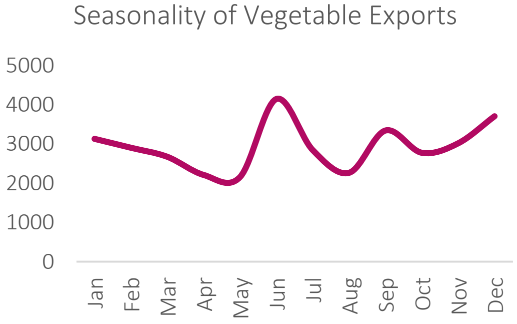
- Price of storage must be proportional to usage
- Ability to achieve cost savings

Sustainability

- Achieve cost savings in energy usage and other externalities
- Open to sharing networks

Seasonality gives rise to an opportunity to provide flexible solutions for SMEs





High cost of development creates a need for low-cost refrigerated warehouse

SMEs experience low ROI from building cold storage facilities

Due to a lack of cold storage infrastructure, leasing is costly

Hence, this has negative externalities for the community

240%

Increase in CO_2 emissions in cold chain by 2030

29%

Of fruit and vegetables are wasted in the supply chain

Strategy

Global Cold Chain Alliance 2018

South and Central America present key growth opportunities

EVA Consulting

Outside the Box: Small to Medium Enterprises and Opportunities

Key Customer

Small to Medium Size

89%

Seasonality gives rise to an opportunity to provide flexible solutions for SMEs

Seasonality of Meat Exports

1500

Seasonality of Vegetable Exports

5000

How can MAERSK build a storage capability that is flexible, low-cost and sustainable?

Flexibility

- Flexible contracts terms and prices
- Short term partnerships
- Ability to switch key processes

Low-cost

- Price of storage must be proportional to usage
- Ability to achieve cost savings

Sustainability

- Achieve cost savings in energy usage and other externalities
- Open to sharing networks

Hence, this has negative externalities for the community

240%

Increase in CO_2 emissions in cold chain by 2030

29%

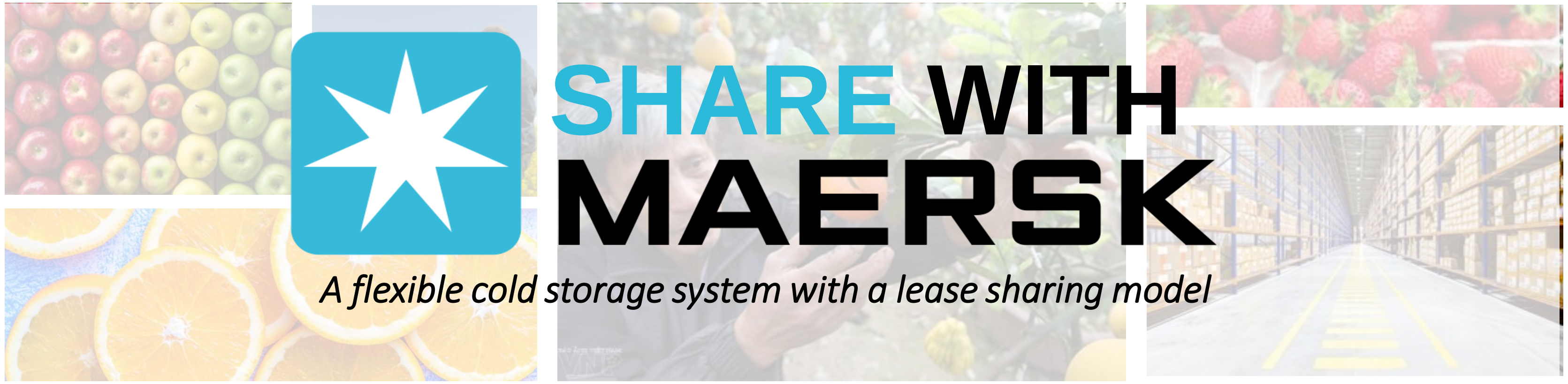
Of fruit and vegetables are wasted in the supply chain

Strategy

Global Cold Chain Alliance 2018

Share With Maersk will enable Maersk to enter the cold storage market

Outside the Box: Introducing Share with Maersk



How does it work?

Peak: Jan - Mar

Peak: April - Dec

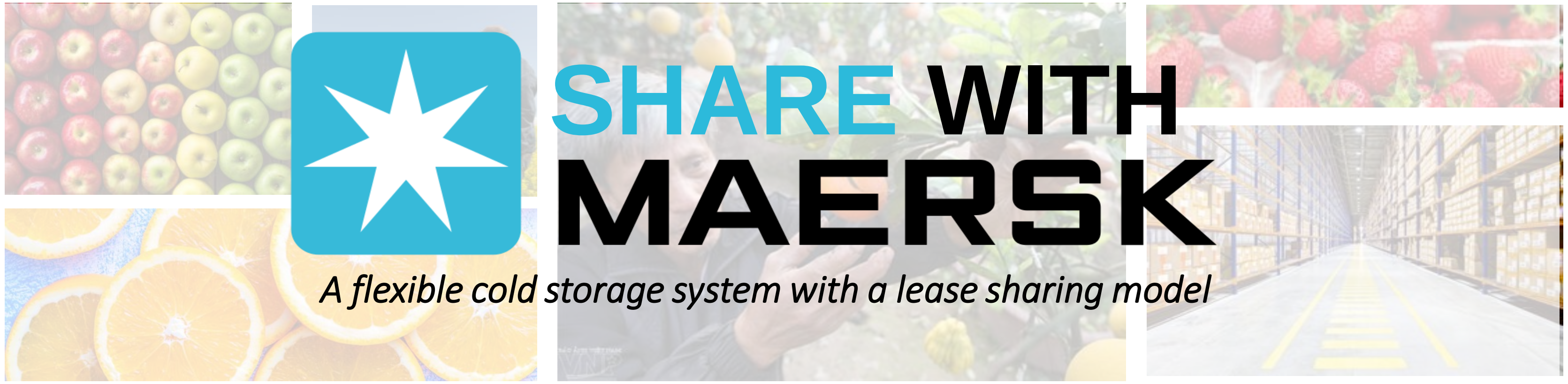
Single Shared Contract

Lower price of rent through sharing lease costs

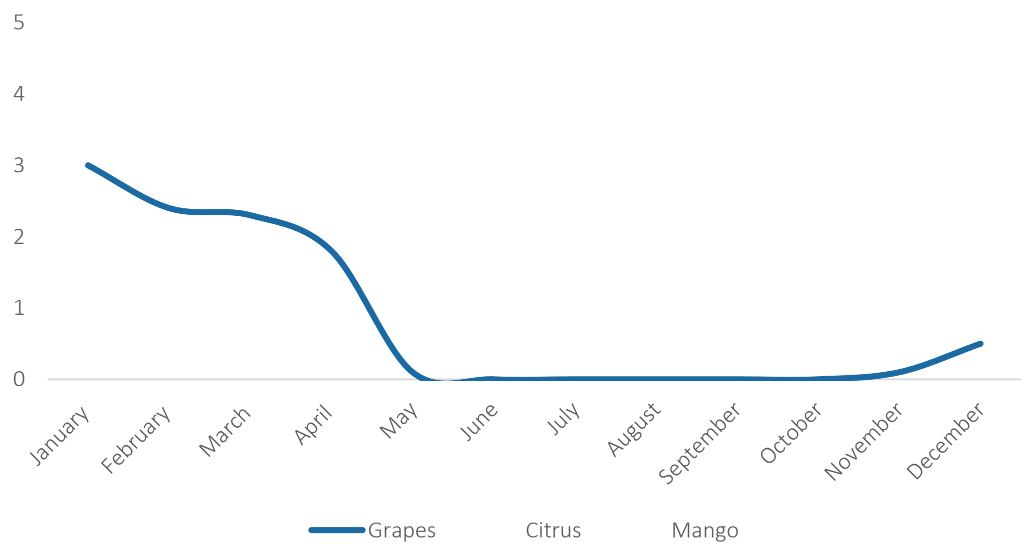
Reduced costs through sharing energy prices

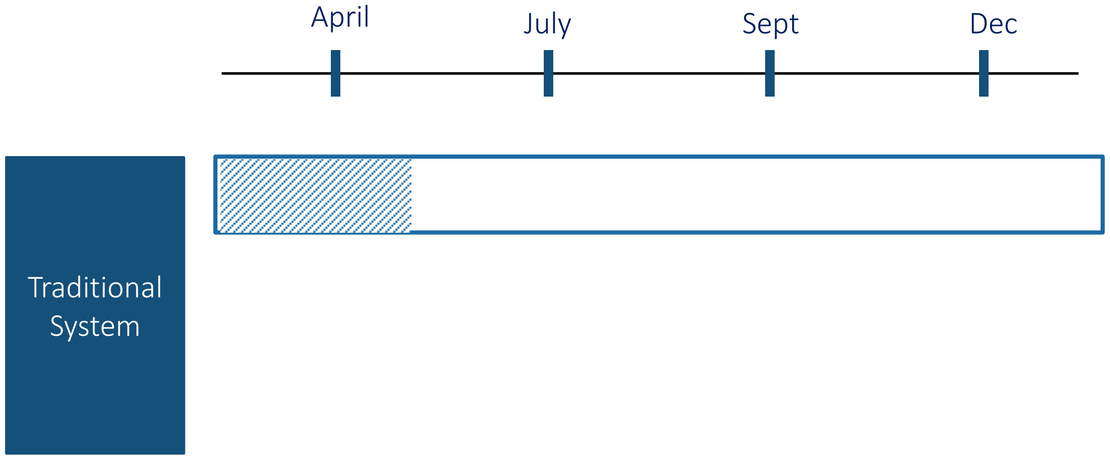
Share With Maersk will enable Maersk to enter the cold storage market

Outside the Box: Introducing Share with Maersk



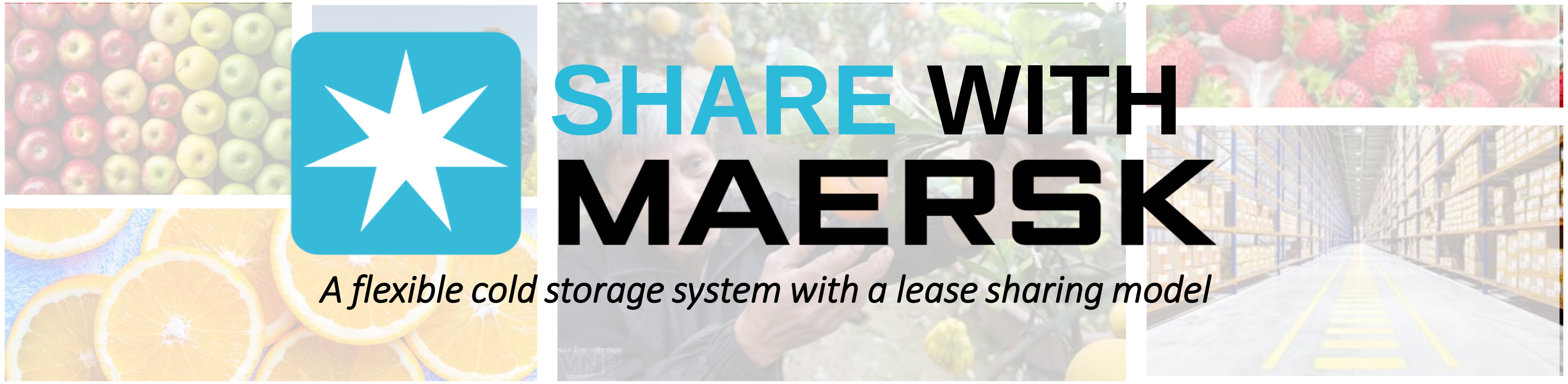
Seasonality of Fruits in Brazil



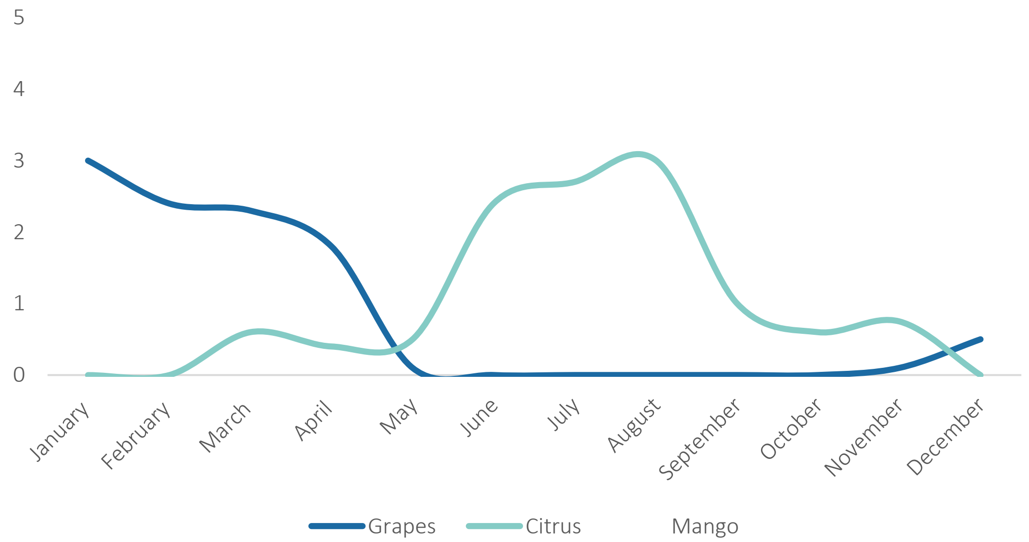


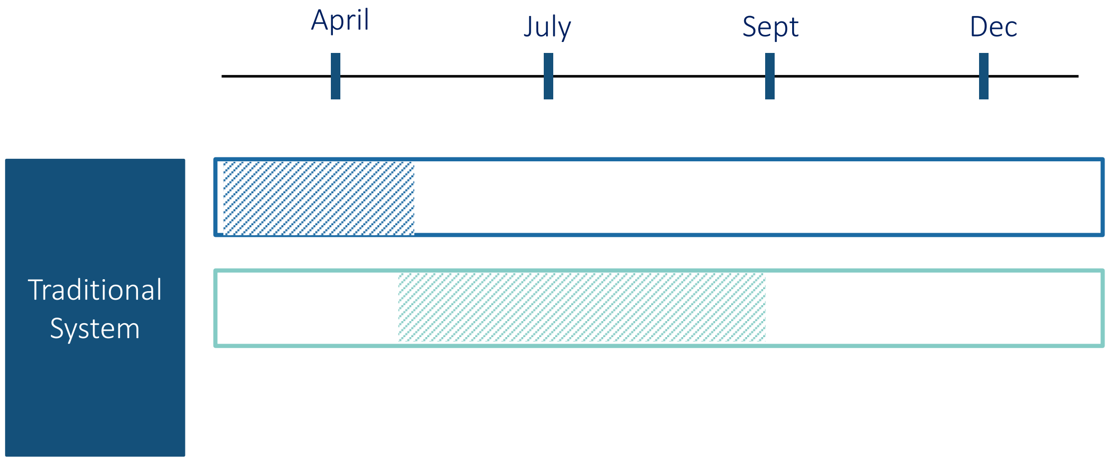
Share With Maersk will enable Maersk to enter the cold storage market

Outside the Box: Introducing Share with Maersk



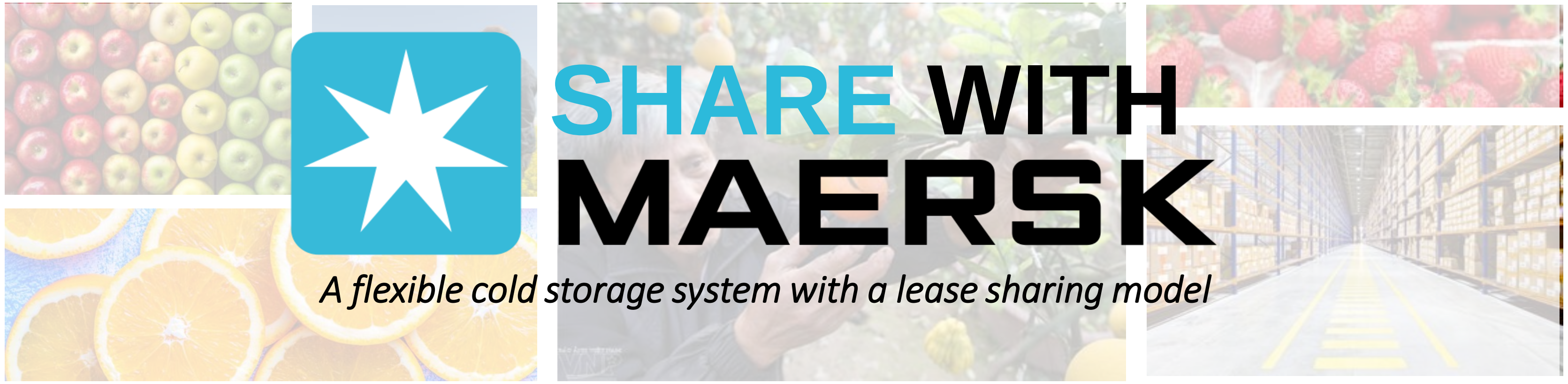
Seasonality of Fruits in Brazil



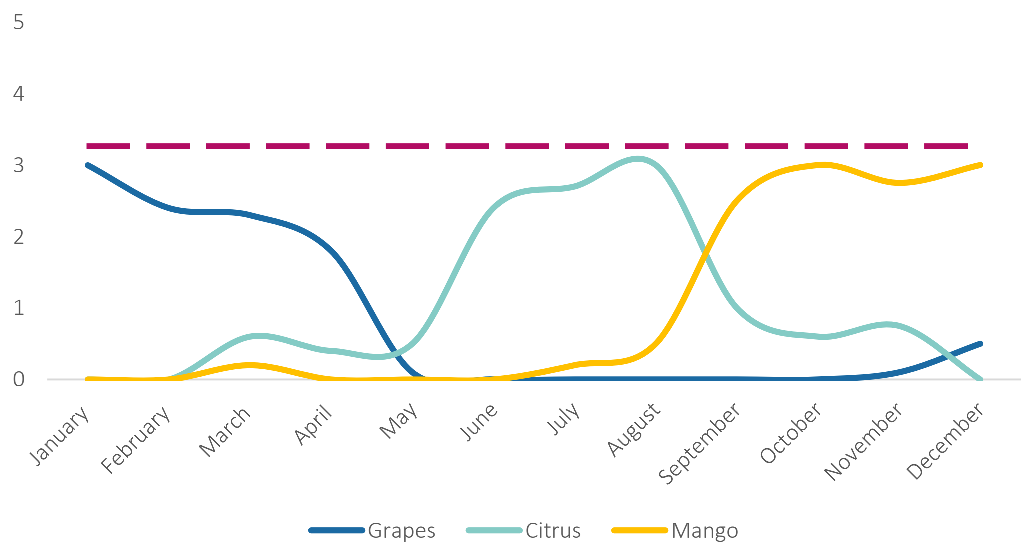


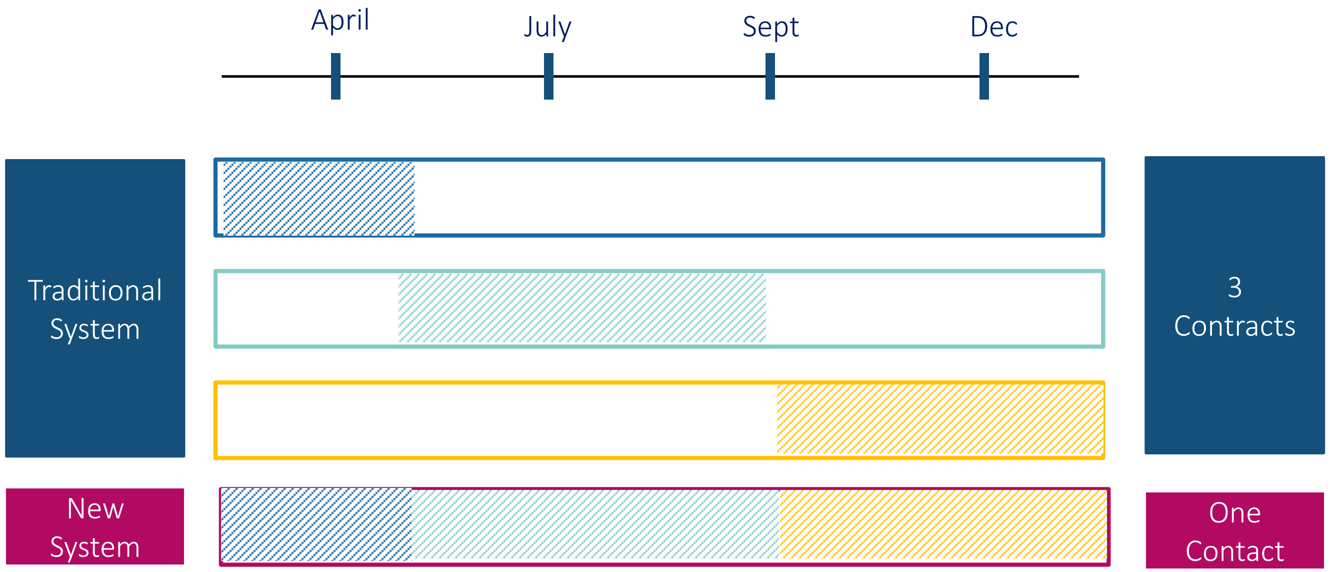
Share With Maersk will enable Maersk to enter the cold storage market

Outside the Box: Introducing Share with Maersk



Seasonality of Fruit Exports in Brazil





Maersk needs to examine key criteria to determine which market to pilot this program

EVA Consulting

MAERSK

Where to Play?

Criteria	Mexico	United States	Brazil	China	Ecuador	Costa Rica	India	Colombia
Forcasted demand of cold chains	Rank 3	Rank 1	Rank 6	Rank 2	Rank 36	Rank 46	Rank 10	Rank
Level of TEUs	16782839	2223219	1959223	1461561	741681	823106	1133232	1070734
Level of Infrastructure (/5)	4.4/5	4.8/5	3/5	4.7/5	2.5/5	2.5/5	3.8/5	3.7/5
Level of regulatory risk (/5)	3/5	0.3/5	2/5	2.5/5	3.5/5	3.5/5	0.9/5	2/5
Cold Storage Capacity	5.0 M/m^3	131 M/m^3	7.6 M/m^3	76 M/m^3	N/A	N/A	131 M/m^3	N/A
Number of Facilities	1200	1154	435	5694	150	231	7654	1 per city
Competitors	Some large players (Hutchinson Ports)	Many players (Americold, AGRO merchants group)	Some major players	Large players (Cosco, Hutichinson Ports)	Fragmented market	Fragmented market	Many players (Snowman Logistics, Cold star logistics)	Almagrio and few cold chain providers
Recommendation	2nd Wave	Pending success	Pilot	Pending success	Assess risk	Assess risk	Pending success	2nd Wave

International Trade Administration

Maersk can develop cold storage capabilities through partnerships and organic growth

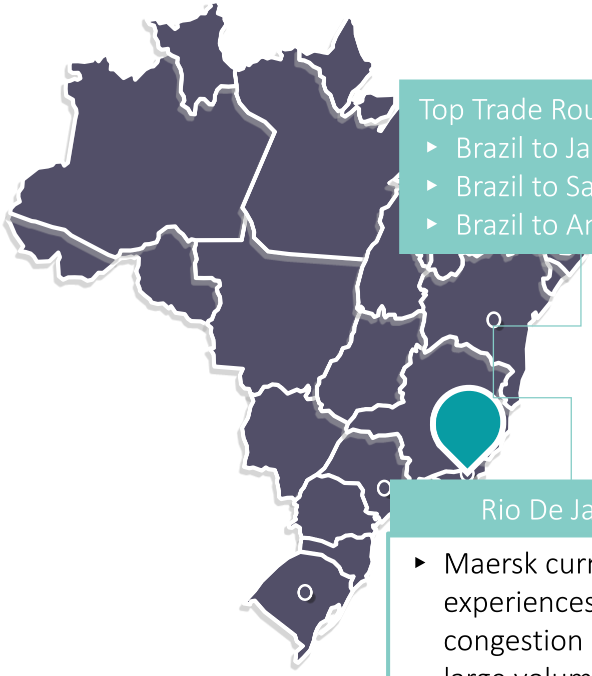
MAERSK

EVA Consulting

Outside the Box: Partnerships and Building Capabilities

Country Profile

Brazil
South America
Population: 209.3 million



Top Trade Routes
- Brazil to Japan
- Brazil to Saudi Arabia
- Brazil to America

Rio De Janeiro
- Maersk currently experiences congestion due to large volumes of incoming exports from competitors

Continuous growth is necessary to ensure the Share with Maersk is scalable across its entire business model

The Five Year Plan	Monday Morning (March 2019)	6 months (September 2019)	12 months (March 2020)	2 years (March 2021)	5 years (March 2024)
How to attain capability?	Partner	Build own warehouses			Expand to rest of the business model
Next Steps	brado logística - Reaching out to key personnel - Discuss synergies	- Target areas with high trade routes currently not covered by Maersk or Brado Logistica - Examples: Colombia and Mexico - Growing trade routes, however have a low penetration of - Conduct a pilot test to determine success of the project - 4 x Project Teams - 10 x Sales people to reach out to clients - Recruit dedicated architects and experts			- Examine dry refrigeration and other parts of the business - Conduct a roll out plan including: research, pilot program and incentives for clients to test products
Number	2 facilities by the end of 22	24 facilities			35 cold storage facilities, 12 countries, 5 new routes

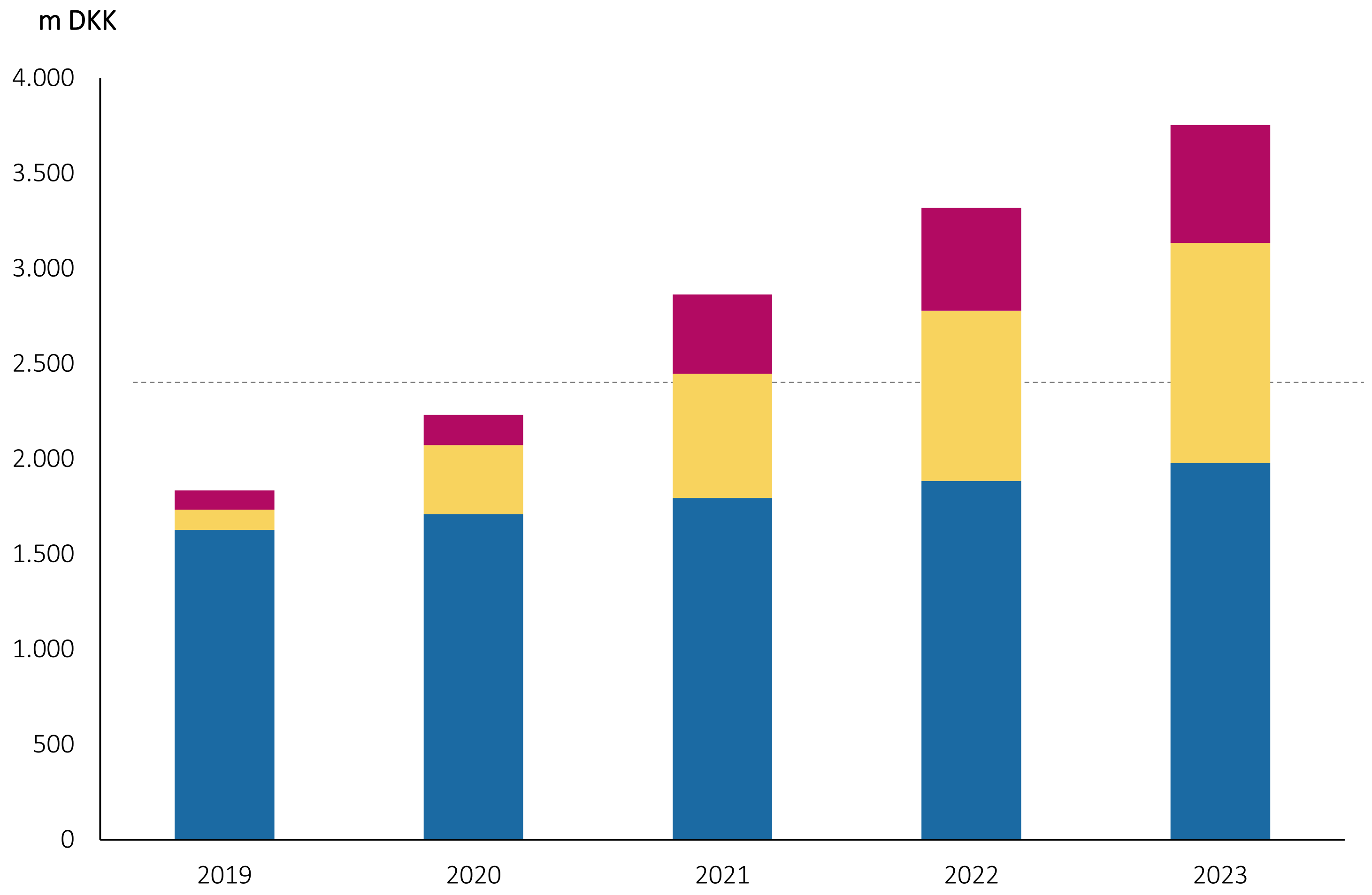
World Economic Forum and Global Cold Chain Alliance

Financials

Maersk will be able to unlock significant top line growth in the cold chain industry

Strategy Impact: Revenue

Revenue Growth, 2019F - 2023F



Status Quo

- Maersk strong leadership within reefer industry with 6.7% of total shipments in the space
- Steady growth projected in standard operations

Connecting the Dots

- Integration of end to end solution to provide significant value proposition for customers
- Revenue streams driven by brokerage capability and flat, premium revenue from secured customers

Outside the Box

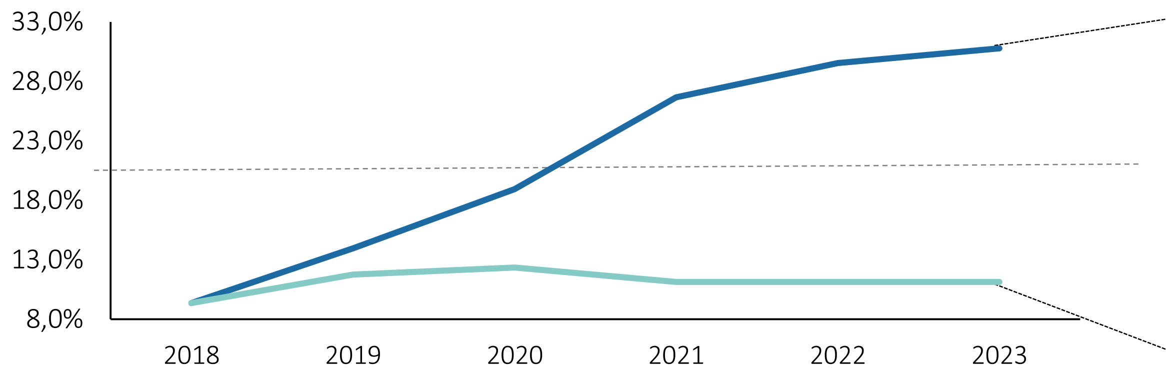
- Building cold storage capability in key emerging markets based on existing strengths
- Sharing model to maximise customer base and promote growth

1.8b DKK annual Incremental Revenue in 5 years

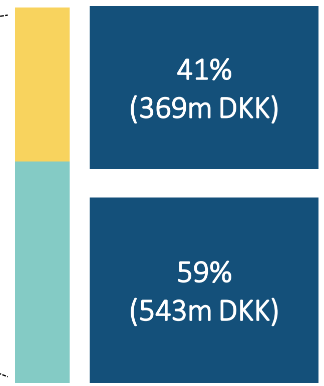
Connecting the Dots and Outside the Box will allow Maersk to improve profitability through offering a premium service

Strategy Impacts – Overall Margin

EBITDA Margin – Overall Strategy vs. Status Quo



Key Drivers



Connect the Dots

- More premium end to end solution
- Leads to stickier customers and higher margins

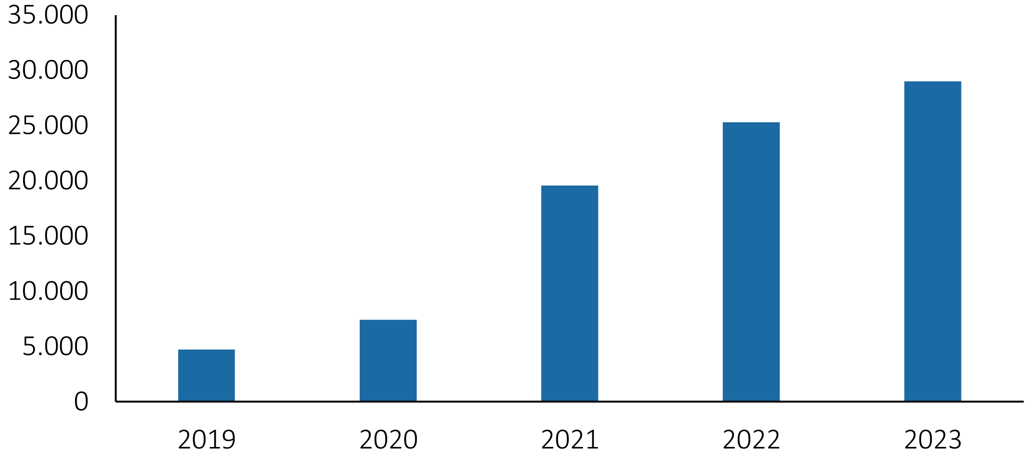
Outside the Box

- Primarily rent inclusion into end-to-end cost for customers
- Costs associated with initial set up

Key Drivers

Initial Partnership in 2019	Staggered acquisition plan in key markets
Terminal 430,000sqm lettable storage area	1 TEU = 160 ft squared

Cold Storage Capacity (TEUs)

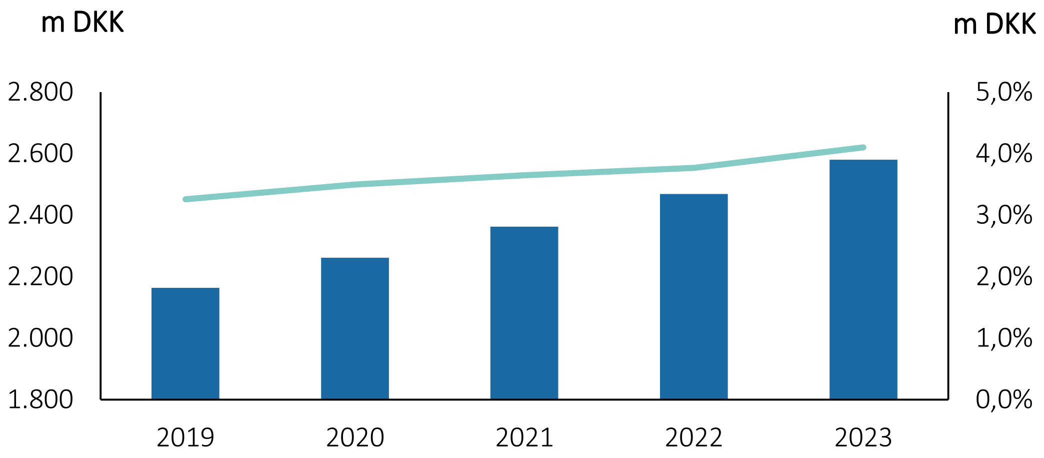


Impact

The upfront cost of the strategies provide a sustainable and scalable means of capability growth

Overall Impact - Cost Analysis

End to End Capability Production



430,000sqm
of cold storage capacity globally

30,000
TEU's stored

Strategy Cost Structure

	Connect the Dots	Outside the Box
Drivers	Cost of building digital capability and platform Integration with existing infrastructure and human capital Hiring of Point of contacts for solutions	Cost of initial partnerships with regional players Cost of acquisition and redevelopment of cold storage
Costs	140.0m DKK	115.0m DKK

255m DKK upfront cost

Impact

Implementing our strategy will allow Maersk to achieve its strategic impacts

Summary and Rationale

Achieved

1 Increased Stickiness for Customers

- Providing a unique value-adding data driven service beyond mere logistics
- Significant barriers to switching created through reliance on seamless experience

30,000 TEUs of Cold Storage Capacity

2 Increased Margins

- Provision of premium product allow higher capacity for pricing
- Value creation for customers allow for sizable value based pricing in line with Pilot

EBITDA Margin of 31% by 2023

3 Increased Top Line Revenue

- Vertical Integration into Cold Chain
- Provision of additional services to extend revenue streams and opportunity for up selling, FFW and conversion

1.8b DKK in Incremental Revenue

The recommendations involve some risk for Maersk…

Risk Analysis

Key risks matrix

1 Low uptake from customers

2 Shift in organisational focus

3 Inability to secure partners for cold storage roll out

4 Significant cost overruns in building cold storage capability

5 Political inhibitions on cold storage strategy

6 Competitors undertaking similar solution

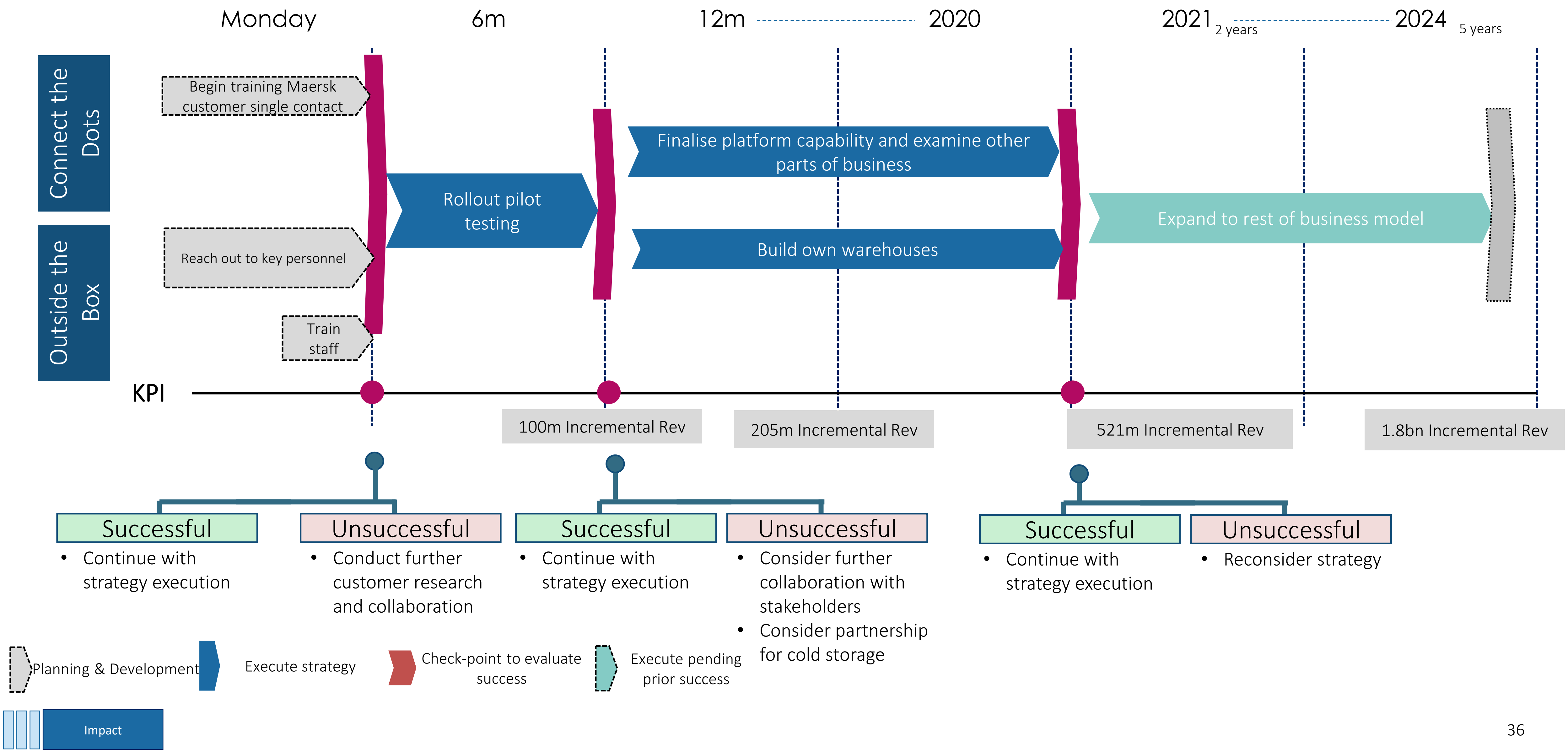
...but Maersk is capable of managing all of these

Risk analysis

Silo	Rank	Risk	Inbuilt mitigations	Recommended contingency response
Strategic	1	Low uptake from customers	• Dynamic and value based pricing will allow customers to see value prior to upfront cost	• Creation of tiered system to assign insights and capability based on capacity to pay
Strategic	2	Shift in organisational focus	• FTE training programs • Stakeholder management meetings	• Continued investment in core business lines
Strategic	6	Competitors undertaking similar solution	• Long-term partnerships limit likelihood • First mover advantage to build on existing strengths	• Maintain R&D investment to further innovate harder to replicate technology
Strategic	3	Inability to secure partners for cold storage roll out	• End to end solution to provide significant value to partners with built in sales force	• Acquisition strategy to buy and build
Operational	4	Significant cost overruns in building cold storage capability	• Insurance inclusions within foreseen costs to prevent bill shock	• Access cash on hand or low D/E ratio to maximise financing capacity
Regulatory	5	Political inhibitions on cold storage strategy	• Origin countries open to building greater export channels	• Mitigatory expansion strategies based on other factors as outlined above

Implementation timeline

EVA
Consulting



Executive Summary

Analysis: Executive Summary

Question

How can Maersk leverage its global leadership in reefer shipping to pioneer the global cold chain market in the next 5 years?

Considerations

1. Threat of commoditisation in cold chain
2. Poor customer experience within existing value chain
3. Lack of end-to-end capability

Strategy

Connecting the Dots
- Digital analytics solutions across an end-to-end integrated supply chain
- New MyMaersk Metrics feature built onto the existing MyMaersk platform

Outside the Box
- Developing cold-storage capabilities in origin destinations
- Lease sharing model targeted towards SME customers within region

Impact

30,000 TEU
Cold Storage Capacity

31% EBITDA Margin
by 2023

1.8b DKK annual
Incremental Revenue

Q&A

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Financials I – Status Quo

EVA Consulting

Maersk - Strategy Model (m DKK)	2018	2019F	2020F	2021F	2022F	2023F
1. Status Quo	**2018**	**2019F**	**2020F**	**2021F**	**2022F**	**2023F**
Total Revenue	256,788.33	266,237.11	288,829.57	306,398.26	323,250.17	336,180.18
% Revenue Growth		3.7%	8.5%	6.1%	5.5%	4.0%
Gross Profit	24,027.63					
% Gross Margin	9.4%					
EBITDA	24,027.63	31,295.25	35,682.42	34,179.52	36,059.40	37,501.77
% EBITDA	9.4%	11.8%	12.4%	11.2%	11.2%	11.2%
EBIT	2,145.44	13,186.40	15,145.75	11,267.50	14,949.37	15,179.59
% EBIT	0.8%	5.0%	5.2%	3.7%	4.6%	4.5%
Net Income	20,855.54	6,933.18	11,676.66	15,083.55	15,913.15	16,549.67
% Net Income	8.1%	2.6%	4.0%	4.9%	4.9%	4.9%
Number of Outstanding Shares	20.76	20.76	20.76	20.76	20.76	20.76
Diluted EPS	**-63.13**	**433.81**	**555.96**	**726.71**	**766.68**	**797.34**

Financials II – Connect the Dots Rvenue

2.	Connect the Dots	2018	2019F	2020F	2021F	2022F	2023F
2.1.	**Connect the Dots - Strategy Revenue**	**2018**	**2019F**	**2020F**	**2021F**	**2022F**	**2023F**
		TEUs (per Week)	**Value per Week (b DKK)**				
	South & Central America-Europe	2,302,936	970.6	1,014	1,060	1,108	1,157
	% Growth		4.5%	4.5%	4.5%	4.5%	4.5%
	South & Central America-North America	1,792,590	755.5	789	825	862	901
	% Growth		4.5%	4.5%	4.5%	4.5%	4.5%
	South & Central America-Far East	1,038,611	437.7	457	478	500	522
	% Growth		4.5%	4.5%	4.5%	4.5%	4.5%
	Total Access	**Total Access**	**2163.8**	**2261.1**	**2362.9**	**2469.2**	**2580.3**

Commodities (Americas)	**TEUs**	**Average Value**	**Total Value in DKK (m)**
Fruit and nuts	1,613,422	65,000	686,914
Vegetables	50,222	65,000	21,382
Beverages	30,324	65,000	12,910
Berries	26,277	65,000	11,187
Fish	23,701	50,000	7,762
Meat	22,038	50,000	7,217
Shellfish	15,636	50,000	5,121
Foodstuff	9,934	40,000	2,603
Plants	291	40,000	76
Dairy products	81	65,000	34
Pharmaceuticals	35	65,000	15
Tile, stone and glass	28	40,000	07
Oils and fats	28	40,000	07
Average Shipment Value		**421,446**	**58,095**
Brokerage Fees	2.75%		

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Financials III – Connect the Dots Revenue

EVA Consulting

Brokerage Fees	2.75%						*Average Customs Brokerage Fee less one price discount*
Total Addressable Market		20,775,675,760	49,602,111,836	62,181,033,072	64,979,179,560	67,903,242,641	*Total Annual Fees * Transported Weight*
Conversion through Entry Locations	0.00%	1.00%	2.00%	3.00%	4.00%	5.00%	
% of Goods Flow in Location X							
% of Goods Flow in Location Y							
Serviceable Addressable Market		207,756,758	992,042,237	1,865,430,992	2,599,167,182	3,395,162,132	
Cost of Brokerage as % of Total SC Costs							
Overlap & Pricing		33.00%	33.00%	33.00%	33.00%	33.00%	
Market Value of Goods (per Rfr)		65,000.00	65,000.00	65,000.00	65,000.00	65,000.00	
Raw Value of Goods		20,000.00	20,000.00	20,000.00	20,000.00	20,000.00	
Value Added		45,000.00	45,000.00	45,000.00	45,000.00	45,000.00	
Prevention of Loss		25.00%	25.00%	25.00%	25.00%	25.00%	
Number of Serviced TEUs		3,196.26	3,196.26	3,196.26	3,196.26	3,196.26	
Integrated Incremental Revenue		**104,517,630**	**363,331,838**	**651,550,128**	**893,683,071**	**1,156,361,404**	
Addressable Cold Chain Size		50,000,000,000	55,000,000,000	60,500,000,000	66,550,000,000	73,205,000,000	
% Growth		10%	10%	10%	10%	10%	
Implied Incremental Market Share		0%	1%	1%	1%	2%	
Existing Reefer Revenue		1,628,033,000	1,709,434,650	1,794,906,383	1,884,651,702	1,978,884,287	
Growth		5%	5%	5%	5%	5%	
Total Maersk Line		24,299.00	24,299.00	24,299.00	24,299.00	24,299.00	
Assumed % Reefer		6.70%	6.70%	6.70%	6.70%	6.70%	
Implied Total Market Share		**3.3%**	**3.5%**	**3.7%**	**3.8%**	**4.1%**	

Financials IV – Connect the Dots Costs

EVA Consulting

2.2. Connect the Dots - Strategy Costs	2018	2019F	2020F	2021F	2022F	2023F	
Development Costs							
FTE Costs		11,250,000	12,375,000	13,500,000	14,625,000	15,750,000	
Number of FTEs		50	55	60	65	70	
Average Developer Salary		225,000	225,000	225,000	225,000	225,000	
Server and Testing Costs		60,000	183,600	187,272	191,017	194,838	
Server Bandwidth Requirement		2	3	3	3	3	
Cost per Server Unit		5,000	5,100	5,202	5,306	5,412	
Months Operating		6	12	12	12	12	
Total Development Costs		**11,310,000**	**12,558,600**	**13,687,272**	**14,816,017**	**15,944,838**	
Integration Costs							
Equipment Integration Costs		28,000,000	38,934,260	27,927,772	15,322,583	35,000,000	
Consolidation Costs		3,500,000	3,500,000	3,500,000	3,500,000	3,500,000	
Number of Consolidation Points		8	11.12	7.98	4.38	10.00	
Installation Costs		13,750,000	15,125,000	16,500,000	17,875,000	19,250,000	
Number of FTEs		50	55	60	65	70	
FTE Salaries		275,000	275,000	275,000	275,000	275,000	
Total Integration Costs		**41,750,000**	**54,059,260**	**44,427,772**	**33,197,583**	**54,250,000**	
Operating Costs							
FTE Costs		17,500,000	21,875,000	26,250,000	30,625,000	35,000,000	
Number of Brokerage FTEs		100	125	150	175	200	*Vandergrift Case Study*
Average FTE Salary		175,000	175,000	175,000	175,000	175,000	*Glassdoor*
Average Customs Brokerage Margin		18%	18%	18%	18%	18%	
Indicative Customs Charges		65%	65%	65%	65%	65%	
Total Operating Costs		**85,436,460**	**258,040,695**	**449,757,583**	**611,518,996**	**786,634,913**	
Total CapEx		53,060,000	66,617,860	58,115,044	48,013,600	70,194,838	
Total OpEx		85,436,460	258,040,695	449,757,583	611,518,996	786,634,913	
Total Strategy Cost		**138,496,460**	**324,658,555**	**507,872,627**	**659,532,596**	**856,829,750**	

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Financials V – Outside the Box Revenue

EVA Consulting

3.	Outside the Box	2018	2019F	2020F	2021F	2022F	2023F
3.1.	Outside the Box - Strategy Revenue	2018	2019F	2020F	2021F	2022F	2023F
	South & Central America						
	Capacity Utilisation Rate						
	Upside	95%					
	Neutral	80%					
	Downside	65%					
	Brazil						
	Yield	5%	5%	5%	5%	5%	5%
	Capitalisation Rate	5%	5%	5%	5%	5%	5%
	Total Facilities		2	3	5	5	5
	Average NLA (sqm)		20,000	20,000	20,000	20,000	20,000
	Rent per sqm	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00
	Net Income		**57,600,000**	**86,400,000**	**144,000,000**	**144,000,000**	**144,000,000**
	Colombia						
	Yield	6%	6%	6%	6%	6%	6%
	Capitalisation Rate	6%	6%	6%	6%	6%	6%
	Total Facilities			1	3	4	5
	Average NLA (sqm)		20,000	20,000	20,000	20,000	20,000
	Rent per sqm	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00
	Net Income	**0**	**0**	**28,800,000**	**86,400,000**	**115,200,000**	**144,000,000**

Financials VI – Outside the Box Revenue II

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EVA Consulting

Mexico						
Yield	6%	6%	6%	6%	6%	6%
Capitalisation Rate	6%	6%	6%	6%	6%	6%
Total Facilities	1	1	1	2	3	3
Average NLA (sqm)		30,000	30,000	30,000	30,000	30,000
Rent per sqm	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00
Net Income	**0**	**43,200,000**	**43,200,000**	**86,400,000**	**129,600,000**	**129,600,000**
China						
Yield	4.50%	4.50%	4.50%	4.50%	4.50%	4.50%
Capitalisation Rate	4.50%	4.50%	4.50%	4.50%	4.50%	4.50%
Total Facilities		0	0	2	3	4
Average NLA (sqm)		35,000	35,000	35,000	35,000	35,000
Rent per sqm	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00	1,800.00
Net Income	**0**	**0**	**0**	**100,800,000**	**151,200,000**	**201,600,000**
Total Net Income	**0**	**100,800,000**	**158,400,000**	**417,600,000**	**540,000,000**	**619,200,000**
Total sqm of cold storage	**Total sqm of cold storage**	70,000	110,000	290,000	375,000	430,000
	Total TEU Cold Storage Capability	4,721	7,419	19,558	25,291	29,000

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Financials VI – Outside the Box Costs

EVA Consulting

3.2. Outside the Box - Strategy Costs	2018	2019F	2020F	2021F	2022F	2023F	
Market Entry Costs							
Sales Force							
FTE Costs		3,750,000	3,750,000	3,750,000	3,750,000	3,750,000	
Number of FTEs		25	25	25	25	25	
Average FTE Salary		150,000	150,000	150,000	150,000	150,000	
Acquisition Costs		52,500,000	87,500,000	210,000,000	262,500,000	297,500,000	
Land Acquisitions		10,000,000	10,000,000	10,000,000	10,000,000	10,000,000	
Cost of Repurposing		7,500,000	7,500,000	7,500,000	7,500,000	7,500,000	
Number		3	5	12	15	17	
Data Integration Costs		2,150,000	2,460,000	1,950,000	1,850,000	1,750,000	
Monthly Cost		215,000	205,000	195,000	185,000	175,000	
Months		10	12	10	10	10	
Total Entry Costs		58,400,000	93,710,000	215,700,000	268,100,000	303,000,000	
Cold Storage Operating Costs							
Maintenance							
Maintenance as % of Purchase Price	*4%*	*4%*	*4%*	*4%*	*4%*	*4%*	*ResearchGate, Apicol*
Average WALE	*15 years*						*EY*
Average Lifetime	*25 years*						*EY*
Operating Ex		6,552,000	9,108,000	18,009,000	17,465,625	21,672,000	
% of Revenue		*6.50%*	*6%*	*4%*	*3%*	*3.50%*	
Partnership Costs		*50%*	*30%*	*13%*	*10%*	*9%*	
Total Cold Storage Operating Costs		56,952,000	56,628,000	70,209,000	71,465,625	76,307,294	
Total CapEx		58,400,000	93,710,000	215,700,000	268,100,000	303,000,000	
Total OpEx		56,952,000	56,628,000	70,209,000	71,465,625	76,307,294	
Total Strategy Cost		**115,352,000**	**150,338,000**	**285,909,000**	**339,565,625**	**379,307,294**	

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Financials VII – Incremental Impact & DCF

EVA Consulting

4.	Incremental Impact (m)	2019F	2020F	2021F	2022F	2023F	
	Revenue						
	Status Quo	1,628,033,000	1,709,434,650	1,794,906,383	1,884,651,702	1,978,884,287	
	Connect the Dots	104,517,630	363,331,838	651,550,128	893,683,071	1,156,361,404	
	Outside the Box	100,800,000	158,400,000	417,600,000	540,000,000	619,200,000	
	Total Revenue	**1,833,350,630**	**2,231,166,488**	**2,864,056,510**	**3,318,334,772**	**3,754,445,691**	
	EBITDA						
	Status Quo	191,369,663	211,186,025	200,226,470	210,237,793	220,749,683	
	Connect the Dots	19,081,171	105,291,143	201,792,545	282,164,075	369,726,491	41%
	Outside the Box	43,848,000	101,772,000	347,391,000	468,534,375	542,892,706	59%
	Total EBITDA	**254,298,834**	**418,249,168**	**749,410,015**	**960,936,243**	**1,133,368,880**	
		14%	**19%**	**26%**	**29%**	**30%**	

5.	Discounted Cash Flow	2018F	2019F	2020F	2021F	2022F
	Revenue	205,317,630	521,731,838	1,069,150,128	1,433,683,071	1,775,561,404
=	EBITDA	62,929,171	207,063,143	549,183,545	750,698,450	912,619,197
-	CapEx	111,460,000	160,327,860	273,815,044	316,113,600	373,194,838
	FCFF	**-48,530,829**	**46,735,284**	**275,368,501**	**434,584,849**	**539,424,359**
	NPV FCFF	**756,432,014**				

Financials VIII – WACC Calculation

EVA Consulting

6. WACC Calculation

A.P. Moller - Maersk

Total Debt/Equity	35.6%
Total Debt/Capital	26.2%

Comparable Firms	Unlevered Beta	Cost of Equity	After Tax Cost of Debt	D/E Ratio	Debt/Value	Weighted Average Cost of Capital
DSV A/S	0.68	27.20%	5.23%	44%	31%	
COSCO Shipping	0.99	12.38%	5%	124%	55%	
A.P. Moller - Maersk	0.85	12.80%	7.50%	35.6%	26.2%	11.41%
Risk Premium	10%					
Effective WACC	**12.5%**					

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Financials IX – Sensitivity Analysis

EVA Consulting

7. Sensitivity Analysis

NPV by WACC and Occupancy of Cold Storage

Occupancy Rate	WACC 7.5%	10.0%	12.5%	15.0%	17.5%
95%	1,131.4	1,028.6	937.3	856.1	783.6
85%	989.2	898.3	817.7	745.9	682.0
80%	918.1	833.2	757.8	690.9	631.1
75%	847.0	768.0	698.0	635.8	580.3
65%	704.8	637.8	578.4	525.6	478.6

NPV by Terminal Conversion and Brokerage Rate

Terminal Conversion	Brokerage Rate 2.25%	2.50%	2.75%	3.00%	3.25%
3.00%	394.0	432.9	471.7	510.5	549.3
4.00%	510.5	562.3	614.1	665.8	717.6
5.00%	627.0	691.7	756.4	821.1	885.9
6.00%	743.5	821.1	898.8	976.5	1,054.1
7.00%	860.0	950.6	1,041.2	1,131.8	1,222.4

Appendix – DMAIC Optimisation Process

Define
Analyse the process to identify the cause-effect relationships between inputs and outputs

Measure
Collect relevant data about the process and the problem

Analyse
Define the problem and the ideal target to be achieved

Improve
Establish standards and controls to sustain improvements in the long run

Control
Determine the optimum values for key contributing process inputs

Source: Lean Six Sigma Methodology

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Appendix – Capital Raising Process

Mandated Financial Advisor → Financial Close

Materials Preparation	Investor Meetings	Term Sheet Negotiations	Investor Due Diligence	Finalised Legal Agreements
Creation of investor package – Lender Presentation – Financial Model Strategy and business model enhancement	Make introductions Facilitate investor discussions Provide access to company information Secure investors and indicative term sheets	Negotiate higher valuation/more favourable terms Manage process Begin sourcing company counsel	Prepare Management team Present details and clarify investor queries	Work with company counsel for more favourable terms and undertake legal analysis Manage overall process
1 Month	1 Month	2 Weeks	2 Weeks	4 Weeks

4 Months

Source: EY Transaction Advisory Services

Appendix

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Appendix – Funding Options

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Priority

Risk

Capital Stack	Considerations	Key Metrics
Cash	Leads to reduction in liquid assets Requires consideration of working capital and future growth expenditure requirements Cheapest source of capital vs. debt and equity	Cash on Hand Future Cash Requirements
Senior Debt	Revolving/Term Loans Collateralized with tangible assets May impose covenants, i.e Net Leverage Ratio, Interest Coverage Ratio	Unencumbered Assets EBITDA & Headroom
Mezzanine Debt	2nd Lien term loans Require positive cash flow Long term capital with flexible covenants Includes warrants but is less dilutive than equity	Interest Service Capacity (EBITDA/Interest Expense) Potential Covenant Headroom
Preferred Equity	Complicated provisions incl. liquidation, participation, voting rights Generally carries a dividend Long term, non-amortising capital Highly dilutive	Existing and Desired Ownership Cost of Capital Implications
Common Equity	Highly dilutive and results in loss of control Board seats Long term, non-amortising capital	Existing and Desired Ownership Cost of Capital Implications

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Appendix – Blockchain

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Should Blockchain be used?

Decision Point	Outcome (if No)	Rationale
Does the use case involve a database?	Do not use Blockchain	Unnecessary
Yes		
Will there be numerous users updating my database?	Use a centralised database	Costly and inefficient
Yes		
Do these users need to trust each other?	Use multiple copies of centralised database	Cheaper to establish and maintain
Yes		
Are there problems caused by the use of a central/third party entity?	Use a third party/intermediary	Minimise R&D and time requirements
Yes		
Do transactions depend on/interact with each other?	Use a Master/Slave Database	Easier to administer and control
Yes	Use Blockchain	Requires decentralised blockchain technology

Appendix – Change Management

- Establishing performance goals will allow for clearer relationship management between existing and new tenants/clients

4 Pillars to Successful Integration

Intent	Interaction	Integration	Insight
Strong and well articulated intent in conjunction with a well designed approach to information management	Well understood history in conjunction with interactive analysis and management Regularly incorporated feedback mechanisms	A common digital platform log in across the online platform and dash board to monitor progress or successful care workers by region	Integrate with external data sources including client data to predict future needs and behaviors

Source: Strategic Project Partners – Driving Impact through Operations (2015)

MAERSK

EVA Consulting

Traditional international carriers offer similar product offerings

	Container shipping	Container terminals	Logistics	Bulk shipping	Cruise shipping
MAERSK					
msc					
CMA CGM					
COSCO SHIPPING					

MAERSK
- Global integrator of container logistics
- Investments into blockchain network

MSC
- Investing in inland assets including warehouses, trucks, locomotives, and barges under Medlog brand

CMA CGM / COSCO SHIPPING
- Collaborated with COSCO and nine other carriers to launch new blockchain initiative as a Global Network
- Customers can digitise and organise dangerous goods documents
- Automatically connect with relevant parties to streamline the approval process

Complexity of reefers shipping has a risk profile for businesses and customers

Risk for shipping companies

1. Low control over delays within the chain
2. Lack of global network and scaiability
3. Significant technological requirements

Risk for customers (producers)

1. Significant impact in the event of food loss during the chain